

A Digital Twin for the Microstructure of Irradiated Zirconium Alloys

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Abstract

A self-contained, modular cluster-dynamics (CD) framework is presented for the long-time evolution of radiation-induced defects in zirconium and its alloys, with detailed parameterization for Zircaloy-4 and pure α -Zr. The formulation is built bottom-up from a spectrum-dependent description of in-cascade production of the mono-atomic and clustered self-interstitial atom (SIA) and vacancy populations, with explicit basal and prism-plane components and 1-D glide of small SIA clusters along $\langle 11\bar{2}0 \rangle$. A full set of master equations, with explicit Frenkel-pair mass-conservation laws, tracks four cluster populations: prismatic interstitial and vacancy $\langle a \rangle$ -loops on $\{10\bar{1}0\}$ habit planes with Burgers vector $\frac{1}{3}\langle 11\bar{2}0 \rangle$, basal stacking-fault pyramids, and faulted vacancy $\langle c \rangle$ -component loops on $\{0001\}$ basal planes with Burgers vector $\frac{1}{6}\langle 20\bar{2}3 \rangle$, each with distinct capture-bias factors. State-space reduction uses a logarithmic-bin-moment method with two complementary closures (piecewise-constant and Galerkin shape-function), preserving mass conservation while reducing system size by orders of magnitude, and the implementation employs a tailored Woodbury-style preconditioner that exploits the bordered-banded sparsity of the cluster-dynamics Jacobian together with adaptive stiffness management. The framework is realized in the open-source RADCLUSTER code, with full Zircaloy-4 and pure- α -Zr parameter tables, conservation-law diagnostics, and verification against analytical limits. Applied across LWR cladding, CANDU pressure-tube, and pure-Zr conditions, the model reproduces the measured dose dependence of $\langle a \rangle$ -loop number density and mean size and captures the delayed threshold appearance of $\langle c \rangle$ -

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component loops that drives breakaway irradiation growth. Predicted growth strains agree with reactor and test-reactor data to within experimental scatter, supporting the use of the framework for quantitative microstructure prediction over component lifetimes.

Keywords: cluster dynamics, radiation damage, zirconium alloys, Zircaloy, $\langle a \rangle$ -loops, $\langle c \rangle$ -component loops, anisotropic diffusion

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1. Introduction

Zirconium and its alloys—Zircaloy-2 and Zircaloy-4 in light-water reactor (LWR) fuel cladding, Zr-2.5Nb in CANDU pressure tubes, and the more recent low-tin variants M5 and ZIRLO in pressurized-water reactor (PWR) service—operate under prolonged neutron exposure at displacement rates of 10^{-8} to 10^{-7} dpa/s and accumulate total damage of 5–30 dpa over a typical fuel-cycle lifetime [1, 2]. The microstructural changes that result—nucleation and growth of $\langle a \rangle$ prismatic loops, the comparatively delayed appearance of $\langle c \rangle$ -component basal vacancy loops, dislocation channeling under tensile load, anisotropic irradiation growth and creep, and stress-driven hydride reorientation—are the principal source of property degradation in Zr-based components, manifested as hardening, loss of uniform elongation, axial growth of fuel assemblies, lateral creep-down of pressure tubes, and delayed hydride cracking [3–6]. Predicting these property changes from first principles, across the joint envelope of dose, temperature, alloy composition, texture, and applied stress, is the central quantitative challenge of fuel-element and pressure-tube qualification.

The cluster-dynamics (CD) framework, introduced in its modern form during the late 1970s and early 1980s [7–11], remains the workhorse mesoscopic theory for radiation-induced microstructure evolution. CD treats each defect species and size as a separate population whose concentration evolves under a coupled set of master rate equations balancing in-cascade production, diffusion-controlled reactions, thermal emission, and absorption at fixed

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sinks [12–14]. Its three defining advantages are (i) direct access to time scales (10^0 – 10^8 s) that lie far beyond molecular dynamics [15, 16]; (ii) explicit resolution of the cluster-size distribution—and therefore of every quantity that enters the macroscopic property models, from sink-strength to growth-strain calculations [17, 18]; and (iii) modest computational cost relative to spatially resolved kinetic Monte Carlo or atomistic simulation, which together make CD the natural meeting point between atomistic-scale parameterization [19, 20] and engineering-scale prediction [5, 21].

In zirconium specifically, CD has been used to follow the dose evolution of $\langle a \rangle$ -loop number density and mean size, to test nucleation criteria for $\langle c \rangle$ -component basal vacancy loops, and to convert resolved cluster-size distributions into anisotropic growth strain through models of the Holt DAD/EID class [18, 19, 22]. The persistent challenge is that the principal defect populations in hcp Zr—prismatic interstitial $\langle a \rangle$ -loops on $\{10\bar{1}0\}$ habit planes with Burgers vector $\frac{1}{3}\langle 11\bar{2}0 \rangle$, and faulted vacancy $\langle c \rangle$ -component loops on $\{0001\}$ basal planes with Burgers vector $\frac{1}{6}\langle 20\bar{2}3 \rangle$ —differ in both habit plane and Burgers vector, with strongly anisotropic point-defect diffusivities and size-dependent capture biases that are not well captured by isotropic single-loop-family CD treatments inherited from body-centered cubic metals [23–25].

SIA migration in α -Zr is strongly anisotropic, with prism-plane mobility exceeding basal mobility by roughly an order of magnitude [19, 20]. Vacancy diffusion is similarly anisotropic, though more weakly. Most CD treatments of zirconium retain isotropic diffusion constants taken from the BCC literature, conflating the underlying anisotropy with effective sink-bias factors and obscuring the relationship between atomic-scale parameters and observed dimensional change.

Beyond a critical nucleus size, individual cluster equations are typically replaced by a Fokker–Planck (FP) representation [26, 27] or a grouping/binning scheme [17, 21]. Each closure has known limitations: FP closures lose accuracy at the discrete-to-continuous transition; grouping closures introduce mass-conservation errors unless the inter-bin flux is explicitly designed; and few formulations rigorously analyze the *accuracy* of the closure in a way that allows the user to trade fidelity for system size with quantified error.

The primary irradiation defects in hcp zirconium are vacancies, self-interstitial atoms, and their associated clusters. Because the hcp lattice is elastically and crystallographically anisotropic, defect clusters cannot be classified only by size. Their Burgers-vector character and habit plane also

determine their contribution to irradiation growth, hardening, and defect sink evolution. The experimentally observed dislocation-loop population in irradiated Zr and Zr alloys is commonly grouped into $\langle a \rangle$ loops and c -component loops [28–30]. The $\langle a \rangle$ loops have Burgers vector

$$\mathbf{b}_a = \frac{1}{3}\langle 11\bar{2}0 \rangle, \quad (1)$$

whereas the c -component loops contain a Burgers-vector component parallel to the hcp c axis.

For the cluster-dynamics formulation used here, the full crystallographic loop population is reduced to four principal cluster families:

$$I_n^a, \quad V_n^a, \quad S_n, \quad V_n^c. \quad (2)$$

Here, I_n^a denotes an interstitial $\langle a \rangle$ loop containing n excess self-interstitial atoms, V_n^a denotes a vacancy $\langle a \rangle$ loop containing n missing atoms, S_n denotes a basal stacking-fault pyramid (SFP) of n aggregated vacancies, and V_n^c denotes a vacancy-type c -component loop. This notation separates the defect character (interstitial or vacancy) from the crystallographic character ($\langle a \rangle$, basal pyramidal, or c -component).

The $\langle a \rangle$ loops are associated primarily with prismatic or near-prismatic habit planes. Atomistic simulations indicate that $\langle a \rangle$ loops may occur as either interstitial or vacancy loops, and that the energetics of alternative prismatic habit planes can be close; relaxation and rotation between prismatic variants can therefore occur during loop evolution [31]. In contrast, c -component loops in Zr are generally associated with basal habit planes and are commonly treated as vacancy-type loops. Atomistic studies and continuum arguments suggest that vacancy aggregation into basal platelets provides a natural precursor for the formation of basal c -component dislocation loops [32, 33]. Recent transmission electron microscopy observations of nanoscale vacancy platelets support this picture and provide direct experimental evidence for a vacancy-mediated pathway to basal loop formation [33].

This three-population representation is a deliberate simplification. It does not attempt to track every possible loop habit plane, fault structure, or mixed Burgers-vector state. Instead, it retains the four cluster classes most relevant to irradiation growth: prismatic $\langle a \rangle$ interstitial and vacancy loops, basal stacking-fault pyramids, and basal c -component vacancy loops. The two $\langle a \rangle$ populations set the balance of defect storage and recovery on the prismatic

systems; the SFPs collect cascade vacancy debris on the basal plane and, on unfauling, feed the c -component loops, which in turn carry the delayed basal damage responsible for high-fluence irradiation growth. The model state vector for the loop sector is therefore written as

$$\mathcal{C}_{\text{loop}} = \{I_n^a, V_n^a, S_n, V_n^c\}_{n=1}^{n_{\text{max}}} . \quad (3)$$

Figure 1 illustrates the reduced classification. The hcp crystal is represented schematically as a hexagonal prism, with hidden or interior prism edges shown by dashed lines. The prismatic $\langle a \rangle$ interstitial loop, I_n^a , is drawn as a solid platelet to indicate its interstitial character. The prismatic $\langle a \rangle$ vacancy loop, V_n^a , is drawn as a dotted platelet with similar crystallographic orientation but opposite defect character. The basal c -component vacancy loop, V_n^c , is drawn as a dotted basal platelet, and the basal stacking-fault pyramid, S_n , is drawn as a dotted three-dimensional pyramid seated on the upper basal plane. Thus solid outlines denote interstitial-type defects, dotted outlines denote vacancy-type defects, and the platelet or pyramid orientation identifies the dominant habit-plane class.

In the rate equations that follow, each of the four families is assigned separate formation, migration, transformation, and sink-absorption rates, together with the SFP-to- c -loop unfauling transfer. This lets the model distinguish ordinary $\langle a \rangle$ evolution from the delayed nucleation and growth of basal c -component damage while keeping the internal state vector small enough for large-scale parametric calculations.

The present formulation is designed to address each of these gaps in a single, modular framework whose components can be selected, replaced, or specialized depending on the alloy of interest and the target irradiation environment. The principal objectives are:

1. to develop a *general, transparent* CD formulation in which every reaction rate, every binding energy, and every closure approximation is traceable to its physical origin and its calibration data, and in which the formulation can be specialized to a particular Zr alloy (here Zircaloy-4 and pure α -Zr as the principal examples) without modifying the algorithmic core;
2. to provide a *multi-population loop treatment* (Sections ?? and ??) in which prismatic interstitial and vacancy $\langle a \rangle$ -loops, basal stacking-fault

Four loop classes in hcp Zr

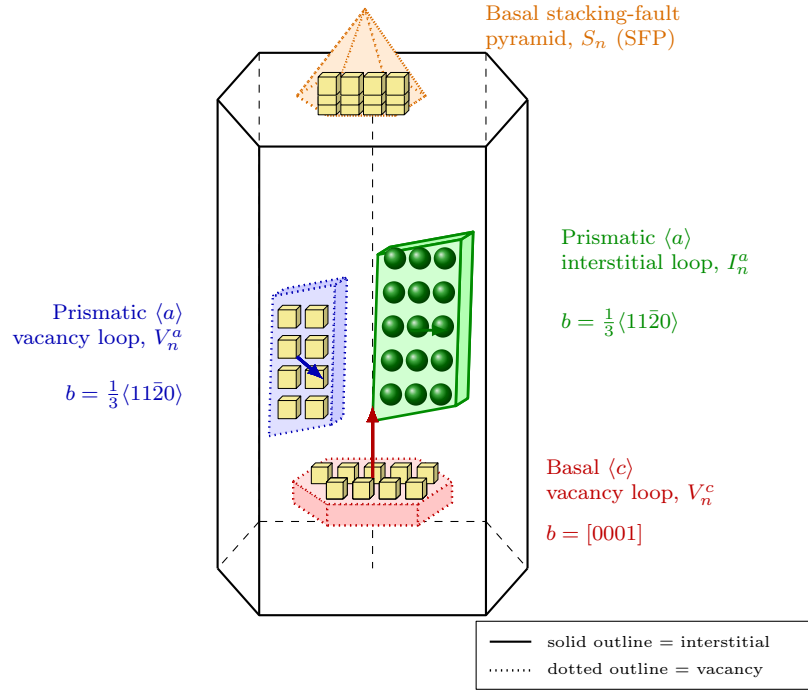


Figure 1: Schematic of the reduced loop and cluster classes tracked for irradiated hcp Zr: a basal stacking-fault pyramid S_n (orange, dotted), a prismatic $\langle a \rangle$ interstitial loop I_n^a (solid green platelet), a prismatic $\langle a \rangle$ vacancy loop V_n^a (blue dotted platelet), and a basal $\langle c \rangle$ vacancy loop V_n^c (red dotted platelet). The SFP is a three-dimensional pyramidal vacancy aggregate seated on the top basal plane that acts as the metastable precursor to the basal $\langle c \rangle$ loops. Hidden/interior edges of the hexagonal prism are shown dashed. Solid outlines denote interstitial-type defects; dotted outlines denote vacancy-type defects.

pyramids, and basal $\langle c \rangle$ -component vacancy loops are tracked as separate species with distinct Burgers vectors, habit planes, and capture-bias factors, coupled through the SFP-to- $\langle c \rangle$ unfauling transfer, so that their joint nucleation and growth produce anisotropic dimensional change consistent with the DAD/EID convention of [18];

3. to incorporate *anisotropic point-defect transport* explicitly through separate basal and prismatic SIA and vacancy diffusivities (Section Appendix C) calibrated against DFT and atomistic data [19, 20];
4. to implement two complementary *closures of the cluster-size distribution* via a logarithmic-bin-moment method—a piecewise-constant (zeroth-order) closure and a Galerkin shape-function closure—with explicit accuracy and mass-conservation analysis, so the user can trade system size against fidelity in a controlled manner;
5. to ground the framework in a *rigorous numerical strategy* that exploits the bordered-banded Jacobian structure through a Sherman–Morrison–Woodbury preconditioner, with mass-conservation laws used as live diagnostics during time integration; and
6. to deliver the formulation in an open-source code, RADCLUSTER, that ships with a complete Zircaloy-4 and pure-Zr parameter set, verification against analytical limits, and demonstration applications to LWR cladding and CANDU pressure-tube irradiation conditions.

The present framework targets bulk Zr-alloy property prediction over component lifetimes, prioritizes auditable approximations, and conservation diagnostics. The reviews of [5] and [1] provide a useful recent map of the methodological and experimental landscape into which the present formulation inserts itself, and the experimental anchors of [23–25, 34] provide the data against which the model is calibrated. The modularity of the formulation is best illustrated by three representative use cases.

(a) LWR fuel cladding (Zircaloy-4 or M5). The dual-loop-family is enabled with anisotropic SIA diffusion of Section Appendix C active. Cascade defect spectra of Section Appendix A are loaded with thermal-fission-spectrum power-law exponents (s_i, s_v) and clustering fractions $(f_i^{\text{cl}}, f_v^{\text{cl}})$. The Galerkin shape-function bin closure of is used at large $\langle a \rangle$ -loop sizes to track loop coarsening and the eventual saturation of the loop population while preserving mass conservation [5, 34].

(b) CANDU pressure tube (Zr–2.5 Nb). The same network is run with $\langle c \rangle$ -loop nucleation enabled to capture the dose-threshold appearance of basal vacancy loops and the resulting transition from steady to breakaway irradiation growth [18, 22]. The applied hoop and axial stresses are coupled through the SIPA and climb–glide creep contributions of Section Appendix E.

(c) Pure α -Zr model alloy (validation). Solute trapping is disabled; only the elementary species and the two loop families remain. The framework is validated against in-situ ion-irradiation TEM [24] and post-mortem neutron-irradiation data [23] where the dose, temperature, and pre-irradiation microstructure are all well controlled. In each case, the algorithmic core is unchanged; only the parameter tables, the closures, and the active reduction options are re-selected.

The remainder of the paper is organized as follows. Section Appendix A develops the cascade-spectrum model for defect production, including survival efficiencies, clustering fractions, and power-law exponents for thermal- and fast-spectrum neutron irradiation, with the recommended Zircaloy-4 parameter table. Section Appendix B compiles defect energetics (formation, migration, and binding energies) for the elementary species and small clusters, with explicit DFT and MD provenance and recommended values for α -Zr. Section Appendix C treats defect diffusion on the hcp lattice with explicit basal and prism-plane components, including solute-trapped SIA cluster transport, vacancy and vacancy-cluster diffusion, and the special case of fast 1-D glide of small SIA clusters along $\langle 11\bar{2}0 \rangle$. Section Appendix D derives cluster-dissociation kinetics for $\langle a \rangle$ - and $\langle c \rangle$ -loops, with binding-energy models grounded in atomistic simulation data. Section Appendix E consolidates these inputs into the unified reaction-rate-constant framework with the transition frequency $\omega = D/a^2$, treating 3D and 1D diffusion-limited reactions on equal footing and providing precomputed constants for the Zircaloy-4 application. Section ?? assembles the full master equations for the SIA-cluster, $\langle a \rangle$ -loop, and $\langle c \rangle$ -loop populations, states the bias-factor and sink-strength formulation, derives the mass-conservation laws (Frenkel-pair moments), and documents the irradiation-growth identity and ODE-system structure used by the solver. Section ?? introduces the logarithmic-bin-moment state-space reduction with both piecewise-constant and Galerkin closures and analyzes accuracy, conservation, and reduced system size. Demonstration simulation results for Zircaloy-4 are given in Section ?. The appendices provide rigorous proofs of the conservation laws (Appendix ??), the numerical-method specification including the Sherman–Morrison–Woodbury preconditioner and

stiffness strategy (Appendix Appendix F), and documentation of the RAD-CLUSTER code and its comparison with related tools (Appendix ??).

2. RADCLUSTER Modeling of Zr and its Alloys

The microstructural model used in this work is built with RADCLUSTER, a spatially averaged, mean-field rate-theory code for the quantitative prediction of irradiation-induced microstructure evolution in chemically and structurally complex alloys. This section specialises the code to α -zirconium and Zircaloy-4 under neutron irradiation. It is organised exactly as the companion treatment of ferritic–martensitic steel: §2.1 fixes the size- and character-resolved state space and the rate kernels that act on it; §2.2 assembles the *reaction admissibility graph* (RAG) that enumerates every physically allowed reaction edge and reads off the coupled master equations from a graph walk; and §2.3 describes the uncertainty-quantification workflow used to calibrate and screen the kernel parameters.

2.1. Model Equations and Parameters

All concentrations are atomic fractions (at/at) and all rate constants carry units of s^{-1} . Point-defect transport in hcp Zr is strongly anisotropic; the basal- and prism-plane mobilities, together with the solute-trapping corrections for the alloying and impurity species (Sn, Fe, Cr, Nb, O), are folded into the *effective reaction frequencies*

$$\omega_{\alpha}^{\text{eff}} = \frac{\nu_{\alpha} \exp(-E_m^{\alpha}/k_B T)}{1 + \sum_s z_s c_s \exp(E_b^{\alpha-s}/k_B T)}, \quad \alpha \in \{i, v\}, \quad (4)$$

which replace the bare hop frequencies ω_{α} in every kernel below (Table C.10; §Appendix C).

Cluster populations.. The post-irradiation microstructure of α -Zr is described by *four* size-resolved cluster families, each carrying a distinct Burgers vector, habit plane, and point-defect polarity (Fig. ??; 3, 5, 23, 24, 34), together with the two free monomers:

1. **Prismatic interstitial $\langle a \rangle$ -loops**, I_n^{Ia} , with $\mathbf{b} = \frac{1}{3}\langle 11\bar{2}0 \rangle$ on $\{10\bar{1}0\}$ planes. They are the dominant loop family at all reactor doses, nucleating from cascade SIA clusters and growing by absorbing free interstitials.

2. **Prismatic vacancy $\langle a \rangle$ -loops**, V_n^{Va} , with the same Burgers vector and habit plane but *vacancy* polarity (a removed rather than inserted atomic plane). This minority population coexists with the interstitial $\langle a \rangle$ -loops and grows by absorbing free vacancies.
3. **Basal stacking-fault pyramids (SFP)**, S_n , three-dimensional pyramidal vacancy aggregates bounded by basal $\frac{1}{6}\langle 20\bar{2}3 \rangle$ and prismatic facets. They form directly from collapsed cascade vacancy debris on the basal plane and act as the metastable precursor to the planar $\langle c \rangle$ -loops [3, 23, 34].
4. **Basal vacancy $\langle c \rangle$ -loops**, V_n^{Vc} , with $\mathbf{b} = [0001]$ on $\{0001\}$ planes. They nucleate above a temperature- and dose-rate-dependent threshold—in part by *unfaulting* of supercritical SFPs—and drive breakaway irradiation growth at high fluence [18, 22, 24].

The size index n counts point defects in a cluster (n SIAs for I_n^{Ia} ; n vacancies for V_n^{Va} , S_n , and V_n^{Vc}). The free monomers are tracked as $c^I \equiv c_1^{\text{Ia}}$ (mono-SIA) and c^V (monovacancy). No helium or 3-D void populations are carried: Zr alloys generate a negligible (n, α) helium yield [3, 5], and multi-vacancy debris in α -Zr collapses onto basal (S_n , V_n^{Vc}) or prismatic (V_n^{Va}) habits rather than forming compact voids.

Mobility limits. Following the steel treatment, mobile clusters are bounded by the size limits of Table E.13: interstitial $\langle a \rangle$ clusters are mobile (3-D or 1-D glide) for $1 \leq n \leq i_{\text{mobile}}$, and larger I_n^{Ia} are sessile loops acting only as biased sinks; vacancy clusters are 3-D mobile for $1 \leq n \leq v_{\text{mobile}}$, while larger V_n^{Va} , S_n , and all V_n^{Vc} are sessile and evolve only by single-monomer exchange and the SFP $\rightarrow$ Vc unfaulting transfer.

Bias factors and loop diameters. The elastic field of a sink preferentially attracts interstitials; the capture efficiencies Z_α^X ($X \in \{\text{Ia}, \text{Va}, \text{Vc}\}$, $\alpha \in \{i, v\}$) and the dislocation/grain-boundary/precipitate sink strengths $Z_\alpha^{d,gb,p}$ are listed in Table E.13. A size- n planar loop has diameter

$$d_n^X = \left(\frac{4n\Omega}{\pi b_X} \right)^{1/2}, \quad b_{\text{Ia}} = b_{\text{Va}} = a_0, \quad b_{\text{Vc}} = c_0, \quad (5)$$

with Ω the atomic volume; the pyramidal SFP of n vacancies has an equivalent basal footprint $d_n^S = (4n\Omega/\pi c_0)^{1/2}$. Loop binding energies $E_b^{\alpha,X}(n)$ entering the emission kernels are taken from §Appendix D.

2.2. Reaction Admissibility Graph (RAG)

The coupled kinetics are generated from a directed multigraph—the reaction admissibility graph (RAG)—whose vertices are the admissible cluster states and whose edges are the elementary reactions permitted by the polarity, habit-plane, and mobility constraints of §2.1. The master equations are then obtained by a deterministic *graph walk*: at each vertex the time derivative is the signed sum of the rate kernels of all incoming (gain) and outgoing (loss) edges incident on that vertex.

Admissible vertex set.. Vertices are partitioned by point-defect polarity about a datum that separates interstitial from vacancy character,

$$\mathcal{V}^+ = \{c^I\} \cup \{I_n^{\text{Ia}} : 2 \leq n \leq N_{\text{Ia}}\}, \quad (6)$$

$$\mathcal{V}^- = \{c^V\} \cup \{V_n^{\text{Va}}, S_n, V_n^{\text{Vc}} : 2 \leq n \leq N_X\}, \quad (7)$$

so that interstitial $\langle a \rangle$ -loops lie on the positive-polarity branch and the three vacancy families (prismatic $\langle a \rangle$ -loops, SFPs, basal $\langle c \rangle$ -loops) on the negative-polarity branch.

Admissible edges (process classes).. Every directed edge belongs to one of the process classes catalogued in Table 1. Cross-polarity edges (P1, P3 annihilation) connect $\mathcal{V}^+$ to $\mathcal{V}^-$; same-polarity edges (P2 capture, P3 coalescence, P5 emission) act within a branch; sink and source edges (P4, P6) connect the graph to the external reservoirs; and the Zr-specific transfer edge (P7) carries supercritical SFPs onto the $\langle c \rangle$ -loop ladder; and the network-evolution edges (P8, P9) incorporate supercritical $\langle a \rangle$ - and $\langle c \rangle$ -loops into the dislocation network as it climbs and sweeps past them. The explicit rate-kernel expressions for every class are derived in §Appendix E and collected in Table E.12.

Mobile clusters are bounded by the limits of Table E.13: $\langle a \rangle$ SIA clusters are mobile for $1 \leq n \leq i_{\text{mobile}}$ and sessile (fixed-sink) beyond; vacancy clusters are 3-D mobile for $1 \leq m \leq v_{\text{mobile}}$, while SFPs and $\langle c \rangle$ -loops are sessile at every size and grow or shrink only by single-monomer exchange.

Rate kernels.. The kernels labelling the edges of Table 1 are, for mono-defect recombination (P1),

$$\mathcal{K}_{iv} = 8\pi \xi_{\text{rec}} (\omega_i^{\text{eff}} + \omega_v^{\text{eff}}) \approx 12.4 \omega_i^{\text{eff}}, \quad (8)$$

Table 1: Admissible process classes (edge types) of the irradiated-Zr RAG. Each elementary reaction maps to one abstract edge class and one rate kernel; the explicit kernel expressions are given in §Appendix E and Table E.12.

Process	Abstract edge class	Rate kernel
P1: V–SIA recombination	cross-polarity (annihilation)	$\mathcal{K}_{iv}$
P2: monomer capture by loop/SFP	growth / shrinkage	$\mathcal{K}_{\alpha,n}^X$
P3: cluster–cluster coalescence	same-/cross-polarity	$\mathcal{K}_{n,n'}^{ii}, \mathcal{K}_{n,n'}^{vv,X}, \mathcal{K}_{n,m}^{vi}$
P4: absorption at fixed sink	sink edge	$\mathcal{D}_{\alpha}^d, \mathcal{D}_{\alpha}^{gb}, \mathcal{D}_{\alpha}^p$
P5: thermal monomer emission	dissociation	$\varepsilon_{\alpha}^X(n)$
P6: cascade defect production	source edge	G_n^X
P7: SFP $\rightarrow \langle c \rangle$ unfauling	inter-population transfer	$\Gamma_{\text{unf}}(n)$
P8: $\langle a \rangle$ -loop absorbed by network	network-evolution edge	Λ_n^{net}
P9: $\langle c \rangle$ -loop absorbed by network	network-evolution edge	Λ_n^{net}
P8: monomer absorbed by prismatic network	growing-sink edge	$\mathcal{D}_i^{\text{pr}}, \mathcal{D}_v^{\text{pr}}$
P9: monomer absorbed by basal network	growing-sink edge	$\mathcal{D}_i^{\text{ba}}, \mathcal{D}_v^{\text{ba}}$

since $\omega_i^{\text{eff}} \gg \omega_v^{\text{eff}}$ at reactor temperatures. Monomer capture by a size- n cluster (P2) takes the Smoluchowski form

$$\mathcal{K}_{i,n}^{\text{Ia}} = A_{\text{loop}}^a n^{1/2} Z_i^{\text{Ia}} \omega_i^{\text{eff}}, \quad \text{SIA} \rightarrow I_n^{\text{Ia}}, \quad (9)$$

$$\mathcal{K}_{v,n}^{\text{Ia}} = A_{\text{loop}}^a n^{1/2} Z_v^{\text{Ia}} \omega_v^{\text{eff}}, \quad \text{V} \rightarrow I_n^{\text{Ia}}, \quad (10)$$

$$\mathcal{K}_{v,n}^{\text{Va}} = A_{\text{loop}}^a n^{1/2} Z_v^{\text{Va}} \omega_v^{\text{eff}}, \quad \text{V} \rightarrow V_n^{\text{Va}}, \quad (11)$$

$$\mathcal{K}_{i,n}^{\text{Va}} = A_{\text{loop}}^a n^{1/2} Z_i^{\text{Va}} \omega_i^{\text{eff}}, \quad \text{SIA} \rightarrow V_n^{\text{Va}}, \quad (12)$$

$$\mathcal{K}_{v,n}^S = A_{\text{sfp}} n^{1/3} Z_v^S \omega_v^{\text{eff}}, \quad \text{V} \rightarrow S_n, \quad (13)$$

$$\mathcal{K}_{i,n}^S = A_{\text{sfp}} n^{1/3} Z_i^S \omega_i^{\text{eff}}, \quad \text{SIA} \rightarrow S_n, \quad (14)$$

$$\mathcal{K}_{v,n}^{\text{Vc}} = A_{\text{loop}}^c n^{1/2} Z_v^{\text{Vc}} \omega_v^{\text{eff}}, \quad \text{V} \rightarrow V_n^{\text{Vc}}, \quad (15)$$

$$\mathcal{K}_{i,n}^{\text{Vc}} = A_{\text{loop}}^c n^{1/2} Z_i^{\text{Vc}} \omega_i^{\text{eff}}, \quad \text{SIA} \rightarrow V_n^{\text{Vc}}, \quad (16)$$

where the prefactors $A_{\text{loop}}^{a,c}$ (planar loops) and A_{sfp} (the $n^{1/3}$ three-dimensional pyramid) follow from the capture geometry of each habit (Table E.13). A captured SIA grows I_n^{Ia} and shrinks the vacancy families, and vice versa for a captured vacancy. Cluster–cluster coalescence (P3) among mobile species

retains the i - i , v - v (separately on the prismatic and basal channels), and cross-polarity v - i annihilation kernels

$$\mathcal{K}_{n,n'}^{ii} = 8\pi(\xi_n + \xi_{n'}) (\omega_n^{\text{eff}} + \omega_{n'}^{\text{eff}}), \quad 1 \leq n' \leq n \leq i_{\text{mobile}}, \quad (17)$$

$$\mathcal{K}_{n,m}^{vi} = 8\pi(\xi_n + \xi_m) (\omega_n^{\text{eff}} + \omega_m^{\text{eff}}), \quad n \leq v_{\text{mobile}}, \quad m \leq i_{\text{mobile}}, \quad (18)$$

with $\mathcal{K}_{n,n'}^{vv,a}$ and $\mathcal{K}_{n,n'}^{vv,c}$ of the same form as Eq. (17); cross-habit vacancy coalescence requires a Burgers-vector reorientation and is suppressed. Absorption at fixed sinks (P4) is first order,

$$\mathcal{D}_\alpha = \mathcal{D}_\alpha^d + \mathcal{D}_\alpha^{gb} + \mathcal{D}_\alpha^p, \quad \mathcal{D}_\alpha^d = Z_\alpha^d \rho_d a^2 \omega_\alpha^{\text{eff}}, \quad \mathcal{D}_\alpha^{gb} = \frac{4\pi^2}{d_g^2} a^2 \omega_\alpha^{\text{eff}}, \quad \mathcal{D}_\alpha^p = \frac{8\pi r_p}{a} c_p^N \omega_\alpha^{\text{eff}}. \quad (19)$$

Thermal emission (P5) follows by detailed balance with capture,

$$\varepsilon_i^{\text{Ia}}(n) = A_{\text{loop}}^a (n-1)^{1/2} Z_i^{\text{Ia}} \omega_i^{\text{eff}} e^{-E_b^{i,\text{Ia}}(n)/k_B T}, \quad (20)$$

$$\varepsilon_v^{\text{Va}}(n) = A_{\text{loop}}^a (n-1)^{1/2} Z_v^{\text{Va}} \omega_v^{\text{eff}} e^{-E_b^{v,\text{Va}}(n)/k_B T}, \quad (21)$$

$$\varepsilon_v^S(n) = A_{\text{sfp}} (n-1)^{1/3} Z_v^S \omega_v^{\text{eff}} e^{-E_b^{v,S}(n)/k_B T}, \quad (22)$$

$$\varepsilon_v^{\text{Vc}}(n) = A_{\text{loop}}^c (n-1)^{1/2} Z_v^{\text{Vc}} \omega_v^{\text{eff}} e^{-E_b^{v,\text{Vc}}(n)/k_B T}. \quad (23)$$

Finally, the Zr-specific transfer edge (P7) carries a supercritical SFP onto the basal $\langle c \rangle$ -loop ladder by unfauling,

$$\Gamma_{\text{unf}}(n) = \nu_0 \Theta(n - n_{\text{unf}}) \exp\left(-\frac{E_{\text{unf}}}{k_B T}\right), \quad (24)$$

active only above the unfauling size n_{unf} , with Θ the Heaviside step.

Dislocation network evolution.. The grown-in and irradiation-built dislocation network is more than the static point-defect sink of class P4. As prismatic $\langle a \rangle$ -loops and faulted basal $\langle c \rangle$ -loops coarsen, the largest of them expand until their elastic strain fields overlap, or they physically intersect, a network line and are swept into it. This introduces two additional admissible edges, which we designate P8 ($\langle a \rangle$ -loop absorbed by network) and P9 ($\langle c \rangle$ -loop absorbed by network). Because the absorbed loops are precisely the large, sessile members of each ladder, the incorporation rate cannot be tied to the loop's own (vanishing) mobility; the relevant kinetics are instead

set by the network sweeping or climbing past a stationary loop. Writing ρ_{net} for the line density of the network, the per-loop incorporation rate is

$$\Lambda_n^{\text{net}} = \rho_{\text{net}} v_{\text{net}} w_c P_{\ell d}(n), \quad P_{\ell d}(n) = \min\left(1, R_{\text{int}}(n)/s_d(n)\right), \quad (25)$$

where v_{net} is the effective network climb/glide velocity, $w_c \sim \mathcal{O}(b)$ a capture width, and $P_{\ell d}(n)$ a geometric ledge-detection probability that switches on once a loop's elastic interaction range $R_{\text{int}}(n)$ reaches the mean loop-network spacing $s_d(n) \simeq \rho_{\text{net}}^{-1/2} - d_n^{(c)}/2$. Equation (25) carries no factor of the loop diffusivity, so the channel remains active for the sessile $\langle a \rangle$ - and $\langle c \rangle$ -loops that dominate incorporation. The climb velocity is itself irradiation-driven and reuses the biased point-defect fluxes already formed for the fixed-sink terms,

$$v_{\text{net}} = \frac{b}{\rho_{\text{net}}} (Z_i^{\text{net}} \omega_i^{\text{eff}} c^I - Z_v^{\text{net}} \omega_v^{\text{eff}} c^V), \quad (26)$$

so no new transport physics is introduced. The loop populations lose members to the network at $\dot{c}_n|_{\text{net}} = -\Lambda_n^c c_n$ ($c = \text{pr}$ or ba), applied to the prismatic I_n^{Ia} , V_n^{Va} families (which join ρ_{pr} via Λ_n^{pr}) and the basal V_n^{Vc} family (which joins ρ_{ba} via Λ_n^{ba}) above an incorporation-onset size $n \geq n_{\text{inc}}$; loops below n_{inc} are too small to register on the network and are excluded. Each absorbed loop donates its n defects to the network and adds approximately one loop circumference of line, so line-length and defect-content conservation give

$$\dot{\rho}_{\text{pr}} = \sum_{n \geq n_{\text{inc}}} \pi d_n^{(a)} \Theta(n - n_{\text{inc}}) \Lambda_n^{\text{pr}} (c_n^{\text{Ia}} + c_n^{\text{Va}}) - \mathcal{K}_{\text{rec}}^{\text{pr}} \rho_{\text{pr}}^2, \quad (27)$$

$$\dot{\rho}_{\text{ba}} = \sum_{n \geq n_{\text{inc}}} \pi d_n^{(c)} \Theta(n - n_{\text{inc}}) \Lambda_n^{\text{ba}} c_n^{\text{Vc}} - \mathcal{K}_{\text{rec}}^{\text{ba}} \rho_{\text{ba}}^2, \quad (28)$$

with $\rho_{\text{net}} = \rho_{\text{pr}} + \rho_{\text{ba}}$ the total network density. The two components are distinct growing sinks: prismatic line absorbs both species with efficiencies $Z_{i,v}^{\text{pr}}$ and basal line with $Z_{i,v}^{\text{ba}}$, entering the monomer balances through $\mathcal{D}_{i,v}^{\text{pr}}$ and $\mathcal{D}_{i,v}^{\text{ba}}$ of Eqs. (37)–(38), and the clusters through Λ_n^{pr} and Λ_n^{ba} . Each diffusivity-weighted capture coefficient scales with its own line density, $\mathcal{D}_i^{\text{pr}} = Z_i^{\text{pr}} D_i \rho_{\text{pr}}$ etc., so the basal and prismatic networks build up and bias point-defect flow independently. the recovery sink $\mathcal{K}_{\text{rec}} \rho_{\text{net}}^2$ being mandatory: with a gain term alone ρ_{net} would grow without bound and the model would merely relocate the non-saturation problem from the loops to the network. $\mathcal{K}_{\text{rec}}$ fixes the steady-state network density and is the principal network calibration

parameter. The defect content carried off by absorbed loops is routed into the same cumulative fixed-sink ledger used by the Frenkel-pair diagnostic, so the swelling identity is conserved edge by edge.

Making ρ_{net} a fully implicit state variable would render the P4 sink diagonal state-dependent and break the bordered-banded Jacobian structure exploited by the Woodbury preconditioner of §Appendix F. Because ρ_{net} evolves on the dose timescale—orders of magnitude slower than the point-defect relaxation that sets solver stiffness—it is held piecewise constant within each adaptive integration segment and refreshed between segments by operator splitting: the stiff cluster system is advanced with ρ_{net} frozen while the gain integral of Eqs. (27)–(28) is accumulated by quadrature, ρ_{net} is then advanced explicitly (gain minus recovery), and the rate tables are rebuilt so that the network sink and the new P8/P9 edges pick up the updated density for the next segment. Within a segment the loss term $\Lambda_n^{\text{net}} c_n$ is a diagonal, state-dependent loss on the loop ladders—structurally identical to the existing mobile fixed-sink term and therefore already inside the band—so it adds no new Jacobian coupling. The network channel finally combines with the fixed sinks of Eq. (19) by the standard additive sink-strength rule, the total loss diagonal being the sum of the P4 rate and Λ_n^{net} .

Basal and prismatic network components. For irradiation growth the network cannot be treated as a single isotropic sink, because the prismatic ($\langle a \rangle$) and basal ($\langle c \rangle$) populations differ in habit plane, Burgers vector, and capture bias. We therefore split the network line density into two co-evolving, growing-sink components,

$$\rho_{\text{net}} = \rho_{\text{pr}} + \rho_{\text{ba}}, \quad (29)$$

where ρ_{pr} is fed by prismatic $\langle a \rangle$ edge dislocations and the incorporation of Ia and Va loops beyond the onset size n_{inc} , and ρ_{ba} is fed by the basal $\langle c \rangle$ -component network and the incorporation of unfaulted Vc loops and stacking-fault pyramids. Each component absorbs both species with its own capture efficiency, giving the monomer fixed-sink rates of (37)–(38) (processes P8 and P9 in Table 1). Because $Z_i^{\text{pr}} \neq Z_v^{\text{pr}}$ and $Z_i^{\text{ba}} \neq Z_v^{\text{ba}}$, each component is a net biased sink: the prismatic network preferentially absorbs interstitials and the basal network preferentially absorbs vacancies, so the two channels store unequal interstitial and vacancy content and contribute to growth in the same way as the $\langle a \rangle$ - and $\langle c \rangle$ -loop ladders. The absorbed

content of each component is accumulated and enters the conservation analysis of Appendix Appendix H.

Master equations... Walking the graph of Fig. 2 vertex by vertex—summing the kernels of all incoming and outgoing edges—yields the coupled cluster-dynamics system. For the *interstitial* $\langle a \rangle$ -loops ($n \geq 2$),

$$\begin{aligned} \frac{dc_n^{\text{Ia}}}{dt} = & G_n^{\text{Ia}} + \mathcal{K}_{i,n-1}^{\text{Ia}} c^I c_{n-1}^{\text{Ia}} - \mathcal{K}_{i,n}^{\text{Ia}} c^I c_n^{\text{Ia}} + \mathcal{K}_{v,n+1}^{\text{Ia}} c^V c_{n+1}^{\text{Ia}} - \mathcal{K}_{v,n}^{\text{Ia}} c^V c_n^{\text{Ia}} \\ & + \varepsilon_i^{\text{Ia}}(n+1) c_{n+1}^{\text{Ia}} - \varepsilon_i^{\text{Ia}}(n) c_n^{\text{Ia}} - \mathcal{D}_n^{\text{Ia}} c_n^{\text{Ia}} \\ & + \frac{1}{2} \sum_{n'=1}^{\min(n-1, i_{\text{mobile}})} \mathcal{K}_{n',n-n'}^{ii} c_{n'}^{\text{Ia}} c_{n-n'}^{\text{Ia}} - c_n^{\text{Ia}} \sum_{n'=1}^{i_{\text{mobile}}} \mathcal{K}_{n,n'}^{ii} c_{n'}^{\text{Ia}} \\ & - \Theta(n - n_{\text{inc}}) \Lambda_n^{\text{pr}} c_n^{\text{Ia}}. \end{aligned} \quad (30)$$

For the *prismatic vacancy* $\langle a \rangle$ -loops ($n \geq 2$),

$$\begin{aligned} \frac{dc_n^{\text{Va}}}{dt} = & G_n^{\text{Va}} + \mathcal{K}_{v,n-1}^{\text{Va}} c^V c_{n-1}^{\text{Va}} - \mathcal{K}_{v,n}^{\text{Va}} c^V c_n^{\text{Va}} + \mathcal{K}_{i,n+1}^{\text{Va}} c^I c_{n+1}^{\text{Va}} - \mathcal{K}_{i,n}^{\text{Va}} c^I c_n^{\text{Va}} \\ & + \varepsilon_v^{\text{Va}}(n+1) c_{n+1}^{\text{Va}} - \varepsilon_v^{\text{Va}}(n) c_n^{\text{Va}} - \mathcal{D}_n^{\text{Va}} c_n^{\text{Va}} \\ & + \frac{1}{2} \sum_{n'=1}^{\min(n-1, v_{\text{mobile}})} \mathcal{K}_{n',n-n'}^{vv,a} c_{n'}^{\text{Va}} c_{n-n'}^{\text{Va}} - c_n^{\text{Va}} \sum_{n'=1}^{v_{\text{mobile}}} \mathcal{K}_{n,n'}^{vv,a} c_{n'}^{\text{Va}} \\ & - \Theta(n - n_{\text{inc}}) \Lambda_n^{\text{pr}} c_n^{\text{Va}}. \end{aligned} \quad (31)$$

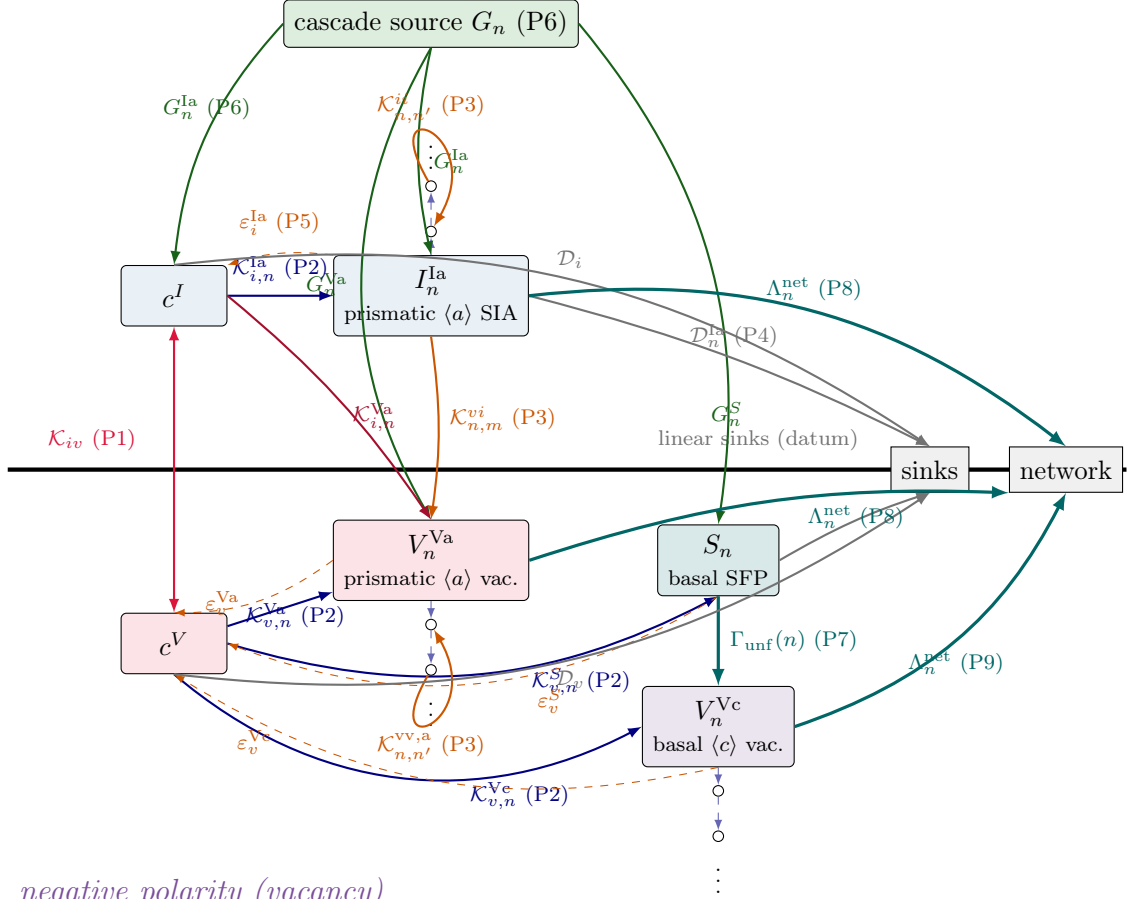
For the *basal stacking-fault pyramids* ($n \geq 2$), which capture and emit single defects, exchange with the mobile vacancy clusters, and lose supercritical members to the $\langle c \rangle$ ladder by unfauling,

$$\begin{aligned} \frac{dc_n^S}{dt} = & G_n^S + \mathcal{K}_{v,n-1}^S c^V c_{n-1}^S - \mathcal{K}_{v,n}^S c^V c_n^S + \mathcal{K}_{i,n+1}^S c^I c_{n+1}^S - \mathcal{K}_{i,n}^S c^I c_n^S \\ & + \varepsilon_v^S(n+1) c_{n+1}^S - \varepsilon_v^S(n) c_n^S - \Gamma_{\text{unf}}(n) c_n^S. \end{aligned} \quad (32)$$

For the *basal vacancy* $\langle c \rangle$ -loops ($n \geq 2$), which receive the unfaulted SFP flux as a source,

$$\begin{aligned} \frac{dc_n^{\text{Vc}}}{dt} = & G_n^{\text{Vc}} + \Gamma_{\text{unf}}(n) c_n^S + \mathcal{K}_{v,n-1}^{\text{Vc}} c^V c_{n-1}^{\text{Vc}} - \mathcal{K}_{v,n}^{\text{Vc}} c^V c_n^{\text{Vc}} \\ & + \mathcal{K}_{i,n+1}^{\text{Vc}} c^I c_{n+1}^{\text{Vc}} - \mathcal{K}_{i,n}^{\text{Vc}} c^I c_n^{\text{Vc}} + \varepsilon_v^{\text{Vc}}(n+1) c_{n+1}^{\text{Vc}} - \varepsilon_v^{\text{Vc}}(n) c_n^{\text{Vc}} \\ & - \Theta(n - n_{\text{inc}}) \Lambda_n^{\text{ba}} c_n^{\text{Vc}}. \end{aligned} \quad (33)$$

positive polarity (SIA)



negative polarity (vacancy)

Figure 2: Reaction admissibility graph (RAG) for irradiated zirconium. Vertices are the four cluster families—interstitial $\langle a \rangle$ loops I_n^{Ia} (positive polarity, top), prismatic vacancy $\langle a \rangle$ loops V_n^{Va} , basal stacking-fault pyramids S_n , and basal vacancy $\langle c \rangle$ loops V_n^{Vc} (negative polarity, bottom)—plus the free monomers c^I, c^V . Directed edges are the admissible processes of Table 1: monomer capture and growth/shrinkage (P2, solid), thermal emission (P5, dashed), V-SIA and loop annihilation (P1, P3), the cascade source (P6) and fixed sinks (P4), and the Zr-specific SFP $\rightarrow \langle c \rangle$ unfaulting transfer (P7). The master equations of §2.2 are read off vertex by vertex as the signed sum of incident edge kernels.

The two free monomers close the system. They are the $n = 1$ special case of the cluster ladders: Eq. (34) is the $n = 1$ member of the interstitial $\langle a \rangle$ -loop family (30), while Eq. (35) is the $n = 1$ member common to the vacancy $\langle a \rangle$ -loop, stacking-fault-pyramid, and $\langle c \rangle$ -loop families (31)–(33). Because a free point defect cannot coagulate, shrink by absorbing its own antidefect into a smaller bound cluster, or carry the in-cascade/RAG channel, the cluster-only terms switch off at $n = 1$ and the monomer balances retain only the source, mutual recombination, net absorption onto all clusters, total emission returned from them, and the fixed-sink loss:

$$\frac{dc^I}{dt} = G_1^I - \mathcal{K}_{iv}c^Ic^V - c^I \sum_X \sum_n \mathcal{K}_{i,n}^X c_n^X + \sum_{n \geq 2} \varepsilon_i^{\text{Ia}}(n) c_n^{\text{Ia}} - (\mathcal{D}_i^{\text{pr}} + \mathcal{D}_i^{\text{ba}})c^I, \quad (34)$$

$$\begin{aligned} \frac{dc^V}{dt} = & G_1^V - \mathcal{K}_{iv}c^Ic^V - c^V \sum_X \sum_n \mathcal{K}_{v,n}^X c_n^X \\ & + \sum_{n \geq 2} [\varepsilon_v^{\text{Va}}(n) c_n^{\text{Va}} + \varepsilon_v^S(n) c_n^S + \varepsilon_v^{\text{Vc}}(n) c_n^{\text{Vc}}] - (\mathcal{D}_v^{\text{pr}} + \mathcal{D}_v^{\text{ba}})c^V, \end{aligned} \quad (35)$$

where in Eqs. (34)–(35) the inner sums run over the loop/SFP families $X \in \{\text{Ia}, \text{Va}, S, \text{Vc}\}$.

The free-monomer source rates $G_1^I \equiv P_1^{(i)}$ and $G_1^V \equiv P_1^{(v)}$ are the survival-corrected mono-defect production rates of Appendix A, Eq. (A.3). Because the interstitial and vacancy cascade spectra differ ($\epsilon_m^{(i)} \neq \epsilon_n^{(v)}$, with the vacancy side biased toward $\langle c \rangle$ -forming clusters that collapse onto the basal plane), more interstitial than vacancy atoms are locked into clusters at birth, so the two free-monomer sources are *unequal*,

$$\beta_p \equiv G_1^V - G_1^I = G \left(\sum_{m \geq 2} m \epsilon_m^{(i)} - \sum_{n \geq 2} n \epsilon_n^{(v)} \right) \neq 0, \quad (36)$$

the *production bias* β_p , which seeds the net interstitial flux to sinks that drives irradiation growth.

The fixed-sink loss of the free monomers is carried by the grown-in and irradiation-built dislocation network, which is resolved into a *prismatic* component (predominantly $\langle a \rangle$ edge dislocations and absorbed $\langle a \rangle$ -loops on $\{10\bar{1}0\}$ habit planes) and a *basal* component (predominantly $\langle c \rangle$ -component network

and absorbed $\langle c \rangle$ -loops on $\{0001\}$ planes). Each component is a growing sink with its own capture efficiency, so the monomer fixed-sink rates split as

$$\mathcal{D}_i^{\text{pr}} = Z_i^{\text{pr}} D_i \rho_{\text{pr}}, \quad \mathcal{D}_i^{\text{ba}} = Z_i^{\text{ba}} D_i \rho_{\text{ba}}, \quad (37)$$

$$\mathcal{D}_v^{\text{pr}} = Z_v^{\text{pr}} D_v \rho_{\text{pr}}, \quad \mathcal{D}_v^{\text{ba}} = Z_v^{\text{ba}} D_v \rho_{\text{ba}}, \quad (38)$$

where ρ_{pr} and ρ_{ba} are the prismatic and basal network line densities ($\rho_{\text{net}} = \rho_{\text{pr}} + \rho_{\text{ba}}$) and $Z_{i,v}^{\text{pr,ba}}$ the corresponding capture efficiencies. The defect content absorbed by each component accumulates in the network and is tracked for the conservation analysis of Appendix Appendix H. The cascade sources partition the in-cascade spectrum $\epsilon_n^{(i)}, \epsilon_n^{(v)}$ of §Appendix A among the families,

$$G_n^{\text{Ia}} = \epsilon_n^{(i)} G, \quad G_1^V = G \left(1 - \sum_{n \geq 2} n \epsilon_n^{(v)} \right), \quad G_n^S = \epsilon_n^{(v)} G \quad (n \geq n_{\text{coll}}), \quad G_n^{\text{Va}} = \epsilon_n^{(v)} G \quad (2 \leq n < n_{\text{coll}}), \quad (39)$$

so that small cascade vacancy clusters feed the prismatic $\langle a \rangle$ family while clusters above the collapse threshold n_{coll} seed the basal SFPs, which subsequently unfault into $\langle c \rangle$ -loops via P7.

Frenkel-pair conservation and the running ledger.. The RAG is constructed so that point-defect number is a *structural* invariant of the graph rather than an emergent property of the calibrated kernels: every admissible edge is examined for its net contribution to the interstitial and vacancy moments before it is added to the master equations. Defining the total SIA and vacancy contents

$$S_I = c^I + \sum_{n \geq 2} n c_n^{\text{Ia}}, \quad S_V = c^V + \sum_{n \geq 2} n (c_n^{\text{Va}} + c_n^S + c_n^{\text{Vc}}), \quad (40)$$

the edge classes of Table 1 fall into three accounting groups.

(i) *Internally conserving edges.* Monomer capture and emission (P_2, P_5) move a single defect between a free monomer and an adjacent ladder rung, so the gain at size n is cancelled exactly by the loss at $n \mp 1$ and the matching monomer term; same-polarity coalescence (P_3) maps $n + n' \rightarrow n + n'$ with no change in $\sum_n n c_n$; the SFP $\rightarrow \langle c \rangle$ unfaulting transfer (P_7) and the network-incorporation edges (P_8, P_9) relabel a cluster's habit or host without creating or destroying vacancies. Each of these edges therefore contributes an equal and opposite gain/loss pair to Eqs. (40) and drops out of dS_I/dt and dS_V/dt identically.

(ii) *Cross-polarity edges.* Recombination (P_1) and the vacancy-SIA annihilation channels of P_3 remove one interstitial and one vacancy per event, decrementing S_I and S_V by the same amount and thus preserving the Frenkel-pair difference $S_I - S_V$ exactly.

(iii) *Source and sink edges.* Only the cascade source (P_6) and the fixed sinks (P_4 , together with the network and grain-boundary losses routed through it) couple the graph to the external reservoirs. The graph structure therefore enforces the exact balance laws

$$\boxed{\frac{dS_I}{dt} = G - \Phi_I, \quad \frac{dS_V}{dt} = G - \Phi_V} \quad (41)$$

where G is the survival-corrected production rate of §Appendix A and Φ_α is the cumulative loss of species α to recombination and to all fixed sinks. Subtracting the two relations gives the stronger statement $d(S_I - S_V)/dt = \Phi_V - \Phi_I$: because the cascade injects interstitials and vacancies in equal number, any net imbalance can only be the difference in their sink-mediated escape, which is precisely the polarity-biased partitioning of point defects between the prismatic $\langle a \rangle$ ladder (interstitial-biased) and the basal ($S_n, \langle c \rangle$) ladders (vacancy-biased). This antisymmetry is the graph-level origin of the anisotropic growth strain quantified in §??.

A live Frenkel-pair ledger. Equation (41) is not merely a formal identity but the primary correctness check on the integrator. The cumulative losses Φ_I, Φ_V are carried as additional state variables advanced by the same time-stepper as the cluster concentrations, so that every defect leaving the explicit cluster inventory through a P_1 or P_4 edge is deposited in the ledger on the same step. The dimensionless residual

$$\delta_{\text{FP}}(t) = \frac{|[S_I(t) - S_I(0)] - [S_V(t) - S_V(0)] - [\Phi_V(t) - \Phi_I(t)]|}{Gt + S_I(0) + S_V(0)} \quad (42)$$

is evaluated at every accepted step and must remain at the level of the integrator's relative tolerance; a drift in δ_{FP} above that floor signals a miscounted edge—most often a coalescence, unfauling, or network-incorporation term that has failed to conserve $\sum_n n c_n$ —and aborts the run. The same ledger furnishes the closure used to verify the input mapping and rate-table rebuilds of §??, so that the anisotropic swelling identity is conserved edge by edge and step by step.

2.3. Uncertainty Quantification

The parameters that enter the RADCLUSTER rate equations for zirconium—the anisotropic migration and binding energies, the loop and cavity capture efficiencies and bias factors $Z_{\text{Ia}}^\alpha, Z_{\text{Va}}^\alpha, Z_{\text{Vc}}^\alpha, Z_d^\alpha$, the solute-trapping coefficients of the Zircaloy additions, the coalescence kernels, and above all the basal collapse threshold n_{coll} —are known only to within physically motivated bounds. The experimental targets against which they are calibrated— $\langle a \rangle$ - and $\langle c \rangle$ -loop number densities, mean loop diameters, and the full size distributions at selected doses and temperatures [18, 22, 34]—carry their own measurement uncertainty. Quantifying how these two sources of uncertainty propagate to the predicted microstructure is an intrinsic part of using the model predictively rather than an optional post-processing step.

A range of established methodologies addresses this problem. The canonical framework for confronting an expensive mechanistic simulator with noisy data is the Bayesian calibration of computer models, in which a physically informed prior over the uncertain parameters is updated into a posterior through a likelihood that accounts for both measurement noise and model discrepancy. Because each RADCLUSTER solve over a full irradiation history is costly, the posterior is not explored on the solver directly; instead a statistical surrogate—most commonly a Gaussian-process emulator—is trained on a designed set of runs and substituted for the forward model. The informativeness of the loop and cavity data is assessed through variance-based global sensitivity analysis, whose Sobol’ indices reveal which parameters and parameter combinations actually control the observables and which are “sloppy” and can be demoted.

Workflow. Rather than applying these techniques as an undifferentiated toolbox, we adopt a single reasoned pipeline whose guiding principle is that *physical inference must never be contaminated by numerical approximation*: every run is carried out at verified integrator tolerances and Frenkel-pair conservation residual δ_{FP} (Eq. (42)) held at the solver floor, so that numerical error cannot masquerade as physics. Uncertainty quantification is then a four-stage sequence, each stage feeding the next.

(i) *Design and screening.* A set of uncertain parameters $\boldsymbol{\theta} \in \mathbb{R}^p$ ($p \approx 10$ – 12), each given a physically bounded prior $p_i(\theta_i)$, is sampled by a space-filling Sobol design and screened with global sensitivity analysis, so that sloppy or regime-inactive parameters are demoted before they can absorb spurious degrees of freedom. For irradiated zirconium the screen consistently retains

a small controlling set—the basal collapse threshold n_{coll} , the $\langle c \rangle$ -loop bias, and the basal-plane vacancy mobility—that dominates the predicted onset of $\langle c \rangle$ -loop nucleation and breakaway growth.

(ii) *Emulation.* The retained parameters are mapped to the observables $\mathbf{y}_{\text{sim}} = \mathcal{M}(\boldsymbol{\theta}, T, d)$ by a *multi-fidelity* Gaussian-process surrogate $\widehat{\mathcal{M}}$ that reports its own predictive variance. A large, space-filling ensemble of *low-fidelity* runs—coarse bin-moment resolution truncated at low dose—cheaply charts the response surface and the early, identifiable observables (incipient $\langle a \rangle$ -loop number density), while a sparse set of *high-fidelity* runs—full size resolution integrated to reactor-relevant dose, capturing the basal SFP collapse and the breakaway $\langle c \rangle$ -loop growth that drives anisotropic strain—anchors the absolute level. The two levels are fused in an auto-regressive multi-fidelity emulator,

$$\widehat{\mathcal{M}}_{\text{HF}}(\boldsymbol{\theta}) = \rho \widehat{\mathcal{M}}_{\text{LF}}(\boldsymbol{\theta}) + \delta(\boldsymbol{\theta}), \quad (43)$$

so that the inexpensive runs ensure parameter-space coverage while the few expensive runs correct the dose-extrapolation bias.

(iii) *Calibration.* The posterior

$$p(\boldsymbol{\theta} \mid \mathbf{y}_{\text{exp}}) \propto p(\mathbf{y}_{\text{exp}} \mid \boldsymbol{\theta}) p(\boldsymbol{\theta}) \quad (44)$$

is obtained on the surrogate, with a log-space likelihood for the loop and cavity densities and a Wasserstein term for the size distributions,

$$-\log p(\mathbf{y}_{\text{exp}} \mid \boldsymbol{\theta}) = \sum_j w_j [\log y_{j,\text{sim}}(\boldsymbol{\theta}) - \log y_{j,\text{exp}}]^2 + w_{\text{dist}} W_1(p_{\text{sim}}(d), p_{\text{exp}}(d)), \quad (45)$$

from which credible intervals on every predicted quantity follow.

(iv) *Active refinement.* New RADCLUSTER runs are requested only where they most reduce posterior uncertainty per unit of predicted CPU cost, closing the loop back to stage (ii). The end product is not a single “best fit” but the calibrated posterior $p(\boldsymbol{\theta} \mid \mathbf{y}_{\text{exp}})$ together with its propagated predictive uncertainty on $\langle a \rangle$ - and $\langle c \rangle$ -loop densities, mean diameters, and distributions. This workflow, summarised in Fig. 3, both furnishes the calibrated parameter set of Table E.13 and attaches a quantified confidence band to the comparisons with experiment presented in §2.

From calibrated model to digital twin. Once the posterior of Eq. (44) is available, RADCLUSTER ceases to be a one-shot forward model and becomes a

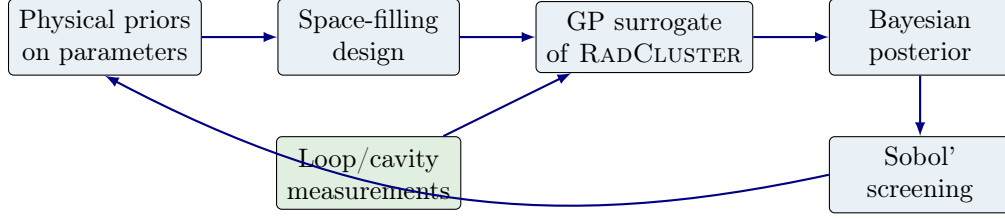


Figure 3: Uncertainty-quantification workflow for RADCLUSTER applied to zirconium. Physical priors and a space-filling design train a Gaussian-process surrogate, which is confronted with the loop and cavity measurements to form a Bayesian posterior; a Sobol’ sensitivity screen identifies the controlling parameters (notably the basal collapse threshold n_{coll}) and feeds back to refine the priors.

digital twin of the evolving irradiated microstructure: a living, uncertainty-aware representation whose state $\mathbf{x}(d, T)$ —the SIA, prismatic and basal vacancy, and SFP cluster populations together with their derived $\langle a \rangle$ - and $\langle c \rangle$ -loop descriptors—is kept consistent with the experimental record as it accumulates. The twin is defined not by a single tuned parameter set but by the calibrated distribution $p(\boldsymbol{\theta} \mid \mathbf{y}_{\text{exp}})$ and the predictive uncertainty it induces. As measurements arrive at successive doses, $\mathbf{z}_k = H(\mathbf{x}_k) + \boldsymbol{\epsilon}_k$, the same Bayesian machinery is run sequentially—naturally as an ensemble Kalman or smoother update—so that the twin assimilates the loop and cavity record across the full irradiation history and continuously reports both its best estimate of the zirconium microstructure and the dose and temperature conditions at which new experiments would most sharpen the prediction.

3. Results and Comparison with Experiments

4. Conclusions

Supplementary Material

Appendix A. Production

Let G denote the *actual* (survival-corrected) defect production rate per unit volume, i.e.

$$G = \eta G_{\text{NRT}}, \quad (\text{A.1})$$

where $\eta = N_d/N_{\text{NRT}}$ is the survival efficiency and G_{NRT} is the NRT displacement rate.

The production rate of interstitial clusters of size m is

$$P_m^{(i)} = \epsilon_m^{(i)} G, \quad m = 2, 3, \dots, i_{\text{cascade}}, \quad (\text{A.2})$$

and the mono-interstitial production rate is

$$P_1^{(i)} = G \left(1 - \sum_{m=2}^{m_{\text{cascade}}} m \epsilon_m^{(i)} \right). \quad (\text{A.3})$$

Analogous expressions hold for vacancies with superscript (v) , cluster size index n , and upper cutoff v_{cascade} .

The total fraction of surviving SIAs born in clusters is

$$f_i^{\text{cl}} = \sum_{m=2}^{m_{\text{cascade}}} m \epsilon_m^{(i)}, \quad (\text{A.4})$$

with the corresponding vacancy quantity f_v^{cl} . Note that the mono-defect fraction is simply $1 - f^{\text{cl}}$.

The fraction of NRT-predicted Frenkel pairs that survive in-cascade recombination is well characterized by MD simulations in α -zirconium [35–38]. Compared with the more extensively studied α -Fe, hcp Zr exhibits comparable Frenkel-pair survival fractions but a notably different intra-cascade clustering behavior: SIA clusters in Zr form predominantly as small glissile groups that quickly assemble into prismatic $\langle a \rangle$ -loops, while vacancy clusters are unstable in 3-D form and instead collapse onto the basal plane to seed faulted $\langle c \rangle$ -component loops at sufficient cascade size [5, 23, 24]. For Zr at reactor-relevant temperatures, the survival efficiency is approximately

$$\eta(E_{\text{PKA}}) \approx 0.80 \left(\frac{E_{\text{PKA}}}{1 \text{ keV}} \right)^{-0.20}, \quad E_{\text{PKA}} \gtrsim 1 \text{ keV}, \quad (\text{A.5})$$

saturating near $\eta \approx 0.25$ above ~ 10 keV.

Table A.2 lists representative values. The “athermal recombination corrected dpa” (arc-dpa) formalism of [38] provides a more refined energy-dependent efficiency function $\xi_{\text{arc}}(E)$ that is recommended for detailed calculations.

Table A.2: Defect survival efficiency $\eta = N_d/N_{\text{NRT}}$ in α -Zr at 100 K from MD cascade simulations [35–37]. Values are approximate; uncertainty is dominated by interatomic-potential choice.

E_{PKA} (keV)	η
0.5	0.93
1	0.80
2	0.69
5	0.55
10	0.45
20	0.36
50	0.29
100	0.25
200	0.22

MD cascade simulations in α -Zr show that the *number* of in-cascade clusters of size m follows an approximate power law [36–38]:

$$N_{\text{cl}}(m) \propto m^{-s}, \quad (\text{A.6})$$

with distinct exponents for SIA clusters (s_i) and vacancy clusters (s_v). In hcp Zr the vacancy-cluster exponent is steeper than the SIA exponent, reflecting the comparatively lower stability of compact vacancy clusters in 3-D form: vacancy “clusters” frequently appear as loose, correlated arrangements at cascade time scales and only later collapse to faulted $\langle c \rangle$ -component basal loops [19, 23, 24].

The production fraction is therefore modeled as

$$\epsilon_m^{(i)} = C_i m^{-s_i}, \quad m = 2, \dots, i_{\text{cascade}}, \quad (\text{A.7})$$

where the normalization constant C_i is fixed by the constraint

$$C_i = \frac{f_i^{\text{cl}}}{\sum_{m=2}^{m_{\text{cascade}}} m^{1-s_i}}. \quad (\text{A.8})$$

An identical prescription applies to vacancies with parameters $(C_v, s_v, v_{\text{cascade}}, f_v^{\text{cl}})$.

The parameters that distinguish thermal-spectrum (LWR) and fast-spectrum (CANDU, fast reactor) neutron irradiation in terms of cascade damage are summarized in Table A.3. Key physical differences:

- **PKA energy spectrum:** Thermal-fission neutrons in an LWR moderator produce a PKA distribution weighted toward lower energies ($\langle E_{\text{PKA}} \rangle \approx 5\text{--}10\text{ keV}$ in a Zircaloy cladding), whereas fast-spectrum reactors and CANDU coolant channels expose Zr–2.5 Nb pressure tubes to harder spectra with $\langle E_{\text{PKA}} \rangle \approx 20\text{--}40\text{ keV}$ [5, 36].
- **Cluster sizes:** Higher PKA energies produce larger in-cascade clusters, so the maximum cluster sizes i_{cascade} and v_{cascade} and the power-law exponents all shift [35, 37].
- **Subcascade fragmentation:** Above $\sim 30\text{ keV}$ PKA, cascades in Zr fragment into multiple subcascades, partially decorrelating the cluster spectrum from the primary recoil energy [36].

Table A.3: Recommended cascade-damage parameters for α -Zr under thermal-spectrum (LWR) and fast-spectrum neutron irradiation, synthesized from [35–38]. Parameter uncertainty is dominated by the interatomic potential and by the limited number of independent Zr cascade studies relative to α -Fe.

Parameter	Thermal (LWR)	Fast / CANDU
$\langle E_{\text{PKA}} \rangle$ (keV)	~ 8	~ 30
η (spectrum-averaged)	0.28	0.24
f_i^{cl} (SIA clustering fraction)	0.50	0.55
f_v^{cl} (vacancy clustering fraction)	0.20	0.25
s_i (SIA power-law exponent)	1.7	1.6
s_v (vacancy power-law exponent)	2.4	2.2
i_{cascade} (max SIA cluster size)	20	40
v_{cascade} (max vacancy cluster size)	10	20

The following closed-form expressions allow direct computation of ϵ_m for

any cluster size. For interstitial clusters,

$$\epsilon_m^{(i)} = \frac{f_i^{\text{cl}} m^{-s_i}}{\sum_{k=2}^{i_{\text{cascade}}} k^{1-s_i}}, \quad m = 2, 3, \dots, i_{\text{cascade}} \quad (\text{A.9})$$

Thermal (LWR): $f_i^{\text{cl}} = 0.50$, $s_i = 1.7$, $i_{\text{cascade}} = 20$, giving $C_i = f_i^{\text{cl}} / \sum_{k=2}^{20} k^{-0.7} \approx 0.105$.

Fast / CANDU: $f_i^{\text{cl}} = 0.55$, $s_i = 1.6$, $i_{\text{cascade}} = 40$, giving $C_i \approx 0.061$.

For vacancy clusters,

$$\epsilon_n^{(v)} = \frac{f_v^{\text{cl}} n^{-s_v}}{\sum_{k=2}^{v_{\text{cascade}}} k^{1-s_v}}, \quad n = 2, 3, \dots, v_{\text{cascade}} \quad (\text{A.10})$$

Thermal (LWR): $f_v^{\text{cl}} = 0.20$, $s_v = 2.4$, $v_{\text{cascade}} = 10$, giving $C_v \approx 0.184$.

Fast / CANDU: $f_v^{\text{cl}} = 0.25$, $s_v = 2.2$, $v_{\text{cascade}} = 20$, giving $C_v \approx 0.149$.

The normalization sums can be pre-computed once:

$$S(s, m_{\text{max}}) = \sum_{k=2}^{m_{\text{max}}} k^{1-s}. \quad (\text{A.11})$$

Then $C = f^{\text{cl}}/S$ and $\epsilon_m = C m^{-s}$.

For large m_{max} , the sum can be approximated by the integral

$$S(s, m_{\text{max}}) \approx \begin{cases} \frac{m_{\text{max}}^{2-s} - 2^{2-s}}{2-s} + \frac{1}{2}(m_{\text{max}}^{1-s} + 2^{1-s}), & s \neq 2, \\ \ln\left(\frac{m_{\text{max}}}{2}\right) + \frac{1}{2}(m_{\text{max}}^{-1} + 2^{-1}), & s = 2. \end{cases} \quad (\text{A.12})$$

For direct implementation in a cluster-dynamics code, the source terms take the form shown in Table A.4.

Table A.5 compares key features of the production spectra under thermal- and fast-spectrum conditions. Fig. A.4 shows the interstitial cluster spectrum under representative conditions in α -Zr, while Fig. A.5 displays the production spectrum for vacancy clusters. The principal effects of the harder fast-reactor spectrum are: (1) a larger fraction of surviving defects born in

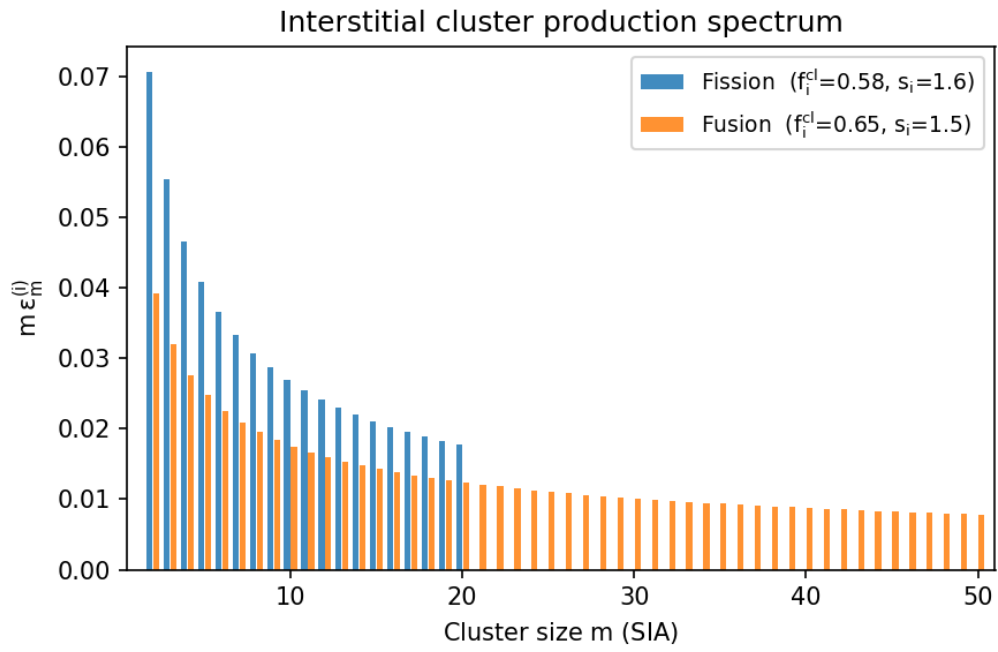


Figure A.4: Interstitial cluster production spectrum under thermal-spectrum (LWR) and fast-spectrum (CANDU/fast-reactor) irradiation conditions in α -Zr.

Table A.4: Rate-theory source-term structure. G is the actual (survival-corrected) defect production rate. The coefficients $\epsilon_m^{(i)}$ and $\epsilon_n^{(v)}$ are read from Table A.3 according to the neutron spectrum.

Species index α	Production rate P_α
1 (mono-SIA)	$P_1^{(i)} = G \left(1 - \sum_{m=2}^{m_{\text{cascade}}} m \epsilon_m^{(i)} \right)$
2	$P_2^{(i)} = \epsilon_2^{(i)} G$
$\vdots$	$\vdots$
i_{cascade}	$P_{m_{\text{cascade}}}^{(i)} = \epsilon_{m_{\text{cascade}}}^{(i)} G$
$N+1$ (mono-vacancy)	$P_1^{(v)} = G \left(1 - \sum_{n=2}^{n_{\text{cascade}}} n \epsilon_n^{(v)} \right)$
$N+2$	$P_2^{(v)} = \epsilon_2^{(v)} G$
$\vdots$	$\vdots$
$N+v_{\text{cascade}}$	$P_{n_{\text{cascade}}}^{(v)} = \epsilon_{n_{\text{cascade}}}^{(v)} G$

clusters; (2) a shallower power-law exponent shifting the distribution toward larger clusters; and (3) extension of the maximum cluster size due to higher-energy cascades and subcascade overlap [36, 37]. The larger vacancy clusters born under fast-spectrum conditions raise the probability of basal collapse, which feeds the $\langle c \rangle$ -component loop nucleation channel responsible for the breakaway irradiation growth observed in Zr–2.5Nb pressure tubes at high fluence [18, 22].

Literature Sources

The Zr cascade-damage literature is considerably less developed than that of α -Fe, and many of the foundational MD techniques and analysis frameworks were developed first in iron and then transferred to zirconium. The principal sources used in the present synthesis are:

- [35]: early MD study of cascade morphology and defect-survival in α -Zr; baseline against which later potentials are compared.
- [36]: comprehensive MD characterization of cascade damage in Zr up to 50 keV, including survival fractions and cluster-size distributions; primary source for the size-resolved Zr spectra used here.

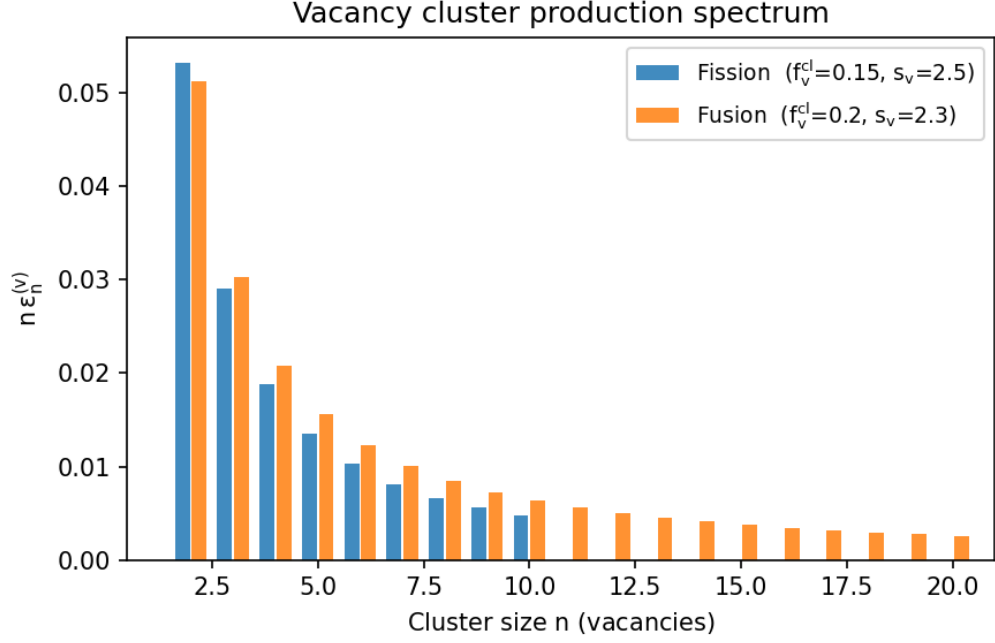


Figure A.5: Vacancy cluster production spectrum under thermal- and fast-spectrum irradiation conditions in α -Zr.

Table A.5: Comparison of defect production characteristics under thermal-spectrum (LWR) and fast-spectrum (CANDU / fast reactor) neutron irradiation in α -Zr.

Quantity	Thermal (LWR)	Fast / CANDU
Mono-SIA fraction, $1 - f_i^{cl}$	0.50	0.45
Mono-vacancy fraction, $1 - f_v^{cl}$	0.80	0.75
Mean SIA cluster size $\langle m \rangle_{cl}$	3.8	5.5
Mean vacancy cluster size $\langle n \rangle_{cl}$	2.5	3.2
Largest SIA cluster $i_{cascade}$	20	40
Largest vacancy cluster $v_{cascade}$	10	20

- [37]: cascade simulations in α -Zr with multiple modern potentials, providing uncertainty bounds on cluster-fraction estimates.
- [23]: post-mortem TEM identification of cascade-collapsed vacancy clusters and their evolution to $\langle c \rangle$ -component loops; provides the experimental anchor linking primary damage to the long-time loop populations.
- [24]: in-situ ion-irradiation TEM at controlled dose and temperature; resolves the small-cluster size distribution and $\langle a \rangle$ -loop nucleation thresholds.
- [19]: DFT-based binding energies and formation energies for small SIA and vacancy clusters in α -Zr; used to constrain the lower end of the in-cascade cluster spectrum.
- [38]: comprehensive review establishing the arc-dpa framework and survival-efficiency parameterization across host metals including Zr.
- [5]: review of cascade damage and the resulting microstructure in zirconium alloys, including a systematic mapping from primary damage to loop populations.

Caveats and Uncertainty

Several sources of uncertainty should be kept in mind:

1. **Interatomic potential:** Predicted cluster fractions depend on the empirical potential used in MD. The Zr literature is dominated by relatively few EAM and modified-embedded-atom potentials, and benchmark comparisons across potentials show cluster-fraction spreads comparable to those reported for Fe [37].
2. **Temperature dependence:** The values here are representative of reactor temperatures (550–600 K). In Zr, interstitial clustering increases slightly and vacancy clustering decreases with increasing temperature, qualitatively similar to Fe [36].
3. **Power-law idealization:** The true size distribution may deviate from a pure power law, particularly at small sizes (where specific magic numbers introduce structure) and at very large sizes (where subcascade fragmentation alters statistics).

4. **Vacancy cluster character:** In hcp Zr, vacancy “clusters” at cascade time scales are often loose, correlated arrangements rather than compact basal loops. Short-range vacancy diffusion subsequently converts a fraction of these into faulted $\langle c \rangle$ -component loops above a cluster-size-dependent threshold [23, 24].
5. **Spectrum averaging:** The values given here use representative average PKA energies; a rigorous treatment would convolve the energy-dependent $\epsilon_m(E_{\text{PKA}})$ over the full PKA recoil spectrum for the specific reactor type and component location [38].

With the production formalism established, we now record the numerical parameters that specify the cascade-damage source terms. Table A.6 collects the cascade-damage production parameters for use in cluster-dynamics modeling of Zircaloy-4 (and, to a good approximation, of pure α -Zr). Zircaloy-4 has a hcp α -Zr matrix with dilute Sn, Fe, and Cr in solid solution, and small amounts of Zr_3Sn / $\text{Zr}(\text{Fe},\text{Cr})_2$ second-phase particles; primary cascade damage is taken to be that of pure α -Zr in the displacement-cascade regime, with solute and second-phase effects deferred to later sections that treat trapping and diffusion. The parameters distinguish thermal- and fast-spectrum neutron irradiation following the synthesis of [5, 36–38].

The production fraction for a cluster of size m is computed as

$$\epsilon_m^{(i)} = C_i m^{-s_i}, \quad \epsilon_n^{(v)} = C_v n^{-s_v}, \quad (\text{A.13})$$

with the mono-defect production rate $P_1 = G(1 - f^{\text{cl}})$.

Appendix B. Energetics

The table below compiles point-defect energetics, elastic constants, and diffusion parameters for hexagonal close-packed (hcp) α -zirconium, as used in rate-theory and cluster-dynamics models of radiation damage in Zr-based fuel-cladding and pressure-tube alloys. Values are from density-functional theory (DFT) calculations unless stated otherwise; experimental values are included where DFT and experiment have been compared in the literature. The Burgers-vector magnitudes for prismatic $\langle a \rangle$ -loops and faulted basal $\langle c \rangle$ -component loops use the experimental lattice parameters $a_0 = 0.3232$ nm and $c_0 = 0.5147$ nm at room temperature.

Table A.6: Defect production input parameters for Zircaloy-4 cluster-dynamics modelling. Values quoted are conservative mid-range estimates; user-specific adjustment within the literature spread is recommended for production runs.

Parameter	Symbol	Thermal	Fast	Units
<i>Survival and dose rate:</i>				
Survival efficiency (spectrum-avg.)	η	0.28	0.24	—
Displacement rate	$\dot{\phi}$	user input	user input	dpa/s
<i>Interstitial cluster production:</i>				
SIA clustering fraction	f_i^{cl}	0.50	0.55	—
SIA power-law exponent	s_i	1.7	1.6	—
Max SIA cluster size	i_{cascade}	20	40	SIAs
SIA normalization constant	C_i	0.105	0.061	—
<i>Vacancy cluster production:</i>				
Vacancy clustering fraction	f_v^{cl}	0.20	0.25	—
Vacancy power-law exponent	s_v	2.4	2.2	—
Max vacancy cluster size	v_{cascade}	10	20	vacancies
Vacancy normalization constant	C_v	0.184	0.149	—

Table B.7: Representative defect and material parameters for hcp α -Zr.

Parameter	Value (range)	Units	References / Notes
Vacancy formation energy, E_v^f	1.75–2.07 (DFT) 1.5 $\pm$ 0.2 (expt.)	eV	DFT: [19, 20]. Experimental positron-annihilation values cluster around 1.5–1.8 eV [5, 39].
Self-interstitial atom formation energy, E_i^f	2.83–3.20 (DFT)	eV	The basal octahedral (BO) and basal split-dumbbell (BS) configurations are nearly degenerate ground states in α -Zr, with formation energy $\approx$ 2.8–3.0 eV [19, 20]. The prismatic crowdion is slightly higher in energy but plays a central role in one-dimensional glide.
Vacancy migration energy (in-plane / out-of-plane), E_v^m	0.51–0.65 (DFT, basal hop) 0.60–0.70 (DFT, c -axis hop)	eV	Anisotropic. DFT gives basal-plane hops $\approx$ 0.55 eV and out-of-plane hops $\approx$ 0.65 eV [19, 20]; experimental resistivity-recovery activation is in the same range [39].
SIA migration energy (prism / basal), E_i^m	0.04–0.10 (DFT, in-plane) 0.25–0.35 (DFT, c -axis)	eV	Strongly anisotropic. In-plane prism-plane migration is nearly athermal ($\sim$ 0.06 eV); c -axis hops are substantially harder ($\sim$ 0.30 eV) [19, 20]. This is the origin of the anisotropic SIA-loop nucleation that dominates Zr microstructure evolution.
Divacancy binding energy (1NN basal), E_{2v}^B	0.15–0.25 (DFT)	eV	DFT divacancy binding is positive but smaller than in α -Fe, indicating weak vacancy clustering in three-dimensional form [19]. Multi-vacancy clusters preferentially collapse onto {0001} basal planes once they exceed a critical size.
Divacancy migration energy, E_{2v}^m	0.65–0.95 (DFT)	eV	Higher than the monovacancy migration barrier; the divacancy is correspondingly slower [19, 20].

Continued on next page

Parameter	Value (range)	Units	References / Notes
Di-interstitial migration energy, E_{2i}^m	$\approx 0.05\text{--}0.15$ (DFT)	eV	Two non-parallel basal SIA dumbbells form a strongly bound di-interstitial that glides along $\langle 11\bar{2}0 \rangle$ with a barrier of 0.05–0.15 eV [19]. This is the elementary mobile species feeding $\langle a \rangle$ -loop nucleation.
Di-interstitial binding energy, E_{2i}^B	0.6–1.0 (DFT)	eV	The di-SIA cluster is strongly bound, with values comparable to those reported in α -Fe. A representative value for the parallel-dumbbell configuration is ≈ 0.8 eV [19].
Isotropic shear modulus, μ	35–36 (RT, expt.)	GPa	For polycrystalline α -Zr, the Voigt–Reuss–Hill average is $\mu \approx 35.5$ GPa at 300 K [40, 41]. It decreases to about 30 GPa at 300 °C. Note that $\mu(\text{Zr}) \approx \frac{1}{2}\mu(\text{Fe})$, which proportionally weakens elastic interactions in Zr and shrinks the bias factors entering the rate equations relative to steel benchmarks.
Burgers-vector magnitude for prismatic $\langle a \rangle$ loops, b_a	0.3232	nm	$b_a = a_0$ for the $\frac{1}{3}\langle 11\bar{2}0 \rangle$ glide direction [5, 40].
Burgers-vector magnitude for basal c -component loops, b_c	0.6173	nm	$b_c = \frac{1}{6}\sqrt{3a_0^2 + 4c_0^2}$ for the $\frac{1}{6}\langle 20\bar{2}3 \rangle$ Burgers vector [5, 23].
Stacking-fault energy on basal $\{0001\}$, $\gamma_{\text{SF}}^{(c)}$	≈ 200 (DFT)	mJ m^{-2}	DFT calculations of the I_2 basal stacking fault give $\gamma_{\text{SF}} \approx 180\text{--}220 \text{ mJ m}^{-2}$ in α -Zr [19, 42]. This sets the formation-energy contribution of faulted basal vacancy c -component loops.
Stacking-fault energy on prismatic $\{10\bar{1}0\}$, $\gamma_{\text{SF}}^{(a)}$	≈ 130 (DFT)	mJ m^{-2}	Lower than the basal fault, consistent with the predominance of prismatic slip and prismatic $\langle a \rangle$ loops [19].

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Parameter	Value (range)	Units	References / Notes
Surface energy, γ_s	1.5–2.0 (DFT) $\approx 1.9 \pm 0.2$ (expt.)	J m^{-2}	DFT gives $\gamma_{\text{basal}} \approx 1.6 \text{ J m}^{-2}$ and $\gamma_{\text{prism}} \approx 1.7\text{--}1.8 \text{ J m}^{-2}$ [43]. The basal surface is the lowest-energy free surface of α -Zr. In the present formulation we do not track three-dimensional voids; the surface energy is retained for completeness.
Pre-exponential factor for vacancy self-diffusion, D_v^0	$\sim 10^{-4}\text{--}10^{-3}$ (DFT-based)	$\text{cm}^2 \text{ s}^{-1}$	DFT-based migration-prefactor estimates from $D_v^0 = f z \nu a^2$ [19, 20]. Anisotropy gives separate prefactors for basal-plane and c -axis components.
Pre-exponential factor for SIA diffusion, D_i^0	$\approx 10^{-4}\text{--}10^{-3}$	$\text{cm}^2 \text{ s}^{-1}$	Estimated from $D_i^0 = f z \nu a^2$ with attempt frequency $\nu \approx 6 \times 10^{12} \text{ s}^{-1}$ and jump distance $a \approx 0.32 \text{ nm}$. In-plane prefactors substantially exceed the c -axis prefactors, mirroring the migration-energy anisotropy [19, 20].

The principal qualitative differences between point-defect energetics in α -Zr and in α -Fe are: (i) substantially smaller μ , which scales the elastic interaction energies and bias factors; (ii) strong anisotropy of both vacancy and SIA migration, absent in cubic Fe and central to the dimensional-change phenomenology of irradiated Zr; and (iii) a marked tendency of multi-vacancy clusters to collapse onto basal planes rather than to form 3-D voids, leading to the $\langle c \rangle$ -component vacancy loops characteristic of Zr at high fluence [19, 23, 24].

The Debye (cut-off) frequency of perfect α -Zr provides the baseline for all defect attempt frequencies. With the experimental Debye temperature $\Theta_D \approx 250\text{--}290 \text{ K}$ [40, 41]:

$$\nu_D = \frac{k_B \Theta_D}{h} \approx 5.2\text{--}6.0 \text{ THz} \approx 4 \times 10^{13} \text{ s}^{-1}. \quad (\text{B.1})$$

The phonon density of states (pDOS) of hcp Zr has prominent transverse-acoustic peaks near $\sim 12\text{--}14 \text{ meV}$ ($\sim 3 \text{ THz}$) and a broad longitudinal peak between $\sim 18\text{--}22 \text{ meV}$ ($\sim 4.5\text{--}5.5 \text{ THz}$) [44, 45].

In Vineyard’s harmonic transition-state theory [46] the jump rate of a

point defect is

$$\Gamma = \nu_0 \exp\left(-\frac{E_m}{k_B T}\right), \quad \nu_0 = \frac{\prod_{i=1}^{3N} \nu_i^{(0)}}{\prod_{i=1}^{3N-1} \nu_i^{(\text{sp})}}, \quad (\text{B.2})$$

where $\nu_i^{(0)}$ and $\nu_i^{(\text{sp})}$ are the real normal-mode frequencies at the stable minimum and the saddle point, respectively. Table B.8 summarizes the key values for the two defect species in α -Zr, with explicit basal and c -axis components where the anisotropy is significant.

Table B.8: Migration energies, attempt frequencies, and local vibrational modes for the vacancy and the self-interstitial atom (SIA) in hcp α -Zr. Anisotropic components are listed separately where the difference between basal-plane and c -axis hops is significant.

Defect	E_m (eV)	Attempt freq. ν_0 (s^{-1})	Local mode (meV / THz)	References
Vacancy (basal hop)	0.51–0.60	$3\text{--}6 \times 10^{12}$	softening of ~ 18 meV LA peak	[19, 20, 46]
Vacancy (c -axis hop)	0.60–0.70	$3\text{--}6 \times 10^{12}$	no localized mode	[19, 20]
SIA (basal split / prism glide)	0.04–0.10 (in-plane)	$\sim 4 \times 10^{12}$	dumbbell stretch $\sim 30\text{--}40$ meV	[19, 20]
SIA (c -axis hop)	0.25–0.35	$\sim 4 \times 10^{12}$	c -axis libration $\sim 15\text{--}20$ meV	[19, 20]

A monovacancy in α -Zr introduces no localized modes *above* the Zr phonon band; its main spectral signature is a softening of the longitudinal acoustic peak at $\sim 18\text{--}22$ meV, proportional to vacancy concentration and confined largely to the first-neighbour shell. The attempt frequency spans a moderate range in the literature: the Debye-based estimate $\nu_0^v \approx 4 \times 10^{12} \text{ s}^{-1}$ is routinely used in rate-theory codes, while DFT Vineyard calculations give somewhat higher values [19, 20].

The ground-state SIA in α -Zr is the basal split-dumbbell configuration, with the basal-octahedral configuration nearly degenerate. In-plane (basal)

migration is essentially athermal ($E_m^{\text{in-plane}} \approx 0.06$ eV), while c -axis hops require crossing a substantially higher barrier (~ 0.30 eV). This single-defect anisotropy propagates through to small clusters and ultimately to the prismatic-loop habit-plane preference observed experimentally [19, 23, 24].

The strongly differing migration energies—SIA in-plane (~ 0.06 eV) $\ll$ SIA c -axis (~ 0.30 eV) $\ll$ vacancy (~ 0.6 eV)—dominate the diffusivity hierarchy over many orders of magnitude. In a uniaxial-stress environment (e.g., Zr–2.5Nb pressure tubes under hoop stress, or fuel cladding under reactor overpressure) the SIA anisotropy gives rise to differential capture biases between $\langle a \rangle$ - and $\langle c \rangle$ -loops, which is the microscopic origin of the SIPA component of irradiation creep [5, 18, 22].

Having defined how formation, migration, and binding energies enter the model, we now tabulate the specific values adopted for Zircaloy-4.

Table B.9 collects the point-defect energetics and material constants for hcp α -Zr / Zircaloy-4 used throughout the rate equations. Solute-modified energetics (Sn, Fe, Cr, Nb trapping) are deferred to Section ??.

Table B.9: Defect energetics input parameters for Zircaloy-4 cluster-dynamics modelling. Where anisotropy is present, both in-plane (basal) and c -axis components are listed.

Parameter	Symbol	Value	Units
<i>Lattice and elastic constants:</i>			
Lattice constant (a)	a_0	0.3232	nm
Lattice constant (c)	c_0	0.5147	nm
c/a ratio	c/a	1.593	—
Atomic volume	Ω	2.32×10^{-29}	m ³
Burgers vector ($\langle a \rangle$ -loop)	b_a	0.3232	nm
Burgers vector ($\langle c \rangle$ -loop)	b_c	0.6173	nm
Shear modulus (RT)	μ	35.5	GPa
Poisson ratio	ν	0.34	—
Basal stacking-fault energy	$\gamma_{\text{SF}}^{(c)}$	200	mJ/m ²
Prism stacking-fault energy	$\gamma_{\text{SF}}^{(a)}$	130	mJ/m ²
Surface energy (basal)	γ_s	1.6	J/m ²
<i>Vacancy energetics:</i>			
Formation energy	E_f^v	1.9	eV
Migration energy (basal)	$E_m^{v,\parallel a}$	0.55	eV
Migration energy (c -axis)	$E_m^{v,\parallel c}$	0.65	eV
Divacancy binding energy	E_B^{2v}	0.20	eV
Divacancy migration energy	E_m^{2v}	0.75	eV
<i>SIA energetics:</i>			
Formation energy (basal split)	E_f^i	2.85	eV
Migration energy (in-plane)	$E_m^{i,\parallel a}$	0.06	eV
Migration energy (c -axis)	$E_m^{i,\parallel c}$	0.30	eV
Di-SIA binding energy	E_B^{2i}	0.80	eV
Di-SIA migration energy	E_m^{2i}	0.10	eV
<i>Attempt frequencies:</i>			
Vacancy	ν_v	5×10^{12}	s ⁻¹
SIA (in-plane)	$\nu_i^{\parallel a}$	5×10^{12}	s ⁻¹
SIA (c -axis)	$\nu_i^{\parallel c}$	4×10^{12}	s ⁻¹

Appendix C. Diffusion

In hcp α -zirconium, point-defect transport is intrinsically anisotropic: hops within the basal plane and hops along the c -axis have distinct migration energies and attempt frequencies, and the asymmetry is much larger for SIAs than for vacancies [19, 20]. The cluster-dynamics rate constants of Section Appendix E require a single effective transition frequency ω_α per species; we obtain it by orientation averaging the anisotropic diffusion tensor and tracking the basal- and c -axis components separately when the bias factors require it.

For an isotropic host, the diffusion coefficient of a random-walking species is

$$D = \frac{1}{6} z \Gamma a^2, \quad (\text{C.1})$$

where z is the coordination of jump sites, Γ is the jump rate, and a is the jump distance. For an anisotropic host, the scalar D is replaced by a diffusion tensor. In hcp Zr, the appropriate decomposition uses two independent components:

$$D_\alpha^{\parallel a} = \frac{1}{4} z_\parallel \Gamma_\alpha^{\parallel a} a_0^2, \quad D_\alpha^{\parallel c} = \frac{1}{6} z_\perp \Gamma_\alpha^{\parallel c} c_0^2, \quad (\text{C.2})$$

with $a_0 = 0.3232$ nm and $c_0 = 0.5147$ nm, $z_\parallel$ counting in-plane (basal) jump partners, and $z_\perp$ counting interlayer (c -direction) partners. Each component follows the Arrhenius form

$$D_\alpha^{\parallel \kappa}(T) = D_{0,\alpha}^{\parallel \kappa} \exp\left(-\frac{E_m^{\alpha,\parallel \kappa}}{k_B T}\right), \quad \kappa \in \{a, c\}. \quad (\text{C.3})$$

The orientation-averaged scalar diffusivity used in 3-D reaction rate constants is

$$\bar{D}_\alpha = \frac{1}{3} (2D_\alpha^{\parallel a} + D_\alpha^{\parallel c}), \quad (\text{C.4})$$

and the orientation-averaged transition frequency $\omega_\alpha = \bar{D}_\alpha / a_0^2$ is what enters the isotropic-equivalent formulas of Section Appendix E. For sinks whose bias depends on direction (e.g., basal $\langle c \rangle$ -loops and prismatic $\langle a \rangle$ -loops), the two components are retained explicitly.

We apply this random-walk framework first to vacancies.

Vacancy migration in α -Zr proceeds by nearest-neighbour hops on the hcp lattice with two distinct barrier energies: a basal-plane hop $E_v^{\parallel a} \approx 0.55$ eV

and a c -axis hop $E_v^{\parallel c} \approx 0.65$ eV [19, 20]. The corresponding components of the vacancy diffusion tensor are

$$D_v^{\parallel a}(T) = \nu_v a_0^2 \exp\left(-\frac{E_v^{\parallel a}}{k_B T}\right), \quad (\text{C.5})$$

$$D_v^{\parallel c}(T) = \nu_v c_0^2/3 \exp\left(-\frac{E_v^{\parallel c}}{k_B T}\right), \quad (\text{C.6})$$

with attempt frequency $\nu_v \sim 5 \times 10^{12} \text{ s}^{-1}$. The modest anisotropy in E_v produces a basal-to-axial diffusivity ratio of order 2–3 at 600 K, weaker than the SIA anisotropy discussed below.

Self-interstitial atoms migrate by a distinct mechanism, which we treat next.

The ground-state SIA configuration in α -Zr is the basal-octahedral or basal split-dumbbell configuration; the two are nearly degenerate within DFT uncertainty [19, 20]. In-plane migration through the prismatic plane carries a near-athermal barrier of $E_i^{\parallel a} \approx 0.06$ eV, while c -axis migration is substantially harder at $E_i^{\parallel c} \approx 0.30$ eV. This order-of-magnitude anisotropy in migration energy produces a basal-to-axial diffusivity ratio that exceeds 10^4 at 600 K and is the microscopic origin of the prismatic habit of $\langle a \rangle$ -loops and the strong texture-dependence of irradiation growth.

The two diffusion components are

$$D_i^{\parallel a}(T) = \nu_i^{\parallel a} a_0^2 \exp\left(-\frac{E_i^{\parallel a}}{k_B T}\right), \quad (\text{C.7})$$

$$D_i^{\parallel c}(T) = \nu_i^{\parallel c} c_0^2/3 \exp\left(-\frac{E_i^{\parallel c}}{k_B T}\right), \quad (\text{C.8})$$

with attempt frequencies $\nu_i^{\parallel a} \approx 5 \times 10^{12} \text{ s}^{-1}$ and $\nu_i^{\parallel c} \approx 4 \times 10^{12} \text{ s}^{-1}$. In a randomly oriented polycrystal, the orientation-averaged diffusivity $\bar{D}_i$ from Eq. (C.4) is dominated by $D_i^{\parallel a}$. In textured Zircaloy or Zr-2.5Nb component microstructures (basal poles oriented at characteristic angles to the rolling/extrusion direction), the texture distributes a non-negligible fraction of the diffusion flux along the c -axis and is one of the parameters that production-grade implementations must adjust per component geometry.

Effective SIA diffusivity in Zircaloy-4. In Zircaloy-4 the principal SIA traps are the substitutional solutes Sn, Fe, and Cr. Each trap reduces the long-range SIA diffusivity by the standard residence-time factor:

$$\omega_i^{\text{eff}} = \frac{\omega_i^{\text{bulk}}}{1 + \sum_{s \in \{\text{Sn, Fe, Cr}\}} z_s c_s \exp(E_b^{i-s}/k_B T)}, \quad (\text{C.9})$$

where z_s is the trapping coordination, c_s is the solute atomic fraction, and E_b^{i-s} is the SIA–solute binding energy. The resulting trapping factor at typical Zircaloy-4 composition and $T \sim 600$ K is of order 5–20, depending on which solutes dominate.

Beyond the mono-interstitial, the mobility of small interstitial clusters must also be specified.

Size-dependent migration mechanisms. Small SIA clusters in α -Zr ($n = 2$ –3) glide rapidly along the $\langle 11\bar{2}0 \rangle$ Burgers-vector direction on prismatic glide planes, with a barrier comparable to that of the single SIA in-plane migration ($E_m^{1D} \lesssim 0.10$ eV). Larger clusters ($n \gtrsim 10$) have progressively reduced glide mobility because the reorientation events that produce 3-D random-walk character become rate-limiting. Above the size at which the assembled prismatic loop carries a well-defined Burgers vector, $n \gtrsim 30$, the cluster is sessile from the cluster-dynamics standpoint and appears only as a sink for free SIAs and vacancies.

1-D diffusion coefficient. The 1-D glide diffusion coefficient of an n -SIA cluster is

$$D_n^{1D}(T) = \nu_0^{1D} a_0^2 \frac{1}{n^s} \exp\left(-\frac{E_m^{1D}}{k_B T}\right), \quad (\text{C.10})$$

where $s \approx 0.5$ –1.0 captures the size-dependence of the glide barrier (atomic-scale studies in Zr suggest $s \approx 0.7$ for $n \leq 10$). The corresponding 1-D transition frequency is $\omega_n^{1D} = D_n^{1D}/a_0^2$.

Mixed 1-D/3-D framework. Occasional Burgers-vector reorientation between distinct $\langle 11\bar{2}0 \rangle$ glide planes converts pure 1-D random walk into mixed 1-D/3-D character. The mean free path L between reorientation events sets the effective 3-D transition frequency

$$\omega_n^{3D, \text{eff}} = \frac{\omega_n^{1D} L}{3\sqrt{3} a_0}, \quad (\text{C.11})$$

which is the form that enters the rate constants of Section Appendix E. In Zircaloy-4, L is shortened by solute trapping to a value of order $L = 50$ –

$100 a_0$, consistent with the in-situ TEM observations of (author?) [24] and the post-mortem analyses of (author?) [25].

Sink reaction rates: 3-D vs. 1-D. The sink reaction rates for 1-D-migrating clusters scale differently with sink size and concentration than the standard 3-D Smoluchowski expressions; see Section Appendix E for the explicit formulae and the Trinkaus–Singh–Golubov interpolation between regimes.

The mobility of vacancy clusters follows by analogous reasoning.

In hcp Zr, multi-vacancy clusters are intrinsically less mobile than their SIA counterparts. Beyond the divacancy stage, vacancy clusters *collapse* onto $\{0001\}$ basal planes to form faulted $\langle c \rangle$ -component loops with Burgers vector $\frac{1}{6}\langle 20\bar{2}3 \rangle$. Once collapsed, a $\langle c \rangle$ -loop is sessile: it cannot glide because its Burgers vector lies normal to the basal habit plane.

The mobile vacancy cluster population is therefore restricted to $1 \leq m \leq v_{\text{mobile}}$, with $v_{\text{mobile}} = 4$ a typical choice for α -Zr. Migration energies for these small clusters are summarised in Table C.10; they exceed the single-vacancy migration energy by 0.05–0.20 eV, reflecting the correlated displacement of multiple vacancies.

Solute effects on vacancy diffusion in Zircaloy-4. The principal vacancy traps in Zircaloy-4 are O (interstitial) and the substitutional solutes Sn and Fe. Vacancy–O binding energies of ~ 0.4 eV are reported by DFT [19]; vacancy–Sn and vacancy–Fe binding energies are weaker (~ 0.10 – 0.20 eV). The effective vacancy frequency follows the same residence-time form as the SIA (Eq. C.9) with the substitutions $i \rightarrow v$ and the appropriate solute set.

Collecting these mechanisms, we now list the diffusion parameters adopted for Zircaloy-4.

Table C.10 collects the diffusion parameters used in cluster-dynamics modelling of Zircaloy-4. Where anisotropy is significant (vacancy and SIA diffusion), both basal-plane and c -axis components are listed.

Implementation note. The orientation-averaged frequencies ω_i and ω_v entering the rate-constant tables are computed at simulation time from the basal and c -axis diffusion components and the assumed texture of the modelled component (random for pure-Zr validation, strongly textured for fuel cladding tubes and pressure tubes). For Zr-2.5Nb pressure tubes, the additional Nb solute trap is added to the trapping sums in Eq. (C.9) with $E_b^{i\text{-Nb}} \approx 0.20$ eV and $E_b^{v\text{-Nb}} \approx 0.10$ eV (approximate values).

Table C.10: Defect-diffusion input parameters for Zircaloy-4 cluster dynamics. Anisotropic components are listed separately for the SIA and the vacancy.

Parameter	Symbol	Value	Units
<i>Lattice parameters:</i>			
a -axis lattice parameter	a_0	0.3232	nm
c -axis lattice parameter	c_0	0.5147	nm
<i>Vacancy diffusion:</i>			
Migration energy (basal)	$E_v^{\parallel a}$	0.55	eV
Migration energy (c -axis)	$E_v^{\parallel c}$	0.65	eV
Attempt frequency	ν_v	5×10^{12}	s^{-1}
<i>SIA diffusion:</i>			
Migration energy (in-plane)	$E_i^{\parallel a}$	0.06	eV
Migration energy (c -axis)	$E_i^{\parallel c}$	0.30	eV
Attempt frequency (in-plane)	$\nu_i^{\parallel a}$	5×10^{12}	s^{-1}
Attempt frequency (c -axis)	$\nu_i^{\parallel c}$	4×10^{12}	s^{-1}
<i>SIA cluster (1-D glide):</i>			
1-D migration energy	E_m^{1D}	0.10	eV
1-D attempt frequency	ν_0^{1D}	5×10^{12}	s^{-1}
Size exponent	s	0.7	—
Max mobile size	i_{mobile}	30	SIAs
<i>Vacancy cluster diffusion:</i>			
Divacancy migration energy	E_m^{2v}	0.75	eV
Tri-vacancy migration energy	E_m^{3v}	0.80	eV
Max mobile size	v_{mobile}	4	vacancies
<i>Solute composition (Zircaloy-4 nominal, wt. %):</i>			
Tin	Sn	1.5	wt. %
Iron	Fe	0.21	wt. %
Chromium	Cr	0.10	wt. %
Oxygen (interstitial)	O	0.12	wt. %
<i>SIA-solute binding energies (DFT, indicative):</i>			
SIA–Sn	$E_b^{i\text{-Sn}}$	0.20	eV
SIA–Fe	$E_b^{i\text{-Fe}}$	0.30	eV
SIA–Cr	$^{46}E_b^{i\text{-Cr}}$	0.25	eV
<i>Vacancy-solute binding energies (DFT, indicative):</i>			
V–O	$E_b^{v\text{-O}}$	0.40	eV
V–Sn	$E_b^{v\text{-Sn}}$	0.15	eV
V–Fe	$E_b^{v\text{-Fe}}$	0.10	eV

Appendix D. Dissociation

The equilibrium thermal vacancy concentration in the bulk is approximated by

$$C_v^0 = \exp\left(-\frac{E_v^f}{k_B T}\right), \quad (\text{D.1})$$

where E_v^f is the vacancy formation free energy or, in the usual approximation, the vacancy formation energy. At the surface of a vacancy cluster of size α , the local equilibrium vacancy concentration is elevated above C_v^0 according to

$$C_v^e(\alpha) = C_v^0 \exp\left(\frac{\mu(\alpha)}{k_B T}\right) = \exp\left[-\frac{E_v^f - \mu(\alpha)}{k_B T}\right], \quad (\text{D.2})$$

where $\mu(\alpha)$ is the vacancy chemical potential of the cluster. For an interstitial cluster of size β , the equilibrium vacancy concentration at the surface is lowered below C_v^0 :

$$C_v^e(\beta) = C_v^0 \exp\left(-\frac{\mu(\beta)}{k_B T}\right) = \exp\left[-\frac{E_v^f + \mu(\beta)}{k_B T}\right]. \quad (\text{D.3})$$

A unified expression for both cluster types follows by defining the binding energy of a vacancy to a vacancy cluster as

$$E_b^{(v)}(\alpha) = E_v^f - \mu(\alpha) \quad (\text{D.4})$$

and the binding energy of a vacancy to an interstitial cluster of size β as

$$E_b^{(i)}(\beta) = E_v^f + \mu(\beta). \quad (\text{D.5})$$

Then the equilibrium vacancy concentration at the surface of either cluster type can be written as

$$C_v^e = \exp\left[-\frac{E_b}{k_B T}\right]. \quad (\text{D.6})$$

The binding energy can be obtained from atomistic calculations (DFT, MD), from experiments, or using continuum theory for large clusters. Explicit forms for α -Zr are derived below. In the hcp framework, both vacancy and interstitial clusters appear as *dislocation loops* rather than as 3-D voids: the dominant vacancy population in irradiated Zr collapses onto basal planes to form faulted $\langle c \rangle$ -component loops with Burgers vector $\frac{1}{6}\langle 20\bar{2}3 \rangle$, while the dominant interstitial population assembles into prismatic $\langle a \rangle$ -loops

with Burgers vector $\frac{1}{3}\langle 11\bar{2}0 \rangle$ on $\{10\bar{1}0\}$ habit planes. The remainder of this section treats the dissociation kinetics of the two loop families.

For a circular dislocation loop of N point defects, with Burgers vector b and habit plane area $A_N = N\Omega/b$, the loop radius is

$$R_L(N) = \sqrt{\frac{A_N}{\pi}} = \sqrt{\frac{N\Omega}{\pi b}}. \quad (\text{D.7})$$

The defect chemical potential at the loop perimeter, including the self-stress field of the dislocation, the stacking-fault energy (faulted loops only), and the externally applied stress, is

$$\mu_L(N) = \frac{\Omega}{b} \left[\frac{\mu_s b^2}{4\pi(1-\nu) R_L(N)} \ln\left(\frac{R_L(N)}{r_0}\right) + \gamma_{\text{SF}} - \sigma_{\text{app}} \right], \quad (\text{D.8})$$

where μ_s is the shear modulus, ν is the Poisson ratio, $r_0 \approx b$ is the dislocation core cutoff, γ_{SF} is the stacking-fault energy of the loop habit plane (zero for perfect loops, $\sim 200 \text{ mJ m}^{-2}$ for faulted basal $\langle c \rangle$ -loops in α -Zr), and σ_{app} is the resolved applied stress in the loop's normal direction.

For the SIA-type prismatic $\langle a \rangle$ -loop, the binding energy of an SIA at the loop perimeter is

$$E_b^{i,(a)}(n) = E_i^f + \mu_L^{(a)}(n), \quad (\text{D.9})$$

with $\mu_L^{(a)}(n)$ evaluated using $b = b_a = a_0$ and $\gamma_{\text{SF}}^{(a)}$ (zero for the perfect $\frac{1}{3}\langle 11\bar{2}0 \rangle$ loop).

For the vacancy-type basal $\langle c \rangle$ -loop, the binding energy of a vacancy at the loop perimeter is

$$E_b^{v,(c)}(m) = E_v^f - \mu_L^{(c)}(m), \quad (\text{D.10})$$

with $\mu_L^{(c)}(m)$ evaluated using $b = b_c = \frac{1}{6}\sqrt{3a_0^2 + 4c_0^2}$ and $\gamma_{\text{SF}}^{(c)} \approx 200 \text{ mJ m}^{-2}$. The $-\mu_L$ sign reflects the vacancy nature of the $\langle c \rangle$ -loop: vacancies are bound (positive E_b) because removing a vacancy at the perimeter requires inflating the faulted region.

The continuum expression above breaks down for the smallest clusters, so we blend it with atomistic data.

The continuum loop expression Eq. (D.8) is accurate for $N \gtrsim 20$ (where the dislocation loop is well-defined) but breaks down for very small clusters where the discrete nature of the defect aggregate dominates. For $N \leq 5$ in α -Zr, the binding energies are taken from atomistic calculations [19, 20]. A

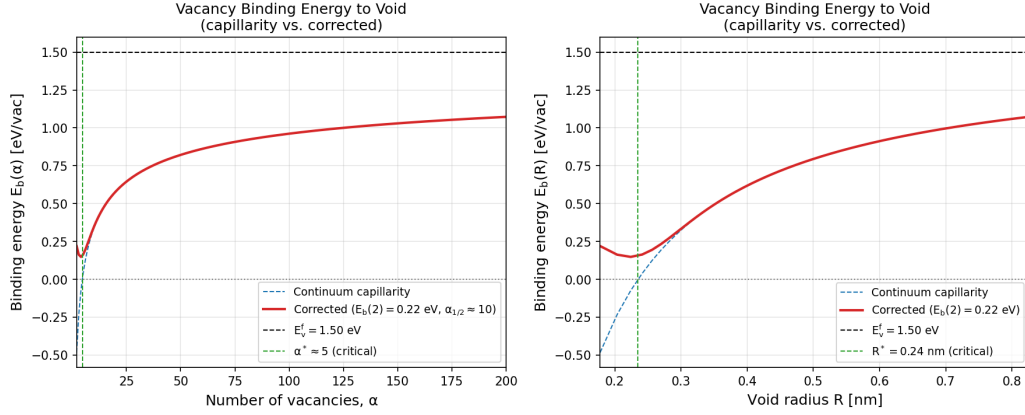


Figure D.6: Binding energy of a vacancy to a vacancy cluster as a function of cluster size, in the continuum approximation (Eqs. D.8–D.10) blended with DFT data for small clusters. The plot is conceptually equivalent to the corresponding void calculation in cubic metals, with μ_L replacing the $2\gamma/R$ Laplace term.

smooth blend across the small-to-large transition is achieved by the same exponential correction used for voids:

$$E_b^{\text{mod}}(N) = E_b^{\text{cont}}(N) + [E_b^{\text{DFT}}(2) - E_b^{\text{cont}}(2)] \exp(-\lambda(N - 2)), \quad (\text{D.11})$$

with the decay constant chosen so the correction is $\sim 1\%$ at $N = 10$: $\lambda = \ln(100)/8 \approx 0.575$.

These binding energies in turn set the rates at which point defects are thermally emitted from clusters.

The thermal emission rate of a point defect from a loop of size N is related to the capture rate by detailed balance. For an SIA emitted from an $\langle a \rangle$ -loop:

$$\varepsilon_i^{(a)}(n) = A_{\text{loop}} (n - 1)^{1/2} Z_i^{(a)} \omega_i^{\text{eff}} \exp\left(-\frac{E_b^{i,(a)}(n)}{k_B T}\right), \quad (\text{D.12})$$

and for a vacancy emitted from a $\langle c \rangle$ -loop:

$$\varepsilon_v^{(c)}(m) = A_{\text{loop}}^{(c)} (m - 1)^{1/2} Z_v^{(c)} \omega_v^{\text{eff}} \exp\left(-\frac{E_b^{v,(c)}(m)}{k_B T}\right). \quad (\text{D.13})$$

The constants A_{loop} , $A_{\text{loop}}^{(c)}$, $Z_i^{(a)}$, and $Z_v^{(c)}$ are defined in Section Appendix E, and the binding energies follow Eqs. (D.9), (D.10), with the small-cluster

blend of Eq. (D.11). Cross-channel emission—SIA from a vacancy $\langle c \rangle$ -loop, vacancy from an $\langle a \rangle$ -loop—is energetically suppressed and is omitted from the master equations.

We close this appendix by tabulating the dissociation parameters used for Zircaloy-4.

Table D.11 collects the binding-energy parameters used in the dissociation kinetics of Zircaloy-4. Material constants (a_0 , c_0 , μ_s , ν , γ_{SF}) are those of Section Appendix B.

Implementation notes. At each time step, $E_b^{i,(a)}(n)$ and $E_b^{v,(c)}(m)$ are evaluated as follows:

1. For $N \leq 5$: use the tabulated DFT value.
2. For $5 < N < 20$: use the blend of Eq. (D.11) with E_b^{cont} from Eq. (D.8).
3. For $N \geq 20$: use the pure continuum expression $E_b^{\text{cont}}(N)$.

Applied stresses (hoop, axial, in-plane) enter through σ_{app} in Eq. (D.8), modifying the loop-perimeter chemical potential, and hence both the binding energy and the emission rate. This stress dependence is the entry point for the SIPA contribution to irradiation creep, and for the hoop-stress-driven hydride reorientation coupling that is treated in extensions of the present formulation.

Appendix E. Reaction Rate Constants

This section derives the rate constants for all reactions between defect clusters in α -Zr, expressed in units of atomic-fraction (s^{-1}) and in terms of reaction frequencies $\omega_\alpha = D_\alpha/a^2$ and the geometric prefactor $\xi(m) = (3m/8\pi)^{1/3}$ introduced in the diffusion section. The hcp framework requires two loop-family-specific rate-constant sets: prismatic $\langle a \rangle$ -loops on $\{10\bar{1}0\}$ habit and faulted basal $\langle c \rangle$ -component loops on $\{0001\}$. The cluster notation and concentration units (at/at) follow the conventions established in Section Appendix C.

For a mobile point defect α absorbed by an immobile spherical cluster of m defects, the Smoluchowski volumetric rate constant is $\mathcal{K}^{\text{vol}} = 4\pi r_m(D_A + D_B)$. Converting to atomic-fraction units ($\mathcal{K} = \mathcal{K}^{\text{vol}}/\Omega$) and substituting $D = \omega a^2$, $r_m = \xi(m)a$, and $\Omega = a^3c/2$:

$$\mathcal{K}_{\alpha,m}^{3D} = 8\pi \xi(m) \omega_\alpha \quad [\text{s}^{-1}]. \quad (\text{E.1})$$

Table D.11: Cluster-dissociation parameters for Zircaloy-4. Atomistic small-cluster binding energies are mid-range estimates from the DFT studies of [19, 20].

Parameter	Symbol	Value	Units
<i>Continuum / capillary parameters:</i>			
Stacking-fault energy on $\{0001\}$	$\gamma_{\text{SF}}^{(c)}$	200	mJ/m ²
Stacking-fault energy on $\{10\bar{1}0\}$	$\gamma_{\text{SF}}^{(a)}$	130	mJ/m ²
Shear modulus	μ_s	35.5	GPa
Poisson ratio	ν	0.34	—
Burgers vector ($\langle a \rangle$ -loop)	b_a	0.3232	nm
Burgers vector ($\langle c \rangle$ -loop)	b_c	0.6173	nm
Core cutoff	r_0	$\approx b$	nm
<i>Atomistic small-cluster binding (DFT):</i>			
$E_b^{i,(a)}(2)$ (di-SIA in $\langle a \rangle$ config.)		0.80	eV
$E_b^{i,(a)}(3)$		1.05	eV
$E_b^{i,(a)}(4)$		1.30	eV
$E_b^{i,(a)}(5)$		1.45	eV
$E_b^{v,(c)}(2)$ (divacancy basal config.)		0.20	eV
$E_b^{v,(c)}(3)$		0.45	eV
$E_b^{v,(c)}(4)$		0.70	eV
$E_b^{v,(c)}(5)$		0.85	eV
<i>Blending parameters:</i>			
Decay constant	λ	0.575	defect ⁻¹
Small-cluster cutoff	N_{small}	5	defects
Onset of continuum regime	N_{cont}	20	defects

For the anisotropic hcp matrix, ω_α is interpreted as the isotropic-equivalent frequency $\omega_\alpha = (\omega_\alpha^{\parallel a})^{2/3} (\omega_\alpha^{\parallel c})^{1/3}$ in the small-cluster limit; for clusters large enough that diffusion to their surface is dominated by the in-plane component, $\omega_\alpha \rightarrow \omega_\alpha^{\parallel a}$ (see Section Appendix C). When both reactants are mobile, the rate constant includes both frequencies:

$$\mathcal{K}_{AB}^{3D} = 8\pi \xi_{\text{rec}} (\omega_A + \omega_B), \quad (\text{E.2})$$

with $\xi_{\text{rec}} = (3/8\pi)^{1/3} \approx 0.492$ for nearest-neighbour capture in hcp Zr at the basal-plane bond distance.

Special cases. Vacancy–SIA recombination ($m = n = 1$, both mobile):

$$\mathcal{K}_{iv} = 8\pi \xi_{\text{rec}} (\omega_i + \omega_v) \approx 12.4 \omega_i, \quad (\text{E.3})$$

since $\omega_i \gg \omega_v$ at all relevant temperatures, with the in-plane SIA frequency dominating because of its very low migration barrier.

Point defect absorption by a prismatic $\langle a \rangle$ -loop of n SIAs (interstitial nature, dominant population in α -Zr) — the target geometry is a circular dislocation loop of radius $r_n = b_a \sqrt{n/\pi}$ on a $\{10\bar{1}0\}$ habit plane:

$$\mathcal{K}_{i,n}^{(a)} = A_{\text{loop}} n^{1/2} Z_i^{(a)} \omega_i, \quad (\text{E.4})$$

$$\mathcal{K}_{v,n}^{(a)} = A_{\text{loop}} n^{1/2} Z_v^{(a)} \omega_v, \quad (\text{E.5})$$

with $A_{\text{loop}} = 8\sqrt{\pi/\sqrt{3}} \approx 10.78$ and $Z_i^{(a)} > Z_v^{(a)}$ to capture the dilatational bias of an SIA loop (Table E.13). SIA absorption grows the loop by one; vacancy absorption shrinks it by one.

Point defect absorption by a basal $\langle c \rangle$ -component loop of m vacancies (faulted, vacancy nature on $\{0001\}$) — the target is a circular loop of radius $r_m = b_c \sqrt{m/\pi}$ with the loop plane perpendicular to c :

$$\mathcal{K}_{v,m}^{(c)} = A_{\text{loop}}^{(c)} m^{1/2} Z_v^{(c)} \omega_v, \quad (\text{E.6})$$

$$\mathcal{K}_{i,m}^{(c)} = A_{\text{loop}}^{(c)} m^{1/2} Z_i^{(c)} \omega_i, \quad (\text{E.7})$$

where $A_{\text{loop}}^{(c)}$ accounts for the basal habit plane through the geometric correction $A_{\text{loop}}^{(c)} = A_{\text{loop}} \sqrt{b_c/b_a} \approx 14.9$, and the bias factors $Z_v^{(c)} > Z_i^{(c)}$ reflect the compressive elastic field of a vacancy basal loop preferentially attracting vacancies. Vacancy absorption grows the $\langle c \rangle$ -loop by one; SIA absorption shrinks it by one.

Point defect absorption by the dislocation network (first-order sink):

$$\left. \frac{dc_\alpha}{dt} \right|_{\text{disl}} = -Z_\alpha^d \rho_d a^2 \omega_\alpha c_\alpha, \quad (\text{E.8})$$

where ρ_d (m^{-2}) is the dislocation line density. In Zr, ρ_d is split into a network component, an $\langle a \rangle$ -loop contribution proportional to $\sum_n 2\pi r_n c_n^{(a)}$, and a $\langle c \rangle$ -loop contribution proportional to $\sum_m 2\pi r_m c_m^{(c)}$.

Small SIA clusters glide one-dimensionally rather than diffusing in three dimensions, which modifies their capture rates as follows.

Small SIA clusters of size $n \geq 2$ in α -Zr migrate by 1-D glide along $\langle 11\bar{2}0 \rangle$ on prismatic glide planes [5, 19], with reaction frequency (from Section Appendix C):

$$\omega_n^{1D} = \frac{3}{2} \cdot \frac{\nu_0}{n^s} \exp\left(-\frac{E_m^{1D}}{k_B T}\right), \quad s \approx 0.5\text{--}1.0, \quad E_m^{1D} \lesssim 0.10 \text{ eV}. \quad (\text{E.9})$$

The lower bound $E_m^{1D} \lesssim 0.10 \text{ eV}$ reflects the in-plane SIA migration barrier in α -Zr; small clusters retain the same near-athermal mobility along $\langle 11\bar{2}0 \rangle$ until they exceed the size at which Burgers-vector reorientation becomes the rate-limiting step (typically $n \sim 10$ to 30).

Mixed 1D/3D regime (finite mean free path). The effective 3D reaction frequency after Burgers-vector rotation between glide planes is:

$$\omega_n^{3D,\text{eff}} = \frac{\omega_n^{1D} L}{3\sqrt{3} a}, \quad (\text{E.10})$$

where L is the mean free path between rotation events. The rate constant for an n -SIA cluster reacting with an immobile $\langle c \rangle$ -loop of m vacancies is then:

$$\mathcal{K}_{n,m}^{\text{mixed}} = A_{\text{loop}}^{(c)} m^{1/2} \omega_n^{3D,\text{eff}} = \frac{A_{\text{loop}}^{(c)}}{3\sqrt{3}} m^{1/2} \hat{L} \omega_n^{1D}, \quad (\text{E.11})$$

where $\hat{L} = L/a$ is the dimensionless mean free path.

Pure 1D regime. For a 1-D-migrating cluster reacting with randomly distributed loop-shaped sinks of size m and atomic fraction $c_m^{(c)}$, the loss rate is:

$$\left. \frac{dc_n^{(a)}}{dt} \right|_{\text{sink}} = -\mathcal{K}_{n,m}^{1D} (c_m^{(c)})^2 c_n^{(a)}, \quad (\text{E.12})$$

where:

$$\mathcal{K}_{n,m}^{1D} = A_{1D} m \omega_n^{1D}, \quad A_{1D} \approx 2.7, \quad (\text{E.13})$$

with the modified m -power exponent reflecting the line-target geometry of the basal loop relative to a 1-D glide channel along $\langle 11\bar{2}0 \rangle$. The stronger sensitivity to sink geometry in the 1-D regime is the physical origin of the enhanced production-bias contribution to anisotropic dimensional change in Zr [18, 22].

Interpolation formula. For intermediate L , the Trinkaus–Singh–Golubov interpolation gives:

$$\mathcal{K}_{n,m}^{\text{eff}} = \frac{A_{\text{loop}}^{(c)} m^{1/2} \omega_n^{1D}}{1 + B_{\text{rot}} \hat{L}^2 m^{-1/2}}, \quad B_{\text{rot}} \approx 2.6. \quad (\text{E.14})$$

In the limits: $\hat{L} \rightarrow 0$ recovers the 3-D rate constant $A_{\text{loop}}^{(c)} m^{1/2} \omega_n^{1D}$; $\hat{L} \rightarrow \infty$ gives a rate proportional to $m \hat{L}^{-2}$, scaling as r^2 as expected for 1-D kinetics.

The reverse of these capture reactions is thermal emission, whose rates we specify next.

Thermal emission of species α from a cluster of size n is related to the capture rate by detailed balance. The daughter cluster has size $(n - 1)$, so the capture rate that enters the detailed-balance relation uses $(n - 1)$.

For a prismatic $\langle a \rangle$ -loop emitting an SIA:

$$\varepsilon_i^{(a)}(n) = A_{\text{loop}} (n - 1)^{1/2} Z_i^{(a)} \omega_i \exp\left(-\frac{E_b^{i,(a)}(n)}{k_B T}\right) \quad [\text{s}^{-1}], \quad (\text{E.15})$$

where $E_b^{i,(a)}(n)$ is the binding energy of the last SIA to the prismatic loop (from Section Appendix D).

For a basal $\langle c \rangle$ -component loop emitting a vacancy:

$$\varepsilon_v^{(c)}(m) = A_{\text{loop}}^{(c)} (m - 1)^{1/2} Z_v^{(c)} \omega_v \exp\left(-\frac{E_b^{v,(c)}(m)}{k_B T}\right). \quad (\text{E.16})$$

Vacancy emission from the basal vacancy-loop family corresponds to un-faulting events at the loop perimeter and is the dominant inverse process to vacancy absorption. The emission enters the master equation as a first-order process:

$$\left. \frac{dc_n}{dt} \right|_{\text{emission}} = +\varepsilon(n + 1) c_{n+1} - \varepsilon(n) c_n. \quad (\text{E.17})$$

Cross-channel emission—SIA emission from a vacancy $\langle c \rangle$ -loop, or vacancy emission from a prismatic $\langle a \rangle$ -loop—is also possible but is energetically strongly suppressed because the emitted point defect must climb the binding energy of the opposite loop family. In practice these channels are negligible at reactor temperatures and are omitted from the master equations.

For efficiency, the rate constants derived above are precomputed and stored, as summarized below.

All reaction-rate geometry is captured by four numerical constants:

$$\begin{aligned}
A_{\text{sph}} &= (48\pi^2)^{1/3} && \approx 7.818 \quad (\text{spherical clusters, Section Appendix E}), \\
A_{\text{loop}} &= 8\sqrt{\pi/\sqrt{3}} && \approx 10.78 \quad (\langle a \rangle \text{ prismatic loops}), \\
A_{\text{loop}}^{(c)} &= A_{\text{loop}}\sqrt{b_c/b_a} && \approx 14.9 \quad (\langle c \rangle \text{ basal loops}), \\
A_{1D} &\approx 2.7 && (\text{pure 1D sinks}), \\
B_{\text{rot}} &\approx 2.6 && (1D/3D \text{ interpolation}).
\end{aligned} \tag{E.18}$$

Every rate constant in the cluster-dynamics system is built from these constants, the cluster size raised to a geometry-dependent power, and the appropriate reaction frequency ω_α . Table E.12 collects the complete set.

Finally, we tabulate the reaction-rate parameters adopted for Zircaloy-4.

Table E.13 collects the rate-constant prefactors, sink parameters, and state-space controls used in the reaction-rate expressions. Bias factors are split between the two loop families, reflecting the qualitatively different elastic response of prismatic interstitial loops and basal vacancy loops to point-defect strain fields.

Implementation notes. Every rate constant is built from one of the geometric constants (A_{sph} , A_{loop} , $A_{\text{loop}}^{(c)}$, A_{1D} , B_{rot}), the cluster size raised to a geometry-dependent power, and the appropriate effective reaction frequency $\omega_\alpha^{\text{eff}}$ (Eq. 4). The binding energies $E_b^{i,(a)}(n)$ and $E_b^{v,(c)}(m)$ for $\langle a \rangle$ - and $\langle c \rangle$ -loops come from Section Appendix D. The two loop families are treated symmetrically in the master equations (Section ??), with their distinct bias factors driving the partition of point-defect flux that ultimately generates anisotropic dimensional change.

Table E.12: Summary of rate-constant expressions for the Zr cluster dynamics formulation. All rate constants are in s^{-1} and all concentrations in at/at. The effective reaction frequencies $\omega_\alpha^{\text{eff}}$ incorporate solute-trapping corrections (Table C.10) and should replace ω_α in all rate-equation implementations.

Process	Rate constant	Rate-eq
V–SIA recombination	$\mathcal{K}_{iv} = 8\pi\xi_{\text{rec}}(\omega_i^{\text{eff}} + \omega_v^{\text{eff}})$	$-\mathcal{K}_{iv} c_1^{(a)}$
i capture by $\langle a \rangle$ -loop n	$\mathcal{K}_{i,n}^{(a)} = A_{\text{loop}} n^{1/2} Z_i^{(a)} \omega_i^{\text{eff}}$	$-\mathcal{K}_{i,n}^{(a)} c_1^{(a)}$
v capture by $\langle a \rangle$ -loop n	$\mathcal{K}_{v,n}^{(a)} = A_{\text{loop}} n^{1/2} Z_v^{(a)} \omega_v^{\text{eff}}$	$-\mathcal{K}_{v,n}^{(a)} c_1^V$
v capture by $\langle c \rangle$ -loop m	$\mathcal{K}_{v,m}^{(c)} = A_{\text{loop}}^{(c)} m^{1/2} Z_v^{(c)} \omega_v^{\text{eff}}$	$-\mathcal{K}_{v,m}^{(c)} c_1^V$
i capture by $\langle c \rangle$ -loop m	$\mathcal{K}_{i,m}^{(c)} = A_{\text{loop}}^{(c)} m^{1/2} Z_i^{(c)} \omega_i^{\text{eff}}$	$-\mathcal{K}_{i,m}^{(c)} c_1^{(a)}$
Dislocation sink	$\mathcal{D}_\alpha^d = Z_\alpha^d \rho_d a^2 \omega_\alpha^{\text{eff}}$	$-\mathcal{D}_\alpha^d c_\alpha$
Grain-boundary sink	$\mathcal{D}_\alpha^{gb} = (4\pi^2/d_g^2) a^2 \omega_\alpha^{\text{eff}}$	$-\mathcal{D}_\alpha^{gb} c_\alpha$
Precipitate sink	$\mathcal{D}_\alpha^p = (8\pi r_p/a) c_p^N \omega_\alpha^{\text{eff}}$	$-\mathcal{D}_\alpha^p c_\alpha$
SIA cluster (mixed 1D/3D, $\langle c \rangle$ -loop sink)	$\mathcal{K}_{n,m}^{\text{eff}} = \frac{A_{\text{loop}}^{(c)} m^{1/2} \omega_n^{1D}}{1 + B_{\text{rot}} \hat{L}^2 m^{-1/2}}$	$-\mathcal{K}_{n,m}^{\text{eff}} c_n^{(c)}$
SIA cluster (pure 1D)	$\mathcal{K}_{n,m}^{1D} = A_{1D} m \omega_n^{1D}$	$-\mathcal{K}_{n,m}^{1D} c_n^{(c)}$
SIA emission from $\langle a \rangle$ -loop n	$\varepsilon_i^{(a)}(n) = A_{\text{loop}} (n-1)^{1/2} Z_i^{(a)} \omega_i^{\text{eff}} e^{-E_b^{i,(a)}/k_B T}$	first-order
V emission from $\langle c \rangle$ -loop m	$\varepsilon_v^{(c)}(m) = A_{\text{loop}}^{(c)} (m-1)^{1/2} Z_v^{(c)} \omega_v^{\text{eff}} e^{-E_b^{v,(c)}/k_B T}$	first-order

Table E.13: Reaction-rate input parameters for Zircaloy-4 cluster dynamics. Bias factors are illustrative midrange values; the literature spread is substantial and the specific choice should be adjusted to match the targeted experimental observable (e.g., neutron-diffraction texture evolution, in-reactor creep rate).

Parameter	Symbol	Value	Units
<i>Geometric prefactors (precomputed constants):</i>			
Spherical cluster	$A_{\text{sph}} = (48\pi^2)^{1/3}$	7.818	—
Prismatic $\langle a \rangle$ -loop	A_{loop}	10.78	—
Basal $\langle c \rangle$ -loop	$A_{\text{loop}}^{(c)}$	14.9	—
Pure 1D sink	A_{1D}	2.7	—
1D/3D interpolation	B_{rot}	2.6	—
V–SIA recombination	$8\pi\xi_{\text{rec}}$	12.4	—
<i>Dislocation sink:</i>			
Network dislocation density	ρ_d	$10^{13}\text{--}10^{14}$	m^{-2}
Interstitial bias factor	Z_i^d	1.05–1.15	—
Vacancy bias factor	Z_v^d	1.00	—
<i>$\langle a \rangle$-loop bias factors:</i>			
SIA capture	$Z_i^{(a)}$	1.10–1.20	—
Vacancy capture	$Z_v^{(a)}$	1.00	—
<i>$\langle c \rangle$-loop bias factors:</i>			
Vacancy capture	$Z_v^{(c)}$	1.10–1.25	—
SIA capture	$Z_i^{(c)}$	0.95–1.05	—
<i>Grain-boundary sink:</i>			
Grain diameter	d_g	5–30	μm
Bias factors	$Z_i^{gb} = Z_v^{gb}$	1.00	—
<i>Second-phase precipitate sink ($\text{Zr}(\text{Fe}, \text{Cr})_2$, $\beta\text{-Nb}$):</i>			
Mean precipitate radius	r_p	20–80	nm
Precipitate number density	N_p	$10^{20}\text{--}10^{21}$	m^{-3}
Bias factors	$Z_i^p = Z_v^p$	1.00	—
<i>Mobility cutoffs:</i>			
Max mobile SIA cluster	i_{mobile}	30	SIAs
Max mobile vacancy cluster	v_{mobile}	4	vacancies
<i>Size grid (user-defined):</i>			
	57		
Max $\langle a \rangle$ -loop tracked	N	user input	SIAs
Max $\langle c \rangle$ -loop tracked	M	user input	vacancies
Grouping transition size	n_{group}	$\sim 50\text{--}200$	—

Appendix F. Numerical Methods

The state vector is

$$\mathbf{y} = \left[\underbrace{c_1^{(a)}, \dots, c_N^{(a)}}_N, \underbrace{c_1^{(c)}, \dots, c_M^{(c)}}_M \right]^T \in \mathbb{R}^{N_{\text{eq}}}, \quad (\text{F.1})$$

where $c_n^{(a)}$ is the concentration of $\langle a \rangle$ -loops (prismatic, interstitial nature) of size n and $c_m^{(c)}$ is the concentration of $\langle c \rangle$ -component loops (basal, vacancy nature) of size m . The mono-defect entries $c_1^{(a)}$ and $c_1^{(c)}$ are identified with the free SIA and free vacancy concentrations, respectively.

The master equations (ME_{SIA} , $\text{ME}_{a\text{-loop}}$, $\text{ME}_{c\text{-loop}}$) from the RadCluster formulation [47] couple the state variables through the following reactions:

1. **SIA–SIA coalescence:** $I_{n'} + I_{n-n'} \rightarrow I_n$ for mobile $n' \leq i_m$.
2. **V–V coalescence:** $V_{m'} + V_{m-m'} \rightarrow V_m$ for mobile $m' \leq v_m$.
3. **V–I annihilation:** $V_{m'} + I_n \rightarrow I_{n-m'}$ for mobile $m' \leq v_m$.
4. **SIA cluster– $\langle c \rangle$ -loop absorption:** $I_n + V_m^{(c)} \rightarrow V_{m-n}^{(c)}$ for mobile $n \leq i_m$.
5. **Thermal emission:** $I_n \rightarrow I_{n-1} + I_1$ from $\langle a \rangle$ -loops and $V_m^{(c)} \rightarrow V_{m-1}^{(c)} + V_1$ from $\langle c \rangle$ -loops (nearest-neighbor coupling).
6. **Fixed-sink loss:** $\mathcal{D}_\alpha(n) c_n$ (diagonal).

The Jacobian $J = \partial \mathbf{f} / \partial \mathbf{y}$ inherits its sparsity pattern from these couplings. We analyze two limiting cases.

We first specialize the Jacobian assembly to the case in which only monomers are mobile.

When only monomers are mobile, coalescence sums collapse to their $n' = 1$ or $m' = 1$ terms, and all cluster–cluster rate constants reduce to the monomer absorption forms $\mathcal{K}^{ii} \rightarrow \mathcal{K}_{i,n}^{\text{loop}}$, $\mathcal{K}^{vv} \rightarrow \mathcal{K}_{\alpha,m}^{\text{cav}}$, $\mathcal{K}^{vi} \rightarrow \mathcal{K}_{iv}$.

Coupling pattern for SIA clusters. For a sessile SIA cluster $c_n^{(i)}$ with $n \geq 2$:

$$\frac{dc_n^{(i)}}{dt} = \underbrace{\mathcal{K}_{i,n-1}^{\text{loop}} c_1^{(i)} c_{n-1}^{(i)}}_{\text{gain: monomer hits } n-1} - \underbrace{\mathcal{K}_{i,n}^{\text{loop}} c_1^{(i)} c_n^{(i)}}_{\text{loss: monomer hits } n} + \underbrace{\varepsilon_i(n+1) c_{n+1}^{(i)}}_{\text{emission gain}} - \underbrace{\varepsilon_i(n) c_n^{(i)}}_{\text{emission loss}} - \underbrace{\mathcal{K}_{iv} c_1^{(v)} c_n^{(i)}}_{\text{V-I annihilation}} - \underbrace{\mathcal{D}_i c_n^{(i)}}_{\text{sinks}}. \quad (\text{F.2})$$

The Jacobian row for equation n depends on exactly four SIA variables:

$$\frac{\partial f_n}{\partial c_{n-1}^{(i)}}, \quad \frac{\partial f_n}{\partial c_n^{(i)}}, \quad \frac{\partial f_n}{\partial c_{n+1}^{(i)}}, \quad \frac{\partial f_n}{\partial c_1^{(i)}}, \quad (\text{F.3})$$

plus one vacancy variable $c_1^{(v)}$. The first three give a *tridiagonal* pattern; the fourth adds a *dense column* at index 1.

The SIA monomer equation ($n = 1$) depends on *all* $c_n^{(i)}$ (coalescence loss $c_1^{(i)} \sum_n c_n^{(i)}$) and all $c_m^{(v)}$ (cavity absorption), giving a *dense row*.

Block Jacobian structure. Partitioning J into SIA and VAC blocks:

$$J = \begin{pmatrix} J_{II} & J_{IV} \\ J_{VI} & J_{VV} \end{pmatrix}, \quad (\text{F.4})$$

each diagonal block is an **arrowhead** (bordered tridiagonal) matrix:

$$J_{II} = \underbrace{T_I}_{\text{tridiagonal}} + \underbrace{\mathbf{u}_1 \mathbf{e}_1^T + \mathbf{e}_1 \mathbf{v}_1^T}_{\text{rank-2 border}}, \quad (\text{F.5})$$

where $\mathbf{u}_1$ is the dense column from $c_1^{(i)}$ appearing in every equation, and $\mathbf{v}_1$ is the dense row from the monomer equation depending on all clusters.

The off-diagonal blocks J_{IV} and J_{VI} are **rank-1**: they couple through the single mobile monomer of the other species.

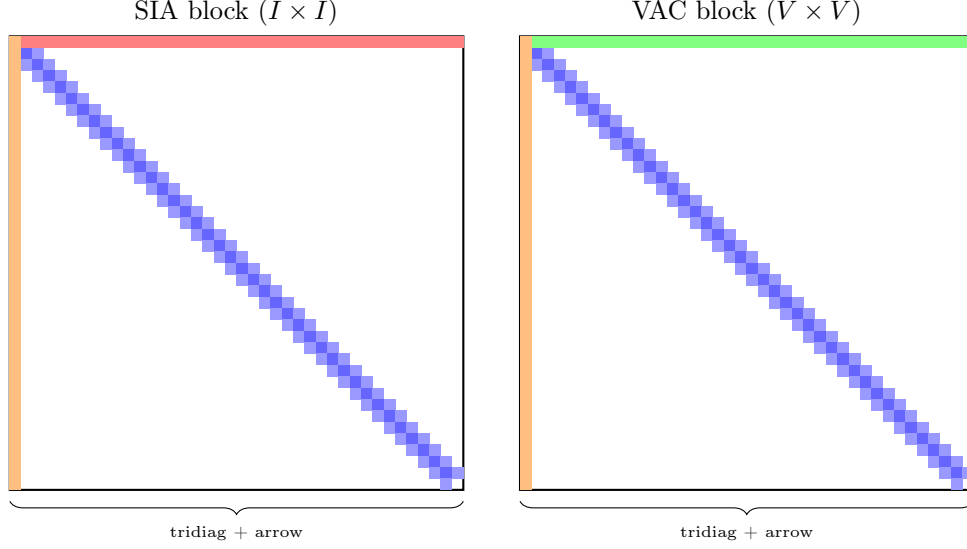


Figure F.7: Jacobian sparsity for monomer-only reactions ($i_m = v_m = 1$). Each diagonal block is an arrowhead matrix (tridiagonal + one dense row and column). Off-diagonal blocks are rank-1. Total non-zeros: $O(N_{\text{eq}})$.

The arrowhead system can be solved in $O(N_{\text{eq}})$ operations using the Sherman–Morrison formula (rank-2 update of a tridiagonal solve) [48].

The more general situation, in which several mobile species coalesce, is treated next.

When multiple mobile species participate in coalescence (e.g., $i_m = 10$, $v_m = 5$), each SIA equation for cluster n acquires the following additional dependencies:

- **Coalescence gain:** $\sum_{k=1}^{\min(n-1, i_m)} \mathcal{K}_{k, n-k}^{ii} c_k^{(i)} c_{n-k}^{(i)} \Rightarrow$ depends on $c_{n-k}^{(i)}$ for $k = 1, \dots, i_m$ (band of width i_m below the diagonal).
- **Coalescence loss:** $c_n^{(i)} \sum_{k=1}^{i_m} \mathcal{K}_{n,k}^{ii} c_k^{(i)} \Rightarrow$ depends on $c_k^{(i)}$ for $k = 1, \dots, i_m$ (i_m dense columns).
- **V–I annihilation gain:** $\sum_{k=1}^{v_m} \mathcal{K}_{k, n+k}^{vi} c_k^{(v)} c_{n+k}^{(i)} \Rightarrow$ depends on $c_{n+k}^{(i)}$ for $k = 1, \dots, v_m$ (band of width v_m above the diagonal).
- **V–I annihilation loss:** $\sum_{k=1}^{v_m} \mathcal{K}_{k, n}^{vi} c_k^{(v)} c_n^{(i)} \Rightarrow$ depends on $c_k^{(v)}$ for $k = 1, \dots, v_m$ (v_m dense columns in the off-diagonal block).

Bordered-banded decomposition. The full Jacobian admits the decomposition:

$$\boxed{J = T + U W V^T} \quad (\text{F.6})$$

where:

- $T \in \mathbb{R}^{N_{\text{eq}} \times N_{\text{eq}}}$ is **banded** with half-bandwidth

$$b = \max(2i_{\text{m}}, 2v_{\text{m}}) + 1, \quad (\text{F.7})$$

capturing emission (nearest-neighbor), local coalescence gain/loss (within the band), and diagonal terms (sinks, self-reaction).

- $U \in \mathbb{R}^{N_{\text{eq}} \times \text{rank}}$ collects the rank dense columns, where

$$\text{rank} = i_{\text{m}} + v_{\text{m}}. \quad (\text{F.8})$$

These arise because each mobile species $c_k^{(i)}$ ($k = 1, \dots, i_{\text{m}}$) and $c_k^{(v)}$ ($k = 1, \dots, v_{\text{m}}$) appears in *every* equation of the opposing (and same-species) block through coalescence loss and annihilation loss.

- $W \in \mathbb{R}^{\text{rank} \times \text{rank}}$ is a small correction matrix, and $V \in \mathbb{R}^{N_{\text{eq}} \times \text{rank}}$ encodes the corresponding row structure.

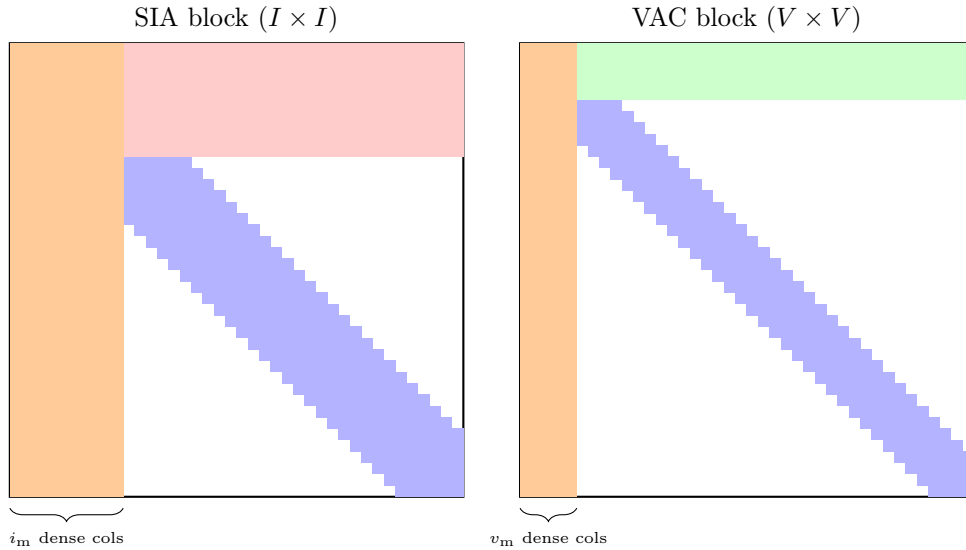


Figure F.8: Jacobian sparsity for coalescence ($i_{\text{m}} = 10$, $v_{\text{m}} = 5$). Each diagonal block is banded (blue) with i_{m} or v_{m} dense columns (red/green) from mobile species. Off-diagonal blocks (orange) have the same rank- i_{m} or rank- v_{m} structure.

Consistency with the monomer case. Setting $i_m = v_m = 1$ in (F.6) recovers the arrowhead structure of Section Appendix F: the band reduces to a tridiagonal ($b = 3$) and the border is rank-2 (rank = 2). The Woodbury preconditioner described in Section Appendix F thus handles both cases with a *single code path*—no special-casing is needed.

The sparsity structure that results motivates a closer look at how the linear systems are solved, beginning with the shortcomings of a generic Krylov approach.

CVODE’s BDF method requires solving the Newton correction system

$$(I - \gamma J) \Delta \mathbf{y} = -\mathbf{g} \quad (\text{F.9})$$

at each time step, where $\gamma = h\beta$ involves the step size h and the BDF coefficient β . The matrix $M = I - \gamma J$ inherits the bordered-banded structure of J .

Dense direct solver. The default dense LU factorization costs $O(N_{\text{eq}}^3)$ per Newton step. For $N_{\text{eq}} = I + V + 2 \approx 20,000$, this is $\sim 8 \times 10^{12}$ floating-point operations per step—entirely impractical for production runs.

Unpreconditioned GMRES. GMRES [49] is an iterative Krylov method that computes $\Delta \mathbf{y} \approx \arg \min_{\mathbf{x} \in \mathcal{K}_k} \|M\mathbf{x} - \mathbf{b}\|_2$ over the Krylov subspace $\mathcal{K}_k(M, \mathbf{b})$. Each iteration costs $O(N_{\text{eq}})$ (one matrix–vector product plus Gram–Schmidt), but the number of iterations depends on the *condition number* $\kappa(M)$.

For the bordered-banded Jacobian, the eigenvalue spectrum is problematic:

1. The dense columns from mobile species couple all N_{eq} equations, creating $O(\text{rank})$ eigenvalues that are well separated from the banded spectrum of T .
2. The stiffness ratio between fast monomer reactions and slow large-cluster evolution spans 10–15 orders of magnitude, producing $\kappa(M) \sim 10^{10}\text{--}10^{15}$.
3. GMRES convergence stalls when the eigenvalue distribution has widely separated outliers [50], which is exactly the pattern created by the low-rank border.

In practice, preconditioned GMRES requires $O(N_{\text{eq}})$ iterations (i.e., the full Krylov subspace), negating its advantage over a direct solver.

Diagonal (Jacobi) preconditioner. The current RadCluster implementation offers a Jacobi (diagonal) preconditioner:

$$P_{\text{Jacobi}} = \text{diag}(M) = \text{diag}(1 - \gamma J_{kk}). \quad (\text{F.10})$$

This captures only the self-reaction rates (sinks, self-coalescence) and completely misses:

- The **banded coupling** from emission and local coalescence ($\sim 2b$ off-diagonal entries per row).
- The **dense-column coupling** from mobile species appearing in every equation (rank dense columns).
- The **cross-species coupling** between SIA and VAC blocks.

For the coalescence case, the Jacobi preconditioner reduces GMRES iteration counts only modestly (from $O(N_{\text{eq}})$ to perhaps $O(N_{\text{eq}}/2)$), which is still far too many for efficiency.

These limitations are overcome by exploiting the bordered-banded structure directly through a Woodbury-based preconditioner.

The Sherman-Morrison-Woodbury identity. For any invertible T and conformable U, W, V , the Sherman-Morrison-Woodbury (SMW) identity [51, 52] states that

$$\boxed{(T + UWV^T)^{-1} = T^{-1} - T^{-1}U(W^{-1} + V^T T^{-1}U)^{-1}V^T T^{-1}} \quad (\text{F.11})$$

Applied to $M = I - \gamma J = (I - \gamma T) - \gamma U W V^T$, we define:

$$\hat{T} = I - \gamma T \quad (\text{banded, bandwidth } b), \quad (\text{F.12})$$

$$\hat{U} = \gamma U, \quad \hat{V} = V, \quad (\text{F.13})$$

so that $M = \hat{T} - \hat{U} W \hat{V}^T$ and

$$M^{-1} = \hat{T}^{-1} + \hat{T}^{-1} \hat{U} (W^{-1} - \hat{V}^T \hat{T}^{-1} \hat{U})^{-1} \hat{V}^T \hat{T}^{-1}. \quad (\text{F.14})$$

Construction of T, U, V, W . The decomposition $J = T + U W V^T$ is constructed as follows:

1. **Band extraction:** T is the restriction of J to entries within bandwidth b of the diagonal. This captures emission, local coalescence, sinks, and all nearest-neighbor coupling. Since J is not available analytically, we approximate T by *banded finite differences*: perturbing groups of state variables that are more than b apart (Curtis–Powell–Reid coloring [53]), requiring only $2b+1$ RHS evaluations instead of N_{eq} .
2. **Dense columns:** U collects the $\text{rank} = i_{\text{m}} + v_{\text{m}}$ dense columns of J corresponding to mobile species indices. Column j of U is approximated by a single finite-difference perturbation of y_j , minus the entries already captured in T (to avoid double-counting).
3. **Row structure:** For the preconditioner we use the simplification $W = I_{\text{rank}}$ and let V^T encode the dense rows. In practice, the dense rows correspond to the same mobile species indices as the dense columns, giving $V = U$ to leading order (the Jacobian sub-blocks involving mobile species are approximately symmetric in their sparsity pattern, though not in their values).
4. **Simplification:** When the dense column/row indices coincide (as they do here), the border can be represented as:

$$J \approx T + \sum_{j=1}^{\text{rank}} (\mathbf{J}_{:,j} - \mathbf{T}_{:,j}) \mathbf{e}_j^T, \quad (\text{F.15})$$

where $\mathbf{J}_{:,j}$ is the j -th column of J (obtained by a single finite-difference probe) and $\mathbf{T}_{:,j}$ is its banded restriction. This is a rank-rank correction with $U = [\mathbf{J}_{:,1} - \mathbf{T}_{:,1}, \dots, \mathbf{J}_{:,\text{rank}} - \mathbf{T}_{:,\text{rank}}]$, $W = I$, $V = [\mathbf{e}_1, \dots, \mathbf{e}_{\text{rank}}]$.

Algorithm.

Algorithm 1 Woodbury preconditioner setup

Require: State $\mathbf{y}$, RHS function $\mathbf{f}$, BDF parameter γ

```
1: Evaluate base RHS:  $\mathbf{f}_0 \leftarrow \mathbf{f}(\mathbf{y})$ 
2: // Band extraction via CPR coloring
3: for color  $c = 1, \dots, 2b + 1$  do
4:   Perturb:  $\tilde{\mathbf{y}} \leftarrow \mathbf{y}$ ;  $\tilde{y}_j \leftarrow y_j + \delta_j$  for all  $j$  with  $j \bmod (2b + 1) = c$ 
5:    $\tilde{\mathbf{f}} \leftarrow \mathbf{f}(\tilde{\mathbf{y}})$ 
6:   for each perturbed index  $j$  do
7:      $T_{k,j} \leftarrow (\tilde{f}_k - f_{0,k})/\delta_j$  for  $|k - j| \leq b$ 
8:   end for
9: end for
10: // Dense column probes for mobile species
11: for  $j \in \{\text{mobile SIA indices}\} \cup \{\text{mobile VAC indices}\}$  do
12:    $\tilde{\mathbf{y}} \leftarrow \mathbf{y}$ ;  $\tilde{y}_j \leftarrow y_j + \delta_j$ 
13:    $\mathbf{u}_j \leftarrow (\mathbf{f}(\tilde{\mathbf{y}}) - \mathbf{f}_0)/\delta_j - T_{:,j}$  ▷ Subtract banded part
14: end for
15: // Form preconditioner matrices
16:  $\hat{T} \leftarrow I - \gamma T$ 
17: Factor  $\hat{T}$  via banded LU (LAPACK dgbtrf)
18: Solve  $\hat{T}^{-1}\hat{U}$  via rank banded back-solves
19: Form  $S \leftarrow I_{\text{rank}} + \gamma V^T \hat{T}^{-1}\hat{U}$  ▷ rank  $\times$  rank Schur complement
20: Factor  $S$  via dense LU (LAPACK dgetrf)
```

Algorithm 2 Woodbury preconditioner solve: $P^{-1}\mathbf{r} = \mathbf{z}$

Require: Residual $\mathbf{r}$, factored $\hat{T}$, factored S , pre-computed $\hat{T}^{-1}\hat{U}$

```
1:  $\mathbf{w}_1 \leftarrow \hat{T}^{-1}\mathbf{r}$  ▷ Banded back-solve,  $O(N_{\text{eq}} b)$ 
2:  $\mathbf{w}_2 \leftarrow V^T \mathbf{w}_1$  ▷ Extract rank components,  $O(\text{rank})$ 
3:  $\mathbf{w}_3 \leftarrow S^{-1} \mathbf{w}_2$  ▷ Dense solve,  $O(\text{rank}^2)$ 
4:  $\mathbf{z} \leftarrow \mathbf{w}_1 - (\hat{T}^{-1}\hat{U}) \mathbf{w}_3$  ▷  $O(N_{\text{eq}} \text{rank})$ 
```

Computational cost.

The Woodbury preconditioner achieves a **speedup of** $\sim 10^5$ over the dense solver and requires only $2b + 1 + \text{rank} = 58$ RHS evaluations for setup (vs. $N_{\text{eq}} = 20,000$ for a full finite-difference Jacobian).

Monomer-case reduction. For $i_m = v_m = 1$: $b = 3$ (tridiagonal), $\text{rank} = 2$.

Table F.14: Operation counts per Newton step. $N_{\text{eq}} \approx 20,000$, $b = 21$, $r = 15$ for $i_{\text{m}} = 10$ and $v_{\text{m}} = 5$.

Method	Setup	Solve	RHS evals	GMRES iters
Dense direct	$O(N_{\text{eq}}^3)$ $\sim 8 \times 10^{12}$	$O(N_{\text{eq}}^2)$ $\sim 4 \times 10^8$	N_{eq} 20,000	— —
Jacobi + GMRES	$O(N_{\text{eq}})$ $\sim 2 \times 10^4$	$O(N_{\text{eq}}k)$ —	1 —	$k \gg 1$ $\sim 1000+$
Woodbury + GMRES	$O(N_{\text{eq}}b^2)$ $\sim 9 \times 10^6$	$O(N_{\text{eq}}b)$ $\sim 4 \times 10^5$	$2b + 1 + r$ 58	1–3 —

- Setup: $O(N_{\text{eq}} \cdot 9 + 8) = O(N_{\text{eq}})$ — effectively a tridiagonal LU plus a 2×2 Schur complement.
- Solve: $O(N_{\text{eq}} \cdot 3 + 2) = O(N_{\text{eq}})$ — one tridiagonal back-solve plus a 2×2 dense solve.
- RHS evaluations for probing: $2 \cdot 3 + 1 + 2 = 9$.

This is the *optimal* solver for arrowhead matrices [48], confirming that the Woodbury preconditioner does not degrade performance in the monomer limit.

One remaining numerical detail concerns the choice of perturbation used to form the finite-difference Jacobian.

The perturbation for column j of the Jacobian is chosen as

$$\delta_j = \sqrt{u_{\text{round}}} \cdot \max(|y_j|, y_{\text{typ}}) \cdot \text{sign}(y_j), \quad (\text{F.16})$$

where $u_{\text{round}} \approx 10^{-16}$ is machine epsilon and y_{typ} is a typical concentration scale (set to the absolute tolerance). This is the same formula used internally by CVODE for its difference-quotient Jacobian [54].

In summary, the numerical strategy combines the following ingredients.

1. The RadCluster Jacobian has a **bordered-banded** structure: banded (bandwidth $b \sim 2 \max(i_{\text{m}}, v_{\text{m}})$) plus a low-rank dense border (rank $\text{rank} = i_{\text{m}} + v_{\text{m}}$) from mobile species coupling.
2. The **monomer case** ($i_{\text{m}} = v_{\text{m}} = 1$) is a special case: the Jacobian reduces to an **arrowhead** matrix (tridiagonal + rank-2 border).

3. **Dense direct** solvers cost $O(N_{\text{eq}}^3)$ and are impractical for $N_{\text{eq}} > 1000$.
4. **Unpreconditioned GMRES** fails because the bordered-banded eigenvalue spectrum has widely separated outliers from the dense border.
5. The **Jacobi preconditioner** captures only diagonal entries, missing the dominant banded and dense-column coupling.
6. The **Woodbury preconditioner** exploits the bordered-banded decomposition via the SMW identity:
 - Setup: banded LU ($O(N_{\text{eq}} b^2)$) + rank $\times$ rank Schur complement ($O(\text{rank}^3)$).
 - Solve: banded back-solve ($O(N_{\text{eq}} b)$) + Woodbury correction ($O(N_{\text{eq}} \text{rank})$).
 - GMRES converges in 1–3 iterations.
 - Speedup $\sim 10^5$ over dense direct.
 - Subsumes the arrowhead case with no code changes.

Appendix G. Irradiated Zirconium Microstructure

Zirconium alloys are the primary structural materials for nuclear fuel cladding and pressure tubes in both light water reactors (LWR) and pressurised water reactors (PWR), selected for their combination of low neutron absorption cross-section, adequate corrosion resistance, and reasonable mechanical strength [1]. The most widely deployed alloys are Zircaloy-2 (Zr–1.5Sn–Fe–Cr–Ni, used in BWR cladding and channel boxes), Zircaloy-4 (Zr–1.5Sn–0.2Fe–0.1Cr, used in PWR cladding), Zr–1Nb (alloy E110, used in VVER reactors), Zr–2.5Nb (used as CANDU pressure tube material), and Low-Sn ZIRLO (Zr–0.9Sn–1.0Nb–0.1Fe, advanced PWR cladding).

Under neutron irradiation, the hexagonal close-packed (hcp) crystal structure of α -Zr leads to a microstructural evolution that is qualitatively different from that of cubic (bcc or fcc) metals:

1. Two distinct families of dislocation loops are created on different habit planes: *$\langle a \rangle$ -type loops* with Burgers vector $\mathbf{b} = \frac{1}{3}\langle 11\bar{2}0 \rangle$ lying on prismatic $\{10\bar{1}0\}$ planes, and *$\langle c \rangle$ -component loops* with Burgers vector $\mathbf{b} = \frac{1}{6}\langle 20\bar{2}3 \rangle$ lying on basal $\{0001\}$ planes.

2. The anisotropy of point-defect migration in the hcp lattice (interstitials migrate preferentially in the basal plane) drives *irradiation-induced growth*—stress-free dimensional change in the absence of a net atomic volume change.
3. Second-phase precipitates (intermetallic particles of $\text{Zr}(\text{Fe},\text{Cr})_2$, $\text{Zr}_2(\text{Fe},\text{Ni})$, and $\beta\text{-Nb}$) undergo progressive amorphization and dissolution, redistributing alloying elements into the matrix.
4. Void/cavity formation is limited to a narrow temperature window (625–775 K) because $\langle c \rangle$ loops compete with voids for vacancies.

This report compiles and analyzes all available quantitative experimental data for defect size, number density, and size distributions in irradiated Zr alloys, following the same structure used for ferritic/martensitic steel databases developed previously [55, 56]. Section Appendix G documents the data sources. Section Appendix G covers $\langle a \rangle$ -type loop density and mean size. Section Appendix G covers $\langle c \rangle$ -component loops. Section Appendix G presents size-distribution data. Section Appendix G discusses void/cavity formation. Section Appendix G covers second-phase particle evolution. Section Appendix G derives empirical scaling laws. Section Appendix G discusses implications for cluster-dynamics modeling. All data are archived in the companion workbook `Irradiation_Microstructure_Database_Zr_Zircaloy.xlsx`.

The picture that emerges rests on a body of experimental measurements, whose principal sources we now describe.

The size distributions of irradiation-induced defects are among the most fundamental quantitative outputs of post-irradiation TEM characterization because they encode the competing kinetics of nucleation, growth, and coarsening that govern the macroscopic consequences—irradiation hardening, embrittlement, and dimensional instability (irradiation growth). Reliable size-distribution data are indispensable for the validation of cluster-dynamics rate-theory models and for the calibration of the dispersed-barrier hardening (DBH) model [55, 56].

Table G.15 summarizes the primary experimental studies from which data were extracted for this database.

We begin with the prismatic $\langle a \rangle$ -type loops, the dominant feature of the irradiated microstructure, considering their density and mean size.

General Morphology and Character.

$\langle a \rangle$ -type loops in $\alpha\text{-Zr}$ form with Burgers vector $\mathbf{b} = \frac{1}{3}\langle 11\bar{2}0 \rangle$ and lie

Table G.15: Primary sources of microstructure data compiled in this study. Abbreviations: N = neutron, P = proton, SI = self-ion, DI = dual-ion, WBDF = weak-beam dark-field TEM, BF = bright-field, STEM = scanning TEM, SXRD = synchrotron X-ray diffraction.

Material	Irrad.	Facility	Dose (dpa)	T (°C)	Method	Reference
Pure Zr / Zircaloy-2	N fission	Round robin	0.5–1dpa	280–400	TEM	(author?)
Zircaloy-2 & -4	N fission	BWR/ATR	~10dpa	~280	TEM	(author?)
Zircaloy-2	P (2 MeV)	Mandeville	2.3–7dpa	350	STEM	(author?)
					+ SXRD	[59]
Zircaloy-2 / ZIRLO	P (2 MeV)	Mandeville	2.3–10dpa	280–450	STEM	(author?)
					+ SXRD	[60]
Zircaloy-4	P (2 MeV) + Zr-ion	CEA Saclay	2–11.5dpa	300–350	TEM	(author?)
Zr-1%Nb (E110)	N (VVER)	VVER re-actor	~5dpa	~330	TEM	(author?)
					STEM	[61]
Pure α -Zr	SI (Zr ⁺ , in-situ)	In-situ TEM	1–5dpa	300	In-situ TEM	(author?)
Zircaloy-2 (BWR HB)	N fission	BWR high burnup	10dpa	260–280	SXRD	(author?)
					+ TEM	[63]
Zircaloy-4	P (2 MeV) / Heavy-ion	CEA Michigan	2–11.5dpa	280–360	TEM	(author?)
					STEM	[64]
Zr alloys (review)	N (BWR/PWR/CANDU)	Various	1–15dpa	260–360	Review	(author?)
						[1]

on prismatic $\{10\bar{1}0\}$ habit planes. Both interstitial (I) and vacancy (V) $\langle a \rangle$ loops coexist during neutron and ion irradiation [57]. Below $\sim 300^\circ\text{C}$, interstitial loops are dominant; above $\sim 360^\circ\text{C}$, vacancy loops become prevalent. A distinctive feature of the Zr microstructure is the tendency of $\langle a \rangle$ loops to arrange in *rows* or *layers* parallel to the trace of the $\{0001\}$ basal plane (the “corduroy” structure), driven by anisotropic interstitial diffusion preferentially in the basal plane [4].

Temperature Dependence at Fixed Dose.

Figure G.9 illustrates the strong temperature dependence of $\langle a \rangle$ -loop density and mean diameter.

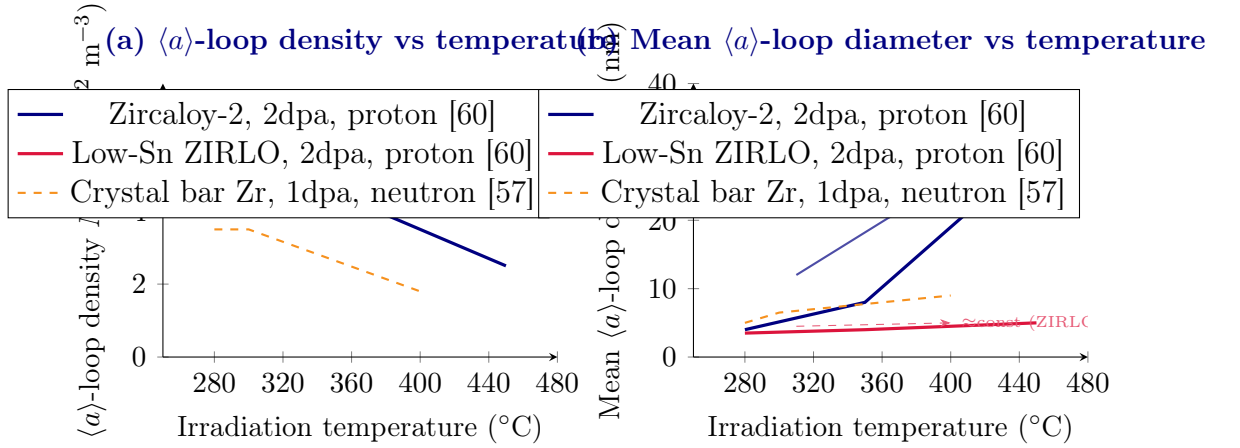


Figure G.9: Temperature dependence of (a) $\langle a \rangle$ -loop number density $N_{\langle a \rangle}$ and (b) mean loop diameter $\bar{d}$ at fixed dose ($\sim 2\text{dpa}$) for Zircaloy-2 (blue circles), Low-Sn ZIRLO (red squares), and crystal bar Zr under neutron irradiation (orange dashed triangles). In Zircaloy-2 (Nb-free), the mean loop diameter increases by a factor of ~ 7.5 from 280 to 450 $^\circ\text{C}$ while density decreases. In Low-Sn ZIRLO (Nb-containing), both density and size are nearly temperature-independent—attributed to Nb nano-cluster pinning of loops [60].

The following quantitative trends are established across all Zr alloys:

1. **Strong temperature dependence in Nb-free alloys.** In Zircaloy-2, the mean $\langle a \rangle$ -loop diameter increases from $\sim 4\text{ nm}$ at 280 $^\circ\text{C}$ to $\sim 30\text{ nm}$ at 450 $^\circ\text{C}$ —a factor of ~ 7.5 —at 2dpa [60]. This is far larger than the factor of ~ 2 observed over a similar temperature range in ferritic/martensitic steels, reflecting the lower binding energies and higher anisotropy of point-defect migration in hcp Zr.

2. **Nb strongly suppresses loop coarsening.** In Low-Sn ZIRLO, the same temperature range produces only a $\sim 1.3\times$ increase in loop diameter (3.5 nm to 5.0 nm) at 2dpa [60]. Nb forms irradiation-induced nano-clusters throughout the matrix that pin $\langle a \rangle$ loops and prevent both their growth and basal-plane alignment. This is the root cause of ZIRLO's superior resistance to irradiation growth compared to Zircaloy-2.
3. **Density decreases with temperature (Nb-free).** In Zircaloy-2, $N_{\langle a \rangle}$ decreases from $\sim 5 \times 10^{22} \text{ m}^{-3}$ at 280 °C to $\sim 2.5 \times 10^{22} \text{ m}^{-3}$ at 450 °C, consistent with thermal coarsening.
4. **Vacancy vs interstitial loop transition.** Below ~ 300 °C, interstitial $\langle a \rangle$ loops dominate. Above ~ 360 °C, vacancy $\langle a \rangle$ loops become prevalent and grow to very large sizes (tens of nm), consistent with the lower thermal stability of interstitial loops at elevated temperatures [57].

Dose Dependence at Fixed Temperature.

Figure G.10 shows the dose evolution of $\langle a \rangle$ -loop density and mean size at fixed temperature.

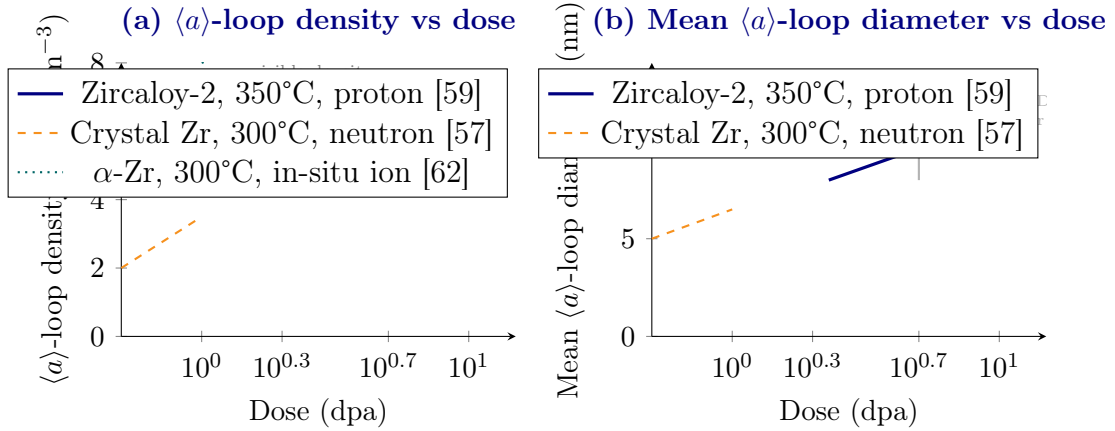


Figure G.10: Dose dependence of (a) $\langle a \rangle$ -loop density and (b) mean diameter at fixed temperature. In Zircaloy-2 at 350 °C (blue circles), the visible loop density approaches saturation by ~ 7 dpa while mean size continues to increase. Synchrotron X-ray diffraction (SXRD) measurements reveal that the total dislocation density is $4\text{--}15\times$ higher than TEM-visible density, indicating a large population of sub-TEM-resolution loops [59, 65].

Key dose-dependence observations:

- In Zircaloy-2 at 350 °C, TEM-visible $\langle a \rangle$ -loop density increases from $4.5 \times 10^{22} \text{ m}^{-3}$ at 2.3dpa to $6.0 \times 10^{22} \text{ m}^{-3}$ at 7.0dpa, approaching saturation [59]. SXR-measured total dislocation density is 4–15 $\times$ higher, indicating that many sub-3 nm loops remain below the TEM detection threshold.
- The $\langle a \rangle$ -loop density saturates at $\sim 10^{22} \text{ m}^{-3}$ for both neutron-irradiated and ion-irradiated Zr alloys under typical LWR conditions [1].
- Mean loop diameter grows continuously with dose following approximately $\bar{d} \propto \phi^{0.3}$ (section Appendix G).

Complementing the $\langle a \rangle$ -loops, the faulted $\langle c \rangle$ -component loops control breakaway growth and are characterized next.

Formation and Character.

$\langle c \rangle$ -component loops have Burgers vector $\mathbf{b} = \frac{1}{6}\langle 20\bar{2}3 \rangle$ and lie in the basal $\{0001\}$ plane. They are of *vacancy type only* and form at doses typically above $\sim 2\text{--}5\text{dpa}$ under reactor-relevant conditions. Their formation is of prime importance because they are responsible for *breakaway growth*—the accelerated irradiation-induced elongation along the $\langle c \rangle$ axis observed in recrystallized Zr alloys at high fluences [4, 30].

$\langle c \rangle$ loops are much larger than $\langle a \rangle$ loops but have a $\sim 10\times$ lower number density. In neutron-irradiated Zircaloy-2 at high burnup, $\langle c \rangle$ loops grow to 100–400 nm in diameter while $N_{\langle c \rangle} \sim 10^{21} \text{ m}^{-3}$.

Dose Dependence and Breakaway Growth Correlation.

Figure G.11 shows the development of $\langle c \rangle$ -loop density and mean diameter with dose in Zircaloy-4 under proton irradiation [30].

1. **Incubation period.** $\langle c \rangle$ loops do not form immediately; they appear after a threshold dose of $\sim 2\text{dpa}$ under proton irradiation at 350 °C, and somewhat later under neutron irradiation where the incubation is related to the dissolution of second-phase particles [4].
2. **Rapid size growth.** Once nucleated, $\langle c \rangle$ loops grow rapidly: from $\sim 20\text{ nm}$ at 3dpa to $\sim 80\text{ nm}$ at 11.5dpa (proton) [30]. Under neutron irradiation, $\langle c \rangle$ loops reach 100–300 nm at 5–10dpa.
3. **Nb strongly suppresses $\langle c \rangle$ loop formation.** Nb-containing alloys (ZIRLO, Zr-1Nb) have much lower $\langle c \rangle$ loop densities than Nb-free

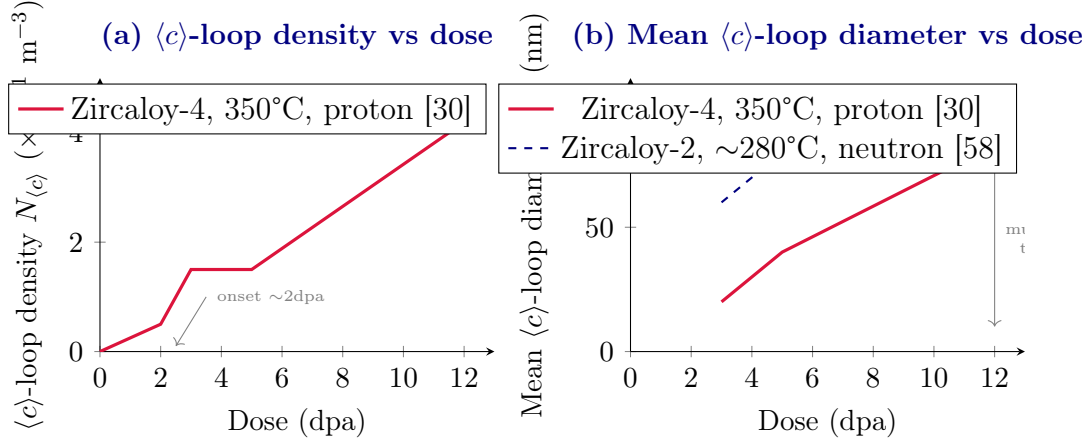


Figure G.11: Dose dependence of (a) $\langle c \rangle$ -loop number density and (b) mean diameter in Zircaloy-4 under proton irradiation at 350°C (red squares, (author?) [30]) and Zircaloy-2 under neutron irradiation at $\sim 280^\circ\text{C}$ (dashed blue circles, (author?) [58]). $\langle c \rangle$ loops appear above ~ 2 dpa and grow rapidly to 50–100 nm by 10dpa. Their number density ($\sim 10^{21} \text{ m}^{-3}$) is $\sim 10\times$ lower than $\langle a \rangle$ -loop density. The formation of $\langle c \rangle$ loops is strongly correlated with the onset of breakaway growth.

alloys at the same dose. In Zr-1Nb, $N_{\langle c \rangle} < 10^{20} \text{ m}^{-3}$ even at 5dpa, compared to $\sim 10^{21} \text{ m}^{-3}$ in Zircaloy-2 [61]. It is proposed that the limited alignment of $\langle a \rangle$ loops along basal traces (due to Nb pinning) slows $\langle c \rangle$ loop nucleation [60].

4. **Temperature dependence.** $\langle c \rangle$ loops form and grow more rapidly at higher temperatures. In the temperature range of commercial reactors ($250\text{--}320^\circ\text{C}$ for LWR), their formation is accelerated by dissolution of Fe-bearing precipitates, which releases Fe into the matrix and lowers the stacking-fault energy [4].

Beyond their densities and mean sizes, the loops exhibit characteristic size distributions.

General Shape.

Figure G.12 illustrates the evolution of the $\langle a \rangle$ -loop size distribution with irradiation temperature, showing the dramatic broadening and shift to larger sizes in Nb-free vs Nb-containing alloys.

Power-Law Size Distributions and Sub-TEM Detection.

A key finding from combined SXR+TEM studies of Zircaloy-2 [59, 65] is that the visible TEM size distribution (typically log-normal, threshold

Zircaloy-2, 280°C Zircaloy-2, 350°C Zircaloy-2, 450°C

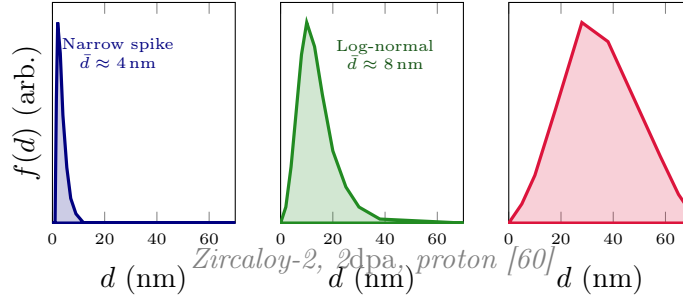


Figure G.12: Schematic $\langle a \rangle$ -loop size distributions in Zircaloy-2 at fixed dose (2dpa) under proton irradiation at three temperatures [60]. At 280 °C, the distribution is a narrow spike centred at ~ 4 nm (interstitial loops dominant). At 350 °C, a log-normal distribution centred at ~ 8 nm develops. At 450 °C, the distribution is very broad with a mean of ~ 30 nm and shifts to the vacancy-loop character.

~ 3 nm) represents only the upper tail of a full power-law distribution that extends to sub-nm sizes:

$$f(d) = A d^{-\alpha}, \quad \alpha \approx 2-3, \quad (\text{G.1})$$

where $f(d)$ is the number density of loops per unit size interval and A is a dose- and temperature-dependent prefactor. Because SXR D detects all loops regardless of size, SXR D-measured total dislocation densities are 4–15 $\times$ higher than TEM values. This implies a large population of 1–3 nm loops that are invisible in conventional TEM but contribute significantly to hardening and the total sink strength for point defects.

Comparison: Neutron vs. Ion Irradiation.

Table G.16 compares key microstructure observables under neutron and ion irradiation.

In contrast to the dislocation loops, voids and cavities play only a limited role in irradiated zirconium.

Void formation in Zr alloys is far less common than in bcc or fcc metals and represents a qualitatively different behaviour from the void swelling observed in ferritic/martensitic steels. The principal reasons are:

1. **Competition with $\langle c \rangle$ loops.** In the temperature window where vacancies are mobile (~ 350 – 500 °C), $\langle c \rangle$ vacancy loops are the preferred sink for vacancies, consuming them before they can form three-dimensional clusters.

Table G.16: Comparison of $\langle a \rangle$ -loop microstructure parameters under neutron and ion irradiation in Zr alloys.

Observable	Neutron irradiation	Proton / ion irradiation
Mean $\langle a \rangle$ -loop diameter	5–10 nm at 280–300 °C; grows with T	Similar or slightly smaller; proton irradiation emulates neutron well
$\langle a \rangle$ -loop density	$\sim (3\text{--}6) \times 10^{22} \text{ m}^{-3}$ (TEM)	Similar; ion higher dose rate $\rightarrow$ comparable microstructure with T shift
$\langle c \rangle$ -loop onset dose	$\sim 2\text{--}5$ dpa (strongly fluence-dependent)	~ 2 dpa at 350 °C (proton); appears earlier than neutron
SPP amorphization	Rim-inward crystalline-to-amorphous; ~ 2 dpa at 285 °C	Homogeneous under heavy ions; ~ 3 dpa at 300 °C (ion)
Corduroy structure	Well-developed loop rows along basal trace	Less-developed; shorter rows at higher dose rate
Sub-TEM loop population	Large; power-law distribution; SXR 4–15 $\times$ higher	Similar power-law; proton emulates neutron SXR distribution

2. **He requirement.** Without insoluble gases (helium from transmutation), small vacancy clusters in Zr collapse to form $\langle a \rangle$ or $\langle c \rangle$ loops rather than cavities. He is necessary to stabilize embryo voids against collapse [4].
3. **Narrow temperature window.** Cavities form only in the range 625–775 K (352–502 °C)—above the threshold for sufficient vacancy mobility and below the temperature where thermal vacancy emission prevents void growth.

When voids do form (typically in high-purity Zr or under conditions of elevated He production), their density is $\sim 10^{21} \text{ m}^{-3}$ and mean diameters are 1–3 nm [4]. Void swelling remains $\lesssim 0.1\%$ even at very high fluences in commercial Zircalloys, making irradiation growth (not swelling) the dominant dimensional change concern [1].

For Zr–2.5Nb pressure tubes (CANDU), He bubbles of ~ 1.5 nm diameter nucleate preferentially at grain boundaries and β -Zr/ α -Zr interfaces at densities of $\sim 5 \times 10^{20} \text{ m}^{-3}$ [18]. These He bubbles contribute to delayed hydride cracking susceptibility.

The alloying additions in Zircaloy give rise to second-phase particles, whose evolution under irradiation is documented next.

Amorphization of Intermetallic Precipitates.

Intermetallic second-phase particles (SPPs) in Zircalloys [Zr(Fe,Cr)₂ and Zr₂(Fe,Ni) in Zircaloy-2; Zr(Fe,Cr)₂ in Zircaloy-4] undergo a progressive crystalline-to-amorphous transformation during neutron irradiation [66, 67]. Figure G.13 illustrates the amorphization mechanism and its temperature dependence.

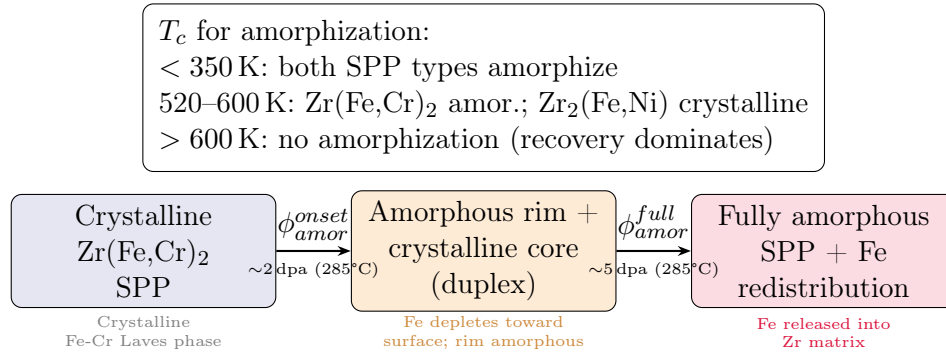


Figure G.13: Schematic of the neutron-irradiation-induced amorphization of Zr(Fe,Cr)₂ intermetallic precipitates in Zircaloy-2 and -4. Amorphization proceeds by a rim-inward mechanism under neutron irradiation, forming a duplex structure [66]. Heavy ions cause homogeneous (bulk) amorphization instead. The critical temperature T_c separates the amorphization regime from the thermal recovery regime [67].

Key quantitative observations on SPP amorphization:

- In Zircaloy-2 at 285 °C (BWR), Zr(Fe,Cr)₂ SPPs are fully amorphous after ~2–3 reactor cycles (equivalent to ~2–3dpa). Zr₂(Fe,Ni) SPPs remain crystalline until ~8–10dpa [66].
- The critical temperature for amorphization is ~600 K (327 °C) for Zr(Fe,Cr)₂. Above this temperature, thermal annealing prevents amorphization [67].
- Under heavy-ion irradiation, amorphization occurs homogeneously (rather than via a rim mechanism) at doses of ~3–8dpa.
- The dissolution of Fe from amorphized SPPs into the Zr matrix eliminates nodular corrosion susceptibility and is linked to the kinetics of $\langle c \rangle$ loop nucleation [4].

Effect of Nb on SPP Stability.

In Nb-containing alloys, the intermetallic precipitates are $\text{Zr}(\text{Fe},\text{Nb})_2$ or $\text{Zr}(\text{Fe},\text{Cr},\text{Nb})_2$. These have the same hcp Laves-phase structure as $\text{Zr}(\text{Fe},\text{Cr})_2$ but undergo partial rather than full amorphization, because Nb modifies the thermodynamics of the amorphous phase [60]. Simultaneously, β -Nb precipitates partially dissolve and reprecipitate, releasing Nb into the matrix as nano-clusters that pin $\langle a \rangle$ loops and suppress $\langle c \rangle$ loop formation [61].

Taken together, these observations can be condensed into a set of scaling laws and empirical correlations.

Power-Law Growth of $\langle a \rangle$ -Loop Mean Diameter.

In the growth/coarsening regime (above the initial accumulation phase), the mean $\langle a \rangle$ -loop diameter follows:

$$\bar{d}_a(\phi) = \bar{d}_0 + A(T) \phi^m, \quad m \approx 0.3\text{--}0.4, \quad (\text{G.2})$$

where $A(T)$ increases strongly with temperature in Nb-free alloys. Fitting the Zircaloy-2 proton data (Balogh 2021) gives $m \approx 0.35$ and $A(350^\circ\text{C}) \approx 5.5 \text{ nm dpa}^{-0.35}$.

Temperature Dependence of Loop Diameter.

An Arrhenius-type relation describes the mean loop diameter at fixed dose:

$$\bar{d}_a(T) \propto \exp\left(\frac{E_{\text{coarsen}}}{k_B T}\right), \quad (\text{G.3})$$

with an apparent activation energy $E_{\text{coarsen}} \approx 0.4\text{--}0.6 \text{ eV}$ for Zircaloy-2 [60], consistent with thermally activated Ostwald ripening. In ZIRLO, Nb effectively eliminates this dependence, reducing the apparent activation energy to $\approx 0.1 \text{ eV}$.

$\langle c \rangle$ -Loop Nucleation Incubation Dose.

The dose at onset of $\langle c \rangle$ -loop nucleation follows:

$$\phi_{\text{incub}}^{\langle c \rangle} \propto \exp\left(\frac{Q}{k_B T}\right), \quad (\text{G.4})$$

with $Q \approx 0.5 \text{ eV}$ [30]. The incubation dose increases from $\sim 2 \text{ dpa}$ at 350°C to $\sim 5 \text{ dpa}$ at 280°C for Zircaloy-4, and is strongly suppressed by Nb alloying.

Irradiation Growth Strain.

The axial irradiation growth strain ε_{zz} (elongation along $\langle a \rangle$ axes) is proportional to the net flux of interstitials to $\langle a \rangle$ loops and vacancies to $\langle c \rangle$

loops:

$$\dot{\varepsilon}_{zz} \propto \frac{1}{3} \rho_a^I \bar{d}_a V_a - \rho_c^V \bar{d}_c V_c, \quad (\text{G.5})$$

where ρ_a^I (ρ_c^V) is the interstitial $\langle a \rangle$ (vacancy $\langle c \rangle$) loop density and $V_{a,c}$ is the loop capture volume. Breakaway growth occurs when $\langle c \rangle$ loops grow sufficiently to dominate the second term, causing rapid elongation along $\langle c \rangle$ axes that is macroscopically observed as rapid axial elongation.

These quantitative trends carry direct implications for the cluster-dynamics model developed in the main text.

The experimental data documented above impose the following constraints on cluster-dynamics (CD) rate-equation models for Zr alloys:

1. **Anisotropic point-defect migration.** The model must implement separate diffusion tensors for interstitials and vacancies in the hcp lattice. Interstitials migrate with a strong in-plane (basal) preference (diffusion anisotropy $D_a/D_c \sim 100$ at LWR temperatures), which drives the formation of prismatic $\langle a \rangle$ loops and prevents void nucleation.
2. **Two independent loop populations.** The CD model must track $\langle a \rangle$ and $\langle c \rangle$ cluster populations separately, with distinct nucleation, growth, and sink-strength contributions. The onset of $\langle c \rangle$ loop formation (~ 2 dpa) and its dose/temperature dependence must be reproduced.
3. **Power-law size distribution.** The experimental finding that the total loop density (from SXRD) is $4\text{--}15\times$ higher than the TEM-visible density requires that the model captures a large population of sub-3 nm loops. The master equation must extend the cluster space to sizes below the TEM detection threshold.
4. **SPP dissolution as a source term.** The progressive amorphization and dissolution of Fe-bearing SPPs provides a time-dependent source of Fe solute in the matrix. This modifies the stacking-fault energy and the nucleation barrier for $\langle c \rangle$ loops. A physically complete CD model must couple the SPP evolution (described by a ballistic mixing model [66]) to the point-defect balance equations.
5. **Nb nano-cluster pinning.** The near-temperature-independence of $\langle a \rangle$ -loop density and size in Nb-containing alloys indicates strong loop

pinning. This can be incorporated as an effective loop-migration energy that is strongly increased in Nb-containing alloys.

We conclude by collecting the principal quantitative trends in a single overview.

Table G.17 condenses the principal quantitative trends.

Quantitative anchors:

- $\langle a \rangle$ -loop density: saturates at $\sim (3\text{--}6) \times 10^{22} \text{ m}^{-3}$ (TEM); total density 4–15 $\times$ higher (SXR).
- $\langle a \rangle$ -loop mean size: 3–5 nm at 280 °C; 8–12 nm at 350 °C; 25–40 nm at 450 °C (Zircaloy-2, $\sim 2\text{dpa}$).
- $\langle c \rangle$ -loop density: $\sim 10^{21} \text{ m}^{-3}$ at $>5\text{dpa}$.
- $\langle c \rangle$ -loop mean size: 20–100 nm at 3–12dpa; up to 300 nm at high neutron burnup.
- SPP amorphization dose: $\sim 2\text{dpa}$ (onset) to $\sim 5\text{dpa}$ (full) at 280–285 °C for $\text{Zr}(\text{Fe},\text{Cr})_2$.
- Void density (when formed): $\sim 10^{21} \text{ m}^{-3}$; mean $\sim 2\text{ nm}$; only at 352–502 °C.

Appendix H. Defect Conservation

We show that the master equations (30)–(35) conserve point-defect mass exactly: every internal channel—ladder absorption and emission, cluster coagulation, mutual recombination, and $\text{SFP} \rightarrow \langle c \rangle$ unfaulting—moves interstitials and vacancies between bins without creating or destroying them, so the total interstitial and vacancy contents change only through the external source and the fixed sinks. The resulting balance is the irradiation-growth identity used in Section ??.

Let each loop or pyramid of class $X \in \{\text{Ia}, \text{Va}, S, \text{Vc}\}$ and size n contain exactly n elementary defects of its own sign. The total numbers of interstitials and vacancies bound in clusters, plus the free monomers, are

$$\mathcal{I} = c^I + \sum_{n \geq 2} n c_n^{\text{Ia}}, \quad \mathcal{V} = c^V + \sum_{n \geq 2} n \left(c_n^{\text{Va}} + c_n^S + c_n^{\text{Vc}} \right). \quad (\text{H.1})$$

Table G.17: Summary of quantitative trends in defect density and mean size as functions of temperature and dose in Zr alloys. $\uparrow/\downarrow$ = increases/decreases.

Quantity	With temperature T (fixed dose)		With dose ϕ (fixed T)	
	Nb-free (Zy-2, Zy-4)	Nb- bearing (ZIRLO, Zr- 1Nb)	$\phi <$ 3dpa	$\phi >$ 5dpa
$\langle a \rangle$ -loop density $N_{\langle a \rangle}$	$\downarrow$ (TEM); all T (SXRD higher)	Nearly constant; T- insensitive	$\uparrow$ rapidly	$\approx$ satu- rated (TEM)
$\langle a \rangle$ -loop mean size $\bar{d}_a$	$\uparrow\uparrow$ ($\times 7.5$ from 280 $\rightarrow$ 450°C)	Nearly constant ($\times 1.3$ only)	$\uparrow$	$\uparrow$ (coars- ening)
$\langle c \rangle$ -loop density $N_{\langle c \rangle}$	$\uparrow$ with T	Very low; T- insensitive	Zero $\rightarrow$ onset ~ 2 dpa	$\uparrow$ (break- away)
$\langle c \rangle$ -loop mean size $\bar{d}_c$	$\uparrow\uparrow$ (100–300 nm at high dose)	Very small (if any)	—	$\uparrow\uparrow$ (10–300 nm)
SPP amorphization	Progressive; Zr(Fe,Cr) ₂ full by 5dpa	Partial; Nb modifies	Rim onset	Full amor- phiza- tion
Irradiation growth	Normal then break- away (> 5 dpa)	Lower; break- away delayed	$\approx$ linear	Breakaway onset

Here c^I and c^V are the $n = 1$ members (Eqs. (34)–(35)), so the sums in (H.1) run over the bound spectrum $n \geq 2$ while the monomers carry the $n = 1$ weight explicitly. The terms N_I^{net} and N_V^{net} are the interstitial and vacancy contents permanently stored in the dislocation network, each resolved into a prismatic and a basal part ($N_X^{\text{net}} = N_X^{\text{pr}} + N_X^{\text{ba}}$, $X = I, V$), so that the absorbed-loop content removed from the ladders by processes P8 and P9 is retained in the bookkeeping rather than lost.

Differentiating (H.1) and inserting the master equations, we collect the internal terms by physical channel.

Ladder absorption and emission.. Weighting the cluster ladder by n and using the telescoping identity $\sum_n n(a_{n-1} - a_n) = \sum_n a_n$ for the absorption pair (and its reverse for emission), the bound interstitial content gains

$$\sum_{n \geq 2} n \left[K_{x,n-1}^X c^x c_{n-1}^X - K_{x,n}^X c^x c_n^X \right] = + c^x \sum_{n \geq 1} K_{x,n}^X c_n^X, \quad (\text{H.2})$$

one bound defect added per capture. This is exactly cancelled by the net-absorption term $-c^x \sum_X \sum_n K_{x,n}^X c_n^X$ that the same capture removes from the monomer bath in (34)–(35). The emission sums $\sum_{n \geq 2} \varepsilon^X(n) c_n^X$ returned to the monomers likewise cancel the $-\varepsilon^X(n) c_n^X$ ladder losses. Absorption and emission are thus mass-neutral.

Coagulation.. The reaction $n+n \rightarrow n+n$ conserves size, $n+n$, so the factor- $\frac{1}{2}$ gain and the loss in (30) (and its Va analogue) cancel identically in $\sum_n n(\cdot)$: coagulation merely repartitions bound defects and contributes nothing to $\dot{\mathcal{I}}$ or $\dot{\mathcal{V}}$.

Recombination.. The mutual term $-K_{iv} c^I c^V$ appears once in each monomer equation, removing one interstitial and one vacancy together. It therefore reduces $\mathcal{I}$ and $\mathcal{V}$ by the *same* amount and drops out of the *difference* $\mathcal{V} - \mathcal{I}$.

SFP unfaulding.. The Zr-specific transfer $\Gamma_{\text{unf}}(n) c_n^S$ leaves the pyramid ladder (32) and enters the $\langle c \rangle$ ladder (33) at the same size n . Both carry weight n , so the loss and gain cancel in $\dot{\mathcal{V}}$: unfaulding conserves vacancy content while reclassifying it from S to Vc .

After every internal channel is canceled, only the external production and

the fixed sinks survive:

$$\frac{d\mathcal{I}}{dt} = \mathcal{G}_I - \mathcal{D}_I - K_{iv}c^I c^V, \quad (\text{H.3})$$

$$\frac{d\mathcal{V}}{dt} = \mathcal{G}_V - \mathcal{D}_V - K_{iv}c^I c^V, \quad (\text{H.4})$$

where $\mathcal{G}_I = G_1^I + \sum_{n \geq 2} n G_n^{\text{Ia}}$ and $\mathcal{G}_V = G_1^V + \sum_{n \geq 2} n (G_n^{\text{Va}} + G_n^S + G_n^{\text{Vc}})$ are the total interstitial and vacancy production rates, and $\mathcal{D}_I = (\mathcal{D}_i^{\text{pr}} + \mathcal{D}_i^{\text{ba}})c^I + \sum_{n \geq 2} n \Theta(n - n_{\text{inc}}) \Lambda_n^{\text{pr}} c_n^{\text{Ia}}$, $\mathcal{D}_V = (\mathcal{D}_v^{\text{pr}} + \mathcal{D}_v^{\text{ba}})c^V + \sum_{n \geq 2} n \Theta(n - n_{\text{inc}}) \Lambda_n^{\text{ba}} c_n^{\text{Vc}}$ collect the fixed-sink (RAG P4) and dislocation-bias losses. The matching network accumulators grow at exactly the rate the monomers and incorporated loops are absorbed, resolved by component,

$$\dot{N}_I^{\text{pr}} = \mathcal{D}_i^{\text{pr}} c^I + \sum_{n \geq 2} n \Theta(n - n_{\text{inc}}) \Lambda_n^{\text{pr}} c_n^{\text{Ia}}, \quad \dot{N}_I^{\text{ba}} = \mathcal{D}_i^{\text{ba}} c^I, \quad (\text{H.5})$$

$$\dot{N}_V^{\text{pr}} = \mathcal{D}_v^{\text{pr}} c^V + \sum_{n \geq 2} n \Theta(n - n_{\text{inc}}) \Lambda_n^{\text{pr}} c_n^{\text{Va}}, \quad \dot{N}_V^{\text{ba}} = \mathcal{D}_v^{\text{ba}} c^V + \sum_{n \geq 2} n \Theta(n - n_{\text{inc}}) \Lambda_n^{\text{ba}} c_n^{\text{Vc}}, \quad (\text{H.6})$$

so that the grand totals $\mathcal{I} + N_I^{\text{net}}$ and $\mathcal{V} + N_V^{\text{net}}$ change only through production and mutual recombination, and every internal absorption is bookkept rather than lost. The same incorporation fluxes drive the basal and prismatic line densities through Eqs. (27)–(28): each absorbed loop adds about one circumference of line to its own component, so ρ_{pr} and ρ_{ba} grow in step with $N_I^{\text{pr}}, N_V^{\text{pr}}$ and $N_I^{\text{ba}}, N_V^{\text{ba}}$ respectively. The growing network thus closes the point-defect balance on two channels at once: it stores the absorbed interstitial and vacancy content (conservation) and, because $Z_i^c \neq Z_v^c$ for each component $c \in \{\text{pr}, \text{ba}\}$, biases the net flux that feeds irradiation growth alongside the $\langle a \rangle$ - and $\langle c \rangle$ -loop ladders.

Under Frenkel-pair production, each cascade injects interstitials and vacancies in equal *total* numbers, $\sum_{n \geq 1} n G_n^{(i)} = \sum_{n \geq 1} n G_n^{(v)} \equiv G$, but they are partitioned differently between free monomers and clusters, so the monomer sources are biased $G_1^V - G_1^I = \beta_p \neq 0$ (the production bias of Eq. (36)). Since the clustered atoms are produced in conserved interstitial/vacancy counts, the net total-production imbalance reduces to this monomer bias, $\mathcal{G}_V - \mathcal{G}_I = \beta_p$. Subtracting (H.3) from (H.4), the production and recomb-

nation terms cancel exactly and we obtain the *irradiation-growth identity*

$$\boxed{\frac{d}{dt}(\mathcal{V} - \mathcal{I}) = \beta_p + \mathcal{D}_I - \mathcal{D}_V = (\mathcal{D}_i c^I - \mathcal{D}_v c^V) + \sum_{n \geq 2} n \Theta(n - n_{\text{inc}}) (\Lambda_n^{\text{pr}} c_n^{\text{Ia}} - \Lambda_n^{\text{ba}} c_n^{\text{Vc}})} \quad (\text{H.7})$$

Equation (H.7) states that the *net retained vacancy excess*, $\mathcal{V} - \mathcal{I}$, grows only through the asymmetry of the sinks: the difference in monomer loss to fixed sinks plus the bias-driven partitioning of supercritical $\langle a \rangle$ - and $\langle c \rangle$ -loops at the biased dislocation network. In the doubly unbiased limit—no production bias ($\beta_p = 0$) and no sink bias ($\mathcal{D}_i^{\text{pr}} + \mathcal{D}_i^{\text{ba}} = \mathcal{D}_v^{\text{pr}} + \mathcal{D}_v^{\text{ba}}$ acting on equal monomer populations, and $c_n^{\text{Ia}} = c_n^{\text{Vc}}$)—the right-hand side vanishes and $\mathcal{V} - \mathcal{I}$ is conserved—no growth. The accumulated excess maps onto the macroscopic growth strain through the partial volumes of the loop families,

$$\varepsilon_g(t) = \frac{1}{\Omega_{\text{at}}} \int_0^t \frac{d}{dt'} (\mathcal{V} - \mathcal{I}) dt', \quad (\text{H.8})$$

so that conservation at the defect level translates directly into the dimensional change reported in Section ??.

Acknowledgments

The author acknowledges support from the U.S. Department of Energy, Office of Fusion Energy Sciences, under Award No. DE-SC0018410:0008 to UCLA.

Declarations

- **Ethics Declaration:** The author declares that this work is original, has been conducted in accordance with accepted standards of research integrity, and does not involve human or animal subjects.
- **Ethics approval and consent to participate:** Not applicable.
- **Consent for publication:** Not applicable.
- **Competing interests:** The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.
- **Author contributions:** Nasr M. Ghoniem conceived the work, performed the analysis, developed the code, and wrote the manuscript.
- **Declaration of generative AI-assisted technologies :** During the preparation of this work, the author(s) used Anthropic Claude in order to research published literature, assist in coding, and prepare figures. After using this service, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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